

CHAPTER 3

Systems of Linear Equations

3.1 Introduction

Systems of linear equations play an important and motivating role in the subject of linear algebra. In fact, many problems in linear algebra reduce to finding the solution of a system of linear equations. Thus, the techniques introduced in this chapter will be applicable to abstract ideas introduced later. On the other hand, some of the abstract results will give us new insights into the structure and properties of systems of linear equations.

All our systems of linear equations involve scalars as both coefficients and constants, and such scalars may come from any number field K . There is almost no loss in generality if the reader assumes that all our scalars are real numbers—that is, that they come from the real field $\mathbf{R}$.

3.2 Basic Definitions, Solutions

This section gives basic definitions connected with the solutions of systems of linear equations. The actual algorithms for finding such solutions will be treated later.

Linear Equation and Solutions

A *linear equation* in unknowns $x_1, x_2, \dots, x_n$ is an equation that can be put in the *standard form*

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b \quad (3.1)$$

where $a_1, a_2, \dots, a_n$, and b are constants. The constant a_k is called the *coefficient* of x_k , and b is called the *constant term* of the equation.

A solution of the linear equation (3.1) is a list of values for the unknowns or, equivalently, a vector u in K^n , say

$$x_1 = k_1, \quad x_2 = k_2, \quad \dots, \quad x_n = k_n \quad \text{or} \quad u = (k_1, k_2, \dots, k_n)$$

such that the following statement (obtained by substituting k_i for x_i in the equation) is true:

$$a_1k_1 + a_2k_2 + \cdots + a_nk_n = b$$

In such a case we say that u *satisfies* the equation.

Remark: Equation (3.1) implicitly assumes there is an ordering of the unknowns. In order to avoid subscripts, we will usually use x, y for two unknowns; x, y, z for three unknowns; and x, y, z, t for four unknowns; they will be ordered as shown.

EXAMPLE 3.1 Consider the following linear equation in three unknowns x, y, z :

$$x + 2y - 3z = 6$$

We note that $x = 5$, $y = 2$, $z = 1$, or, equivalently, the vector $u = (5, 2, 1)$ is a solution of the equation. That is,

$$5 + 2(2) - 3(1) = 6 \quad \text{or} \quad 5 + 4 - 3 = 6 \quad \text{or} \quad 6 = 6$$

On the other hand, $w = (1, 2, 3)$ is not a solution, because on substitution, we do not get a true statement:

$$1 + 2(2) - 3(3) = 6 \quad \text{or} \quad 1 + 4 - 9 = 6 \quad \text{or} \quad -4 = 6$$

System of Linear Equations

A system of linear equations is a list of linear equations with the same unknowns. In particular, a system of m linear equations $L_1, L_2, \dots, L_m$ in n unknowns $x_1, x_2, \dots, x_n$ can be put in the *standard form*

[illegible]

where the a_{ij} and b_i are constants. The number a_{ij} is the *coefficient* of the unknown x_j in the equation L_i , and the number b_i is the *constant* of the equation L_i .

The system (3.2) is called an $m\ n$ (read: m by n) system. It is called a *square system* if $m = n$ —that is, if the number m of equations is equal to the number n of unknowns.

The system (3.2) is said to be *homogeneous* if all the constant terms are zero—that is, if $b_1 = 0$, $b_2 = 0, \dots, b_m = 0$. Otherwise the system is said to be *nonhomogeneous*.

A *solution* (or a *particular solution*) of the system (3.2) is a list of values for the unknowns or, equivalently, a vector u

in K^n , which is a solution of each of the equations in the system. The set of all solutions of the system is called the *solution set* or the *general solution* of the system.

EXAMPLE 3.2 Consider the following system of linear equations:

$$\begin{aligned}x_1 + x_2 + 4x_3 + 3x_4 &= 5 \\2x_1 + 3x_2 + x_3 - 2x_4 &= 1 \\x_1 + 2x_2 - 5x_3 + 4x_4 &= 3\end{aligned}$$

It is a 3 4 system because it has three equations in four unknowns. Determine whether (a) $u = (8, 6, 1, 1)$ and (b) $v = (10, 5, 1, 2)$ are solutions of the system.

(a) Substitute the values of u in each equation, obtaining

$$\begin{array}{llll} -8 + 6 + 4(1) + 3(1) = 5 & \text{or} & -8 + 6 + 4 + 3 = 5 & \text{or} & 5 = 5 \\ 2(-8) + 3(6) + 1 - 2(1) = 1 & \text{or} & -16 + 18 + 1 - 2 = 1 & \text{or} & 1 = 1 \\ -8 + 2(6) - 5(1) + 4(1) = 3 & \text{or} & -8 + 12 - 5 + 4 = 3 & \text{or} & 3 = 3 \end{array}$$

Yes, u is a solution of the system because it is a solution of each equation.

(b) Substitute the values of v into each successive equation, obtaining

$$\begin{array}{llll} -10 + 5 + 4(1) + 3(2) = 5 & \text{or} & -10 + 5 + 4 + 6 = 5 & \text{or} & 5 = 5 \\ 2(-10) + 3(5) + 1 - 2(2) = 1 & \text{or} & -20 + 15 + 1 - 4 = 1 & \text{or} & -8 = 1 \end{array}$$

No, v is not a solution of the system, because it is not a solution of the second equation. (We do not need to substitute v into the third equation.)

The system (3.2) of linear equations is said to be *consistent* if it has one or more solutions, and it is said to be *inconsistent* if it has no solution. If the field K of scalars is

infinite, such as when K is the real field $\mathbf{R}$ or the complex field $\mathbf{C}$, then we have the following important result.

THEOREM 3.1: Suppose the field K is infinite. Then any system $\mathcal{L}$ of linear equations has (i) a unique solution, (ii) no solution, or (iii) an infinite number of solutions.

This situation is pictured in Fig. 3-1. The three cases have a geometrical description when the system $\mathcal{L}$ consists of two equations in two unknowns (Section 3.4).

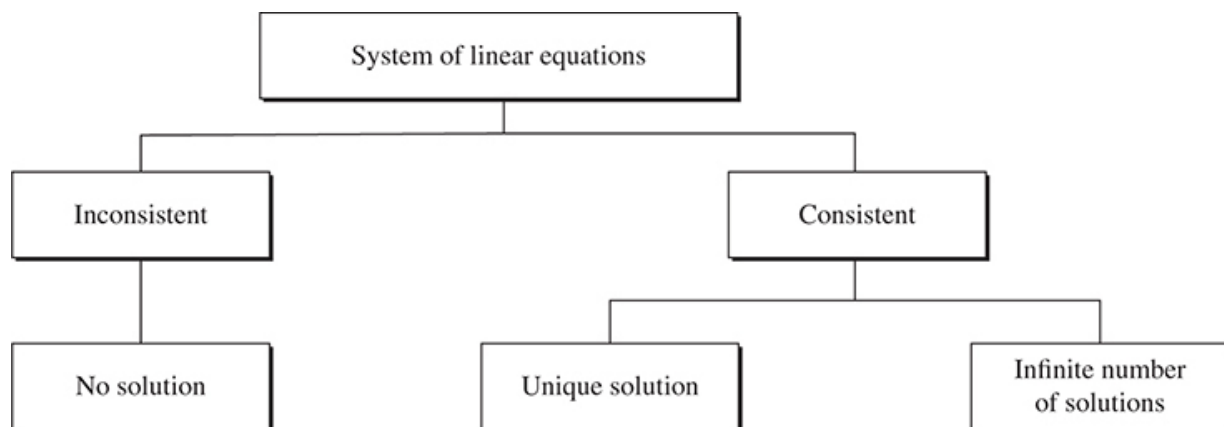


Figure 3-1

Augmented and Coefficient Matrices of a System

Consider again the general system (3.2) of m equations in n unknowns. Such a system has associated with it the following two matrices:

$$M = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_n \end{bmatrix} \quad \text{and} \quad A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

The first matrix M is called the *augmented matrix* of the system, and the second matrix A is called the *coefficient matrix*.

The coefficient matrix A is simply the matrix of coefficients, which is the augmented matrix M without the last column of constants. Some texts write $M = [A, B]$ to emphasize the two parts of M , where B denotes the column vector of constants. The augmented matrix M and the coefficient matrix A of the system in [Example 3.2](#) are as follows:

$$M = \begin{bmatrix} 1 & 1 & 4 & 3 & 5 \\ 2 & 3 & 1 & -2 & 1 \\ 1 & 2 & -5 & 4 & 3 \end{bmatrix} \quad \text{and} \quad A = \begin{bmatrix} 1 & 1 & 4 & 3 \\ 2 & 3 & 1 & -2 \\ 1 & 2 & -5 & 4 \end{bmatrix}$$

As expected, A consists of all the columns of M except the last, which is the column of constants.

Clearly, a system of linear equations is completely determined by its augmented matrix M , and vice versa. Specifically, each row of M corresponds to an equation of the system, and each column of M corresponds to the coefficients of an unknown, except for the last column, which corresponds to the constants of the system.

Degenerate Linear Equations

A linear equation is said to be *degenerate* if all the coefficients are zero—that is, if it has the form

$$0x_1 + 0x_2 + \cdots + 0x_n = b \tag{3.3}$$

The solution of such an equation depends only on the value of the constant b . Specifically,

- (i) If $b \neq 0$, then the equation has no solution.

- (ii) If $b = 0$, then every vector $u = (k_1, k_2, \dots, k_n)$ in K^n is a solution.

The following theorem applies.

THEOREM 3.2: Let $\mathcal{L}$ be a system of linear equations that contains a degenerate equation L , say with constant b .

- (i) If $b \neq 0$, then the system $\mathcal{L}$ has no solution.
- (ii) If $b = 0$, then L may be deleted from the system without changing the solution set of the system.

Part (i) comes from the fact that the degenerate equation has no solution, so the system has no solution. Part (ii) comes from the fact that every element in K^n is a solution of the degenerate equation.

Leading Unknown in a Nondegenerate Linear Equation

Now let L be a nondegenerate linear equation. This means one or more of the coefficients of L are not zero. By the *leading unknown* of L , we mean the first unknown in L with a nonzero coefficient. For example, x_3 and y are the leading unknowns, respectively, in the equations

$$0x_1 + 0x_2 + 5x_3 + 6x_4 + 0x_5 + 8x_6 = 7 \quad \text{and} \quad 0x + 2y - 4z = 5$$

We frequently omit terms with zero coefficients, so the above equations would be written as

$$5x_3 + 6x_4 + 8x_6 = 7 \quad \text{and} \quad 2y - 4z = 5$$

In such a case, the leading unknown appears first.

3.3 Equivalent Systems, Elementary Operations

Consider the system (3.2) of m linear equations in n unknowns. Let L be the linear equation obtained by multiplying the m equations by constants $c_1, c_2, \dots, c_m$, respectively, and then adding the resulting equations. Specifically, let L be the following linear equation:

$$(c_1 a_{11} + \cdots + c_m a_{m1})x_1 + \cdots + (c_1 a_{1n} + \cdots + c_m a_{mn})x_n = c_1 b_1 + \cdots + c_m b_m$$

Then L is called a *linear combination* of the equations in the system. One can easily show ([Problem 3.43](#)) that any solution of the system (3.2) is also a solution of the linear combination L .

EXAMPLE 3.3 Let L_1, L_2, L_3 denote, respectively, the three equations in [Example 3.2](#). Let L be the equation obtained by multiplying L_1, L_2, L_3 by 3, 2, 4, respectively, and then adding. Namely,

$$\begin{array}{rcl} 3L_1: & 3x_1 + 3x_2 + 12x_3 + 9x_4 & = 15 \\ -2L_2: & -4x_1 - 6x_2 - 2x_3 + 4x_4 & = -2 \\ 4L_3: & 4x_1 + 8x_2 - 20x_3 + 16x_4 & = 12 \\ \hline (\text{Sum}) L: & 3x_1 + 5x_2 - 10x_3 + 29x_4 & = 25 \end{array}$$

Then L is a linear combination of L_1, L_2, L_3 . As expected, the solution $u = (8, 6, 1, 1)$ of the system is also a solution of L . That is, substituting u in L , we obtain a true statement:

$$3(-8) + 5(6) - 10(1) + 29(1) = 25 \quad \text{or} \quad -24 + 30 - 10 + 29 = 25 \quad \text{or} \quad 9 = 9$$

THEOREM 3.3: Two systems of linear equations have the same solutions if and only if each equation in each system is a linear combination of the equations in the other system.

Two systems of linear equations are said to be *equivalent* if they have the same solutions. The next subsection shows one way to obtain equivalent systems of linear equations.

Elementary Operations

The following operations on a system of linear equations $L_1, L_2, \dots, L_m$ are called *elementary operations*.

[E₁] Interchange two of the equations. We indicate that the equations L_i and L_j are interchanged by writing:

$$\text{"Interchange } L_i \text{ and } L_j\text{"} \quad \text{or} \quad "L_i \longleftrightarrow L_j"$$

[E₂] Replace an equation by a nonzero multiple of itself. We indicate that equation L_i is replaced by kL_i (where $k \neq 0$) by writing

$$\text{"Replace } L_i \text{ by } kL_i\text{"} \quad \text{or} \quad "kL_i \rightarrow L_i"$$

[E₃] Replace an equation by the sum of a multiple of another equation and itself. We indicate that equation L_j is replaced by the sum of kL_i and L_j by writing

$$\text{"Replace } L_j \text{ by } kL_i + L_j\text{"} \quad \text{or} \quad "kL_i + L_j \rightarrow L_j"$$

The arrow $\rightarrow$ in [E₂ and [E₃ may be read as "replaces."

The main property of the above elementary operations is contained in the following theorem (proved in [Problem 3.45](#)).

THEOREM 3.4: Suppose a system of $\mathcal{M}$ of linear equations is obtained from a system $\mathcal{L}$ of linear equations by a finite sequence of elementary operations. Then $\mathcal{M}$ and $\mathcal{L}$ have the same solutions.

Remark: Sometimes (say to avoid fractions when all the given scalars are integers) we may apply E_2 and $[E_3]$ in one step; that is, we may apply the following operation:

$[E]$ Replace equation L_j by the sum of kL_i and $k' L_j$ (where $k' \neq 0$), written

$$\text{“Replace } L_j \text{ by } kL_i + k'L_j\text{”} \quad \text{or} \quad \text{“} kL_i + k'L_j \rightarrow L_j \text{”}$$

We emphasize that in operations $[E_3]$ and $[E]$, only equation L_j is changed.

Gaussian elimination, our main method for finding the solution of a given system of linear equations, consists of using the above operations to transform a given system into an equivalent system whose solution can be easily obtained.

The details of Gaussian elimination are discussed in subsequent sections.

3.4 Small Square Systems of Linear Equations

This section considers the special case of one equation in one unknown, and two equations in two unknowns. These simple systems are treated separately because their

solution sets can be described geometrically, and their properties motivate the general case.

Linear Equation in One Unknown

The following simple basic result is proved in [Problem 3.5](#).

THEOREM 3.5: Consider the linear equation $ax = b$.

- (i) If $a \neq 0$, then $x = b/a$ is a unique solution of $ax = b$.
- (ii) If $a = 0$, but $b \neq 0$, then $ax = b$ has no solution.
- (iii) If $a = 0$ and $b = 0$, then every scalar k is a solution of $ax = b$.

EXAMPLE 3.4 Solve (a) $4x + 1 = x + 6$, (b) $2x + 5 = x + 3$, (c) $4 + x = 2x + 1$.

- (a) Rewrite the equation in standard form obtaining $3x = 5$. Then $x = \frac{5}{3}$ is the unique solution [[Theorem 3.5 \(i\)](#)].
- (b) Rewrite the equation in standard form, obtaining $0x = 2$. The equation has no solution [[Theorem 3.5 \(ii\)](#)].
- (c) Rewrite the equation in standard form, obtaining $0x = 0$. Then every scalar k is a solution [[Theorem 3.5 \(iii\)](#)].

System of Two Linear Equations in Two Unknowns (2×2 System)

Consider a system of two nondegenerate linear equations in two unknowns x and y , which can be put in the standard form

$$\begin{aligned} A_1x + B_1y &= C_1 \\ A_2x + B_2y &= C_2 \end{aligned} \tag{3.4}$$

Because the equations are nondegenerate, A_1 and B_1 are not both zero, and A_2 and B_2 are not both zero.

The general solution of the system (3.4) belongs to one of three types as indicated in Fig. 3-1. If $\mathbf{R}$ is the field of scalars, then the graph of each equation is a line in the plane $\mathbf{R}^2$ and the three types may be described geometrically as pictured in Fig. 3-2. Specifically,

(1) *The system has exactly one solution.*

Here the two lines intersect in one point [Fig. 3-2 (a)]. This occurs when the lines have distinct slopes or, equivalently, when the coefficients of x and y are not proportional:

$$\frac{A_1}{A_2} \neq \frac{B_1}{B_2} \quad \text{or, equivalently,} \quad A_1 B_2 - A_2 B_1 \neq 0$$

For example, in Fig. 3-2 (a), $1/3 \neq 1/2$.

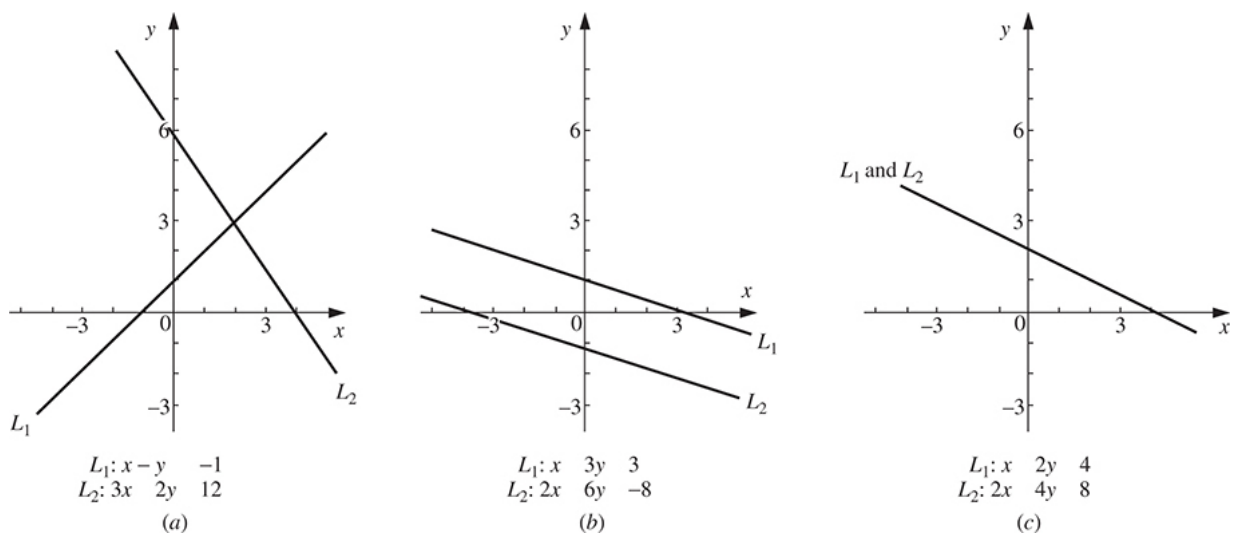


Figure 3-2

(2) *The system has no solution.*

Here the two lines are parallel [Fig. 3-2 (b)]. This occurs when the lines have the same slopes but different y

intercepts, or when

$$\frac{A_1}{A_2} = \frac{B_1}{B_2} \neq \frac{C_1}{C_2}$$

For example, in Fig. 3-2 (b), $1/2 = 3/6 \neq 3/8$.

(3) *The system has an infinite number of solutions.*

Here the two lines coincide [Fig. 3-2 (c)]. This occurs when the lines have the same slopes and same y intercepts, or when the coefficients and constants are proportional, $A_1/A_2 = B_1/B_2 = C_1/C_2$

$$\frac{A_1}{A_2} = \frac{B_1}{B_2} = \frac{C_1}{C_2}$$

For example, in Fig. 3-2 (c), $1/2 = 2/4 = 4/8$.

Remark: The following expression and its value is called a *determinant of order two*:

$$\begin{vmatrix} A_1 & B_1 \\ A_2 & B_2 \end{vmatrix} = A_1B_2 - A_2B_1$$

Determinants will be studied in Chapter 8. Thus, the system (3.4) has a unique solution if and only if the determinant of its coefficients is not zero. (We show later that this statement is true for any square system of linear equations.)

Elimination Algorithm

The solution to system (3.4) can be obtained by the process of elimination, whereby we reduce the system to a single equation in only one unknown. Assuming the system has a unique solution, this elimination algorithm has two parts.

ALGORITHM 3.1: The input consists of two nondegenerate linear equations L_1 and L_2 in two unknowns with a unique solution.

Part A. (Forward Elimination) Multiply each equation by a constant so that the resulting coefficients of one unknown are negatives of each other, and then add the two equations to obtain a new equation L that has only one unknown.

Part B. (Back-Substitution) Solve for the unknown in the new equation L (which contains only one unknown), substitute this value of the unknown into one of the original equations, and then solve to obtain the value of the other unknown.

Part A of Algorithm 3.1 can be applied to any system even if the system does not have a unique solution. In such a case, the new equation L will be degenerate and Part B will not apply.

EXAMPLE 3.5 (Unique Case). Solve the system

$$L_1: 2x - 3y = -8$$

$$L_2: 3x + 4y = 5$$

The unknown x is eliminated from the equations by forming the new equation $L = -3L_1 + 2L_2$. That is, we multiply L_1 by -3 and L_2 by 2 and add the resulting equations as follows:

$$\begin{array}{rcl} -3L_1: & -6x + 9y & = 24 \\ 2L_2: & 6x + 8y & = 10 \\ \hline \text{Addition :} & & 17y = 34 \end{array}$$

We now solve the new equation for y , obtaining $y = 2$. We substitute $y = 2$ into one of the original equations, say L_1 , and solve for the other unknown x , obtaining

$$2x - 3(2) = -8 \quad \text{or} \quad 2x - 6 = -8 \quad \text{or} \quad 2x = -2 \quad \text{or} \quad x = -1$$

Thus, $x = 1$, $y = 2$, or the pair $u = (1, 2)$ is the unique solution of the system. The unique solution is expected, because $2/3 \neq 3/4$. [Geometrically, the lines corresponding to the equations intersect at the point $(1, 2)$.]

EXAMPLE 3.6 (Nonunique Cases)

(a) Solve the system

$$L_1: \quad x - 3y = 4$$

$$L_2: \quad -2x + 6y = 5$$

We eliminated x from the equations by multiplying L_1 by 2 and adding it to L_2 —that is, by forming the new equation $L = 2L_1 + L_2$. This yields the degenerate equation

$$0x + 0y = 13$$

which has a nonzero constant $b = 13$. Thus, this equation and the system have no solution. This is expected, because $1/2 \neq 3/6 \neq 4/5$. (Geometrically, the lines corresponding to the equations are parallel.)

(b) Solve the system

$$L_1: \quad x - 3y = 4$$

$$L_2: \quad -2x + 6y = -8$$

We eliminated x from the equations by multiplying L_1 by 2 and adding it to L_2 —that is, by forming the new equation L

$= 2 L_1 + L_2$. This yields the degenerate equation

$$0x + 0y = 0$$

where the constant term is also zero. Thus, the system has an infinite number of solutions, which correspond to the solutions of either equation. This is expected, because $1/(2) = 3/6 = 4/(8)$. (Geometrically, the lines corresponding to the equations coincide.)

To find the general solution, let $y = a$, and substitute into L_1 to obtain x

$$x - 3a = 4 \quad \text{or} \quad x = 3a + 4$$

Thus, the general solution of the system is

$$x = 3a + 4, y = a \quad \text{or} \quad u = (3a + 4, a)$$

where a (called a *parameter*) is any scalar.

3.5 Systems in Triangular and Echelon Forms

The main method for solving systems of linear equations, Gaussian elimination, is treated in Section 3.6. Here we consider two simple types of systems of linear equations: systems in triangular form and the more general systems in echelon form.

Triangular Form

Consider the following system of linear equations, which is in *triangular form*:

$$\begin{array}{rcl}
2x_1 - 3x_2 + 5x_3 - 2x_4 & = & 9 \\
5x_2 - x_3 + 3x_4 & = & 1 \\
7x_3 - x_4 & = & 3 \\
2x_4 & = & 8
\end{array}$$

That is, the first unknown x_1 is the leading unknown in the first equation, the second unknown x_2 is the leading unknown in the second equation, and so on. Thus, in particular, the system is square and each leading unknown is *directly* to the right of the leading unknown in the preceding equation.

Such a triangular system always has a unique solution, which may be obtained by *back-substitution*. That is,

(1) First solve the last equation for the last unknown to get $x_4 = 4$.

(2) Then substitute this value $x_4 = 4$ in the next-to-last equation, and solve for the next-to-last unknown x_3 as follows:

$$7x_3 - 4 = 3 \quad \text{or} \quad 7x_3 = 7 \quad \text{or} \quad x_3 = 1$$

(3) Now substitute $x_3 = 1$ and $x_4 = 4$ in the second equation, and solve for the second unknown x_2 as follows:

$$5x_2 - 1 + 12 = 1 \quad \text{or} \quad 5x_2 + 11 = 1 \quad \text{or} \quad 5x_2 = -10 \quad \text{or} \quad x_2 = -2$$

(4) Finally, substitute $x_2 = 2$, $x_3 = 1$, $x_4 = 4$ in the first equation, and solve for the first unknown x_1 as follows:

$$2x_1 + 6 + 5 - 8 = 9 \quad \text{or} \quad 2x_1 + 3 = 9 \quad \text{or} \quad 2x_1 = 6 \quad \text{or} \quad x_1 = 3$$

Thus, $x_1 = 3$, $x_2 = 2$, $x_3 = 1$, $x_4 = 4$, or, equivalently, the vector $u = (3, 2, 1, 4)$ is the unique solution of the system.

Remark: There is an alternative form for back-substitution (which will be used when solving a system using the matrix format). Namely, after first finding the value of the last unknown, we substitute this value for the last unknown in all the preceding equations before solving for the next-to-last unknown. This yields a triangular system with one less equation and one less unknown. For example, in the above triangular system, we substitute $x_4 = 4$ in all the preceding equations to obtain the triangular system

$$\begin{aligned} 2x_1 - 3x_2 + 5x_3 &= 17 \\ 5x_2 - x_3 &= -1 \\ 7x_3 &= 7 \end{aligned}$$

We then repeat the process using the new last equation. And so on.

Echelon Form, Pivot and Free Variables

The following system of linear equations is said to be in *echelon form*:

$$\begin{aligned} 2x_1 + 6x_2 - x_3 + 4x_4 - 2x_5 &= 15 \\ x_3 + 2x_4 + 2x_5 &= 5 \\ 3x_4 - 9x_5 &= 6 \end{aligned}$$

That is, no equation is degenerate and the leading unknown in each equation other than the first is to the right of the leading unknown in the preceding equation. The leading unknowns in the system, x_1 , x_3 , x_4 , are called *pivot* variables, and the other unknowns, x_2 and x_5 , are called *free* variables.

Generally speaking, an *echelon system* or a *system in echelon form* has the following form:

$$\begin{array}{rcl}
 a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + a_{14}x_4 + \cdots + a_{1n}x_n & = & b_1 \\
 a_{2j_2}x_{j_2} + a_{2j_2+1}x_{j_2+1} + \cdots + a_{2n}x_n & = & b_2 \\
 \cdots & & \cdots \\
 a_{rj_r}x_{j_r} + \cdots + a_{rn}x_n & = & b_r
 \end{array} \tag{3.5}$$

where $1 < j_2 < \cdots < j_r$ and $a_{11}, a_{2j_2}, \dots, a_{rj_r}$ are not zero. The *pivot* variables are $x_1, x_{j_2}, \dots, x_{j_r}$. Note that $r \leq n$.

The solution set of any echelon system is described in the following theorem (proved in [Problem 3.10](#)).

THEOREM 3.6: Consider a system of linear equations in echelon form, say with r equations in n unknowns. There are two cases:

- (i) $r = n$. That is, there are as many equations as unknowns (triangular form). Then the system has a unique solution.
- (ii) $r < n$. That is, there are more unknowns than equations. Then we can arbitrarily assign values to the $n - r$ free variables and solve uniquely for the r pivot variables, obtaining a solution of the system.

Suppose an echelon system contains more unknowns than equations. Assuming the field K is infinite, the system has an infinite number of solutions, because each of the $n - r$ free variables may be assigned any scalar.

The general solution of a system with free variables may be described in either of two equivalent ways, which we illustrate using the above echelon system where there are $r = 3$ equations and $n = 5$ unknowns. One description is called the “Parametric Form” of the solution, and the other description is called the “Free-Variable Form.”

Parametric Form

Assign arbitrary values, called *parameters*, to the free variables x_2 and x_5 , say $x_2 = a$ and $x_5 = b$, and then use back-substitution to obtain values for the pivot variables x_1 , x_3 , x_4 in terms of the parameters a and b . Specifically,

(1) Substitute $x_5 = b$ in the last equation, and solve for x_4 :

$$3x_4 - 9b = 6 \quad \text{or} \quad 3x_4 = 6 + 9b \quad \text{or} \quad x_4 = 2 + 3b$$

(2) Substitute $x_4 = 2 + 3b$ and $x_5 = b$ into the second equation, and solve for x_3 :

$$x_3 + 2(2 + 3b) + 2b = 5 \quad \text{or} \quad x_3 + 4 + 8b = 5 \quad \text{or} \quad x_3 = 1 - 8b$$

(3) Substitute $x_2 = a$, $x_3 = 1 - 8b$, $x_4 = 2 + 3b$, $x_5 = b$ into the first equation, and solve for x_1 :

$$2x_1 + 6a - (1 - 8b) + 4(2 + 3b) - 2b = 15 \quad \text{or} \quad x_1 = 4 - 3a - 9b$$

Accordingly, the general solution in *parametric form* is

$$x_1 = 4 - 3a - 9b, \quad x_2 = a, \quad x_3 = 1 - 8b, \quad x_4 = 2 + 3b, \quad x_5 = b$$

or, equivalently, $\mathbf{v} = (4 - 3a - 9b, a, 1 - 8b, 2 + 3b, b)$ where a and b are arbitrary numbers.

Free-Variable Form

Use back-substitution to solve for the pivot variables x_1 , x_3 , x_4 directly in terms of the free variables x_2 and x_5 . That is, the last equation gives $x_4 = 2 + 3x_5$. Substitution in the second equation yields $x_3 = 1 - 8x_5$, and then substitution in the first equation yields $x_1 = 4 - 3x_2 - 9x_5$. Accordingly,

$$x_1 = 4 - 3x_2 - 9x_5, \quad x_2 = \text{free variable}, \quad x_3 = 1 - 8x_5, \quad x_4 = 2 + 3x_5, \quad x_5 = \text{free variable}$$

or, equivalently,

$$v = (4 - 3x_2 - 9x_5, x_2, 1 - 8x_5, 2 + 3x_5, x_5)$$

is the *free-variable form* for the general solution of the system.

We emphasize that there is no difference between the above two forms of the general solution, and the use of one or the other to represent the general solution is simply a matter of taste.

Remark: A particular solution of the above system can be found by assigning any values to the free variables and then solving for the pivot variables by back-substitution. For example, setting $x_2 = 1$ and $x_5 = 1$, we obtain

$$x_4 = 2 + 3 = 5, \quad x_3 = 1 - 8 = -7, \quad x_1 = 4 - 3 - 9 = -8$$

Thus, $u = (8, 1, 7, 5, 1)$ is the particular solution corresponding to $x_2 = 1$ and $x_5 = 1$.

3.6 Gaussian Elimination

The main method for solving the general system (3.2) of linear equations is called *Gaussian elimination*. It essentially consists of two parts:

Part A. (Forward Elimination) Step-by-step reduction of the system yielding either a degenerate equation with no solution (which indicates the system has no solution) or an equivalent simpler system in triangular or echelon form.

Part B. (Backward Elimination) Step-by-step back-substitution to find the solution of the simpler system.

Part B has already been investigated in Section 3.4. Accordingly, we need only give the algorithm for Part A, which is as follows.

ALGORITHM 3.2 for (Part A): Input: The $m \times n$ system (3.2) of linear equations.

MINATION STEP: Find the first unknown in the system with a nonzero coefficient (which now must be x_1).

(a) Arrange so that $a_{11} \neq 0$. That is, if necessary, interchange equations so that the first unknown x_1 appears with a nonzero coefficient in the first equation.

(b) Use a_{11} as a *pivot* to eliminate x_1 from all equations except the first equation. That is, for $i > 1$:

(1) Set $m = a_{i1} / a_{11}$;

(2) Replace L_i by $mL_1 + L_i$

The system now has the following form:

$$\begin{array}{rcl} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \cdots + a_{1n}x_n & = & b_1 \\ a_{2j_2}x_{j_2} + \cdots + a_{2n}x_n & = & b_2 \\ \cdots & & \cdots \\ a_{mj_2}x_{j_2} + \cdots + a_{mn}x_n & = & b_n \end{array}$$

where x_1 does not appear in any equation except the first, $a_{11} \neq 0$, and x_{j_2} denotes the first unknown with a nonzero coefficient in any equation other than the first.

(c) Examine each new equation L_i .

(1) If L has the form $0 x_1 + 0 x_2 + \dots + 0 x_n = b$ with $b \neq 0$, then

STOP

The system is *inconsistent* and has no solution.

(2) If L has the form $0 x_1 + 0 x_2 + \dots + 0 x_n = 0$ or if L is a multiple of another equation, then delete L from the system.

RECURSION STEP: Repeat the Elimination Step with each new “smaller” subsystem formed by all the equations excluding the first equation.

OUTPUT: Finally, the system is reduced to triangular or echelon form, or a degenerate equation with no solution is obtained indicating an inconsistent system.

The next remarks refer to the Elimination Step in Algorithm 3.2.

(1) The following number m in (b) is called the *multiplier*:

$$m = -\frac{a_{i1}}{a_{11}} = -\frac{\text{coefficient to be deleted}}{\text{pivot}}$$

(2) One could alternatively apply the following operation in (b):

Replace L_i by $a_{i1} L_1 + a_{11} L_i$

This would avoid fractions if all the scalars were originally integers.

Gaussian Elimination Example

Here we illustrate in detail Gaussian elimination using the following system of linear equations:

$$\begin{array}{lcl}
L_1: & x - 3y - 2z = & 6 \\
L_2: & 2x - 4y - 3z = & 8 \\
L_3: & -3x + 6y + 8z = & -5
\end{array}$$

Part A. We use the coefficient 1 of x in the first equation L_1 as the pivot in order to eliminate x from the second equation L_2 and from the third equation L_3 . This is accomplished as follows:

(1) Multiply L_1 by the multiplier $m = 2$ and add it to L_2 ; that is, “Replace L_2 by $2 L_1 + L_2$.”

(2) Multiply L_1 by the multiplier $m = 3$ and add it to L_3 ; that is, “Replace L_3 by $3 L_1 + L_3$.”

These steps yield

$$\begin{array}{lcl}
(-2)L_1: & -2x + 6y + 4z = -12 & \\
L_2: & 2x - 4y - 3z = 8 & \\
\hline
\text{New } L_2: & 2y + z = -4 & \\
\end{array}
\qquad
\begin{array}{lcl}
3L_1: & 3x - 9y - 6z = 18 & \\
L_3: & -3x + 6y + 8z = -5 & \\
\hline
\text{New } L_3: & -3y + 2z = 13 &
\end{array}$$

Thus, the original system is replaced by the following system:

$$\begin{array}{lcl}
L_1: & x - 3y - 2z = & 6 \\
L_2: & 2y + z = & -4 \\
L_3: & -3y + 2z = & 13
\end{array}$$

(Note that the equations L_2 and L_3 form a subsystem with one less equation and one less unknown than the original system.)

Next we use the coefficient 2 of y in the (new) second equation L_2 as the pivot in order to eliminate y from the (new) third equation L_3 . This is accomplished as follows:

(3) Multiply L_2 by the multiplier $m = \frac{3}{2}$ and add it to L_3 ; that is, "Replace L_3 by $3L_2 + L_3$ " (Alternately, "Replace L_3 by $3L_2 + 2L_3$," which will avoid fractions.)

This step yields

$$\begin{array}{rcl} \frac{3}{2}L_2: & 3y + \frac{3}{2}z = -6 & \\ L_3: & -3y + 2z = 13 & \\ \hline \text{New } L_3: & \frac{7}{2}z = 7 & \end{array} \quad \text{or} \quad \begin{array}{rcl} 3L_2: & 6y + 3z = -12 & \\ 2L_3: & -6y + 4z = 26 & \\ \hline \text{New } L_3: & 7z = 14 & \end{array}$$

Thus, our system is replaced by the following system:

$$\begin{array}{rcl} L_1: & x - 3y - 2z = 6 & \\ L_2: & 2y + z = -4 & \\ L_3: & 7z = 14 \quad (\text{or } \frac{7}{2}z = 7) & \end{array}$$

The system is now in triangular form, so Part A is completed.

Part B. The values for the unknowns are obtained in reverse order, z , y , x , by back-substitution. Specifically,

(1) Solve for z in L_3 to get $z = 2$.

(2) Substitute $z = 2$ in L_2 , and solve for y to get $y = 3$.

(3) Substitute $y = 3$ and $z = 2$ in L_1 , and solve for x to get $x = 1$.

Thus, the solution of the triangular system and hence the original system is as follows:

$$x = 1, \quad y = -3, \quad z = 2 \quad \text{or, equivalently,} \quad u = (1, -3, 2).$$

Condensed Format

The Gaussian elimination algorithm involves rewriting systems of linear equations. Sometimes we can avoid

excessive recopying of some of the equations by adopting a “condensed format.” This format for the solution of the above system follows:

Number	Equation	Operation
(1)	$x - 3y - 2z = 6$	
(2)	$2x - 4y - 3z = 8$	
(3)	$-3x + 6y + 8z = -5$	
(2')	$2y + z = -4$	Replace L_2 by $-2L_1 + L_2$
(3')	$-3y + 2z = 13$	Replace L_3 by $3L_1 + L_3$
(3'')	$7z = 14$	Replace L_3 by $3L_2 + 2L_3$

That is, first we write down the number of each of the original equations. As we apply the Gaussian elimination algorithm to the system, we only write down the new equations, and we label each new equation using the same number as the original corresponding equation, but with an added prime. (After each new equation, we will indicate, for instructional purposes, the elementary operation that yielded the new equation.)

The system in triangular form consists of equations (1), (2'), and (3''), the numbers with the largest number of primes. Applying back-substitution to these equations again yields $x = 1$, $y = 3$, $z = 2$.

Remark: If two equations need to be interchanged, say to obtain a nonzero coefficient as a pivot, then this is easily accomplished in the format by simply renumbering the two equations rather than changing their positions.

EXAMPLE 3.7 Solve the following system:

Number	Equation	Operation
(1)	$x - 3y - 2z = 6$	
(2)	$2x - 4y - 3z = 8$	
(3)	$-3x + 6y + 8z = -5$	
(2')	$2y + z = -4$	Replace L_2 by $-2L_1 + L_2$
(3')	$-3y + 2z = 13$	Replace L_3 by $3L_1 + L_3$
(3'')	$7z = 14$	Replace L_3 by $3L_2 + 2L_3$

We solve the system by Gaussian elimination.

Part A. (Forward Elimination) We use the coefficient 1 of x in the first equation L_1 as the pivot in order to eliminate x from the second equation L_2 and from the third equation L_3 . This is accomplished as follows:

(1) Multiply L_1 by the multiplier $m = 2$ and add it to L_2 ; that is, "Replace L_2 by $2 L_1 + L_2$."

(2) Multiply L_1 by the multiplier $m = 3$ and add it to L_3 ; that is, "Replace L_3 by $3 L_1 + L_3$."

The two steps yield

$$\begin{array}{rcl} x + 2y - 3z = 1 & & \\ y - 2z = 2 & \text{or} & x + 2y - 3z = 1 \\ 2y - 4z = 4 & & y - 2z = 2 \end{array}$$

(The third equation is deleted, because it is a multiple of the second equation.) The system is now in echelon form with free variable z .

Part B. (Backward Elimination) To obtain the general solution, let the free variable $z = a$, and solve for x and y by back-substitution. Substitute $z = a$ in the second equation to obtain $y = 2 + 2a$. Then substitute $z = a$ and $y = 2 + 2a$ into the first equation to obtain

$$x + 2(2 + 2a) - 3a = 1 \quad \text{or} \quad x + 4 + 4a - 3a = 1 \quad \text{or} \quad x = -3 - a$$

Thus, the following is the general solution where a is a parameter:

$$x = -3 - a, \quad y = 2 + 2a, \quad z = a \quad \text{or} \quad u = (-3 - a, 2 + 2a, a)$$

EXAMPLE 3.8 Solve the following system:

$$\begin{aligned} x_1 + 3x_2 - 2x_3 + 5x_4 &= 4 \\ 2x_1 + 8x_2 - x_3 + 9x_4 &= 9 \\ 3x_1 + 5x_2 - 12x_3 + 17x_4 &= 7 \end{aligned}$$

We use Gaussian elimination.

Part A. (Forward Elimination) We use the coefficient 1 of x_1 in the first equation L_1 as the pivot in order to eliminate x_1 from the second equation L_2 and from the third equation L_3 . This is accomplished by the following operations:

(1) “Replace L_2 by $2 L_1 + L_2$ ” and (2) “Replace L_3 by $3 L_1 + L_3$ ”

These yield:

$$\begin{aligned} x_1 + 3x_2 - 2x_3 + 5x_4 &= 4 \\ 2x_2 + 3x_3 - x_4 &= 1 \\ -4x_2 - 6x_3 + 2x_4 &= -5 \end{aligned}$$

We now use the coefficient 2 of x_2 in the second equation L_2 as the pivot and the multiplier $m = 2$ in order to eliminate x_2 from the third equation L_3 . This is accomplished by the operation “Replace L_3 by $2 L_2 + L_3$,” which then yields the degenerate equation

$$0x_1 + 0x_2 + 0x_3 + 0x_4 = -3$$

This equation and, hence, the original system have no solution:

DO NOT CONTINUE

Remark 1: As in the above examples, Part A of Gaussian elimination tells us whether or not the system has a solution—that is, whether or not the system is consistent. Accordingly, Part B need never be applied when a system has no solution.

Remark 2: If a system of linear equations has more than four unknowns and four equations, then it may be more convenient to use the matrix format for solving the system. This matrix format is discussed later.

3.7 Echelon Matrices, Row Canonical Form, Row Equivalence

One way to solve a system of linear equations is by working with its augmented matrix M rather than the system itself. This section introduces the necessary matrix concepts for such a discussion. These concepts, such as echelon matrices and elementary row operations, are also of independent interest.

Echelon Matrices

A matrix A is called an *echelon matrix*, or is said to be in *echelon form*, if the following two conditions hold (where a *leading nonzero element* of a row of A is the first nonzero element in the row):

- (1) All zero rows, if any, are at the bottom of the matrix.

(2) Each leading nonzero entry in a row is to the right of the leading nonzero entry in the preceding row.

That is, $A = [a_{ij}]$ is an echelon matrix if there exist nonzero entries

$$a_{1j_1}, a_{2j_2}, \dots, a_{rj_r}, \quad \text{where } j_1 < j_2 < \dots < j_r$$

with the property that

$$a_{ij} = 0 \quad \text{for} \quad \begin{cases} \text{(i) } i \leq r, & j < j_i \\ \text{(ii) } i > r \end{cases}$$

The entries $a_{1j_1}, a_{2j_2}, \dots, a_{rj_r}$, which are the leading nonzero elements in their respective rows, are called the *pivots* of the echelon matrix.

EXAMPLE 3.9 The following is an echelon matrix whose pivots have been circled:

$$A = \begin{bmatrix} 0 & \textcircled{2} & 3 & 4 & 5 & 9 & 0 & 7 \\ 0 & 0 & 0 & \textcircled{3} & 4 & 1 & 2 & 5 \\ 0 & 0 & 0 & 0 & 0 & \textcircled{5} & 7 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 & \textcircled{8} & 6 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Observe that the pivots are in columns C_2, C_4, C_6, C_7 , and each is to the right of the one above. Using the above notation, the pivots are

$$a_{1j_1} = 2, \quad a_{2j_2} = 3, \quad a_{3j_3} = 5, \quad a_{4j_4} = 8$$

where $j_1 = 2, j_2 = 4, j_3 = 6, j_4 = 7$. Here $r = 4$.

Row Canonical Form

A matrix A is said to be in *row canonical form* (or *row-reduced echelon form*) if it is an echelon matrix—that is, if it satisfies the above properties (1) and (2), and if it satisfies the following additional two properties:

(3) Each pivot (leading nonzero entry) is equal to 1.

(4) Each pivot is the only nonzero entry in its column.

The major difference between an echelon matrix and a matrix in row canonical form is that in an echelon matrix there must be zeros below the pivots [Properties (1) and (2)], but in a matrix in row canonical form, each pivot must also equal 1 [Property (3)] and there must also be zeros above the pivots [Property (4)].

The zero matrix 0 of any size and the identity matrix I of any size are important special examples of matrices in row canonical form.

EXAMPLE 3.10

The following are echelon matrices whose pivots have been circled:

$$\begin{bmatrix} \textcircled{2} & 3 & 2 & 0 & 4 & 5 & -6 \\ 0 & 0 & 0 & \textcircled{1} & -3 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & \textcircled{6} & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} \textcircled{1} & 2 & 3 \\ 0 & 0 & \textcircled{1} \\ 0 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & \textcircled{1} & 3 & 0 & 0 & 4 \\ 0 & 0 & 0 & \textcircled{1} & 0 & -3 \\ 0 & 0 & 0 & 0 & \textcircled{1} & 2 \end{bmatrix}$$

The third matrix is also an example of a matrix in row canonical form. The second matrix is not in row canonical form, because it does not satisfy property (4); that is, there is a nonzero entry above the second pivot in the third column. The first matrix is not in row canonical form, because it satisfies neither property (3) nor property (4); that is, some pivots are not equal to 1 and there are nonzero entries above the pivots.

Elementary Row Operations

Suppose A is a matrix with rows $R_1, R_2, \dots, R_m$. The following operations on A are called *elementary row operations*.

[E₁] (Row Interchange): Interchange rows R_i and R_j . This may be written as

$$\text{“Interchange } R_i \text{ and } R_j\text{”} \quad \text{or} \quad \text{“} R_i \longleftrightarrow R_j \text{”}$$

[E₂] (Row Scaling): Replace row R_i by a nonzero multiple kR_i of itself. This may be written as

$$\text{“Replace } R_i \text{ by } kR_i \text{ (} k \neq 0 \text{)”} \quad \text{or} \quad \text{“} kR_i \rightarrow R_i \text{”}$$

[E₃] (Row Addition): Replace row R_j by the sum of a multiple kR_i of a row R_i and itself. This may be written as

$$\text{“Replace } R_j \text{ by } kR_i + R_j\text{”} \quad \text{or} \quad \text{“} kR_i + R_j \rightarrow R_j \text{”}$$

The arrow $\rightarrow$ in E₂ and E₃ may be read as “replaces.”

Sometimes (say to avoid fractions when all the given scalars are integers) we may apply [E₂] one step; that is, we may apply the following operation:

[E] Replace R_j by the sum of a multiple kR_i of a row R_i and a nonzero multiple $k'R_j$ of itself. This may be written as

$$\text{“Replace } R_j \text{ by } kR_i + k'R_j \text{ (} k' \neq 0 \text{)”} \quad \text{or} \quad \text{“} kR_i + k'R_j \rightarrow R_j \text{”}$$

We emphasize that in operations [E₃] and [E] only row R_j is changed.

Row Equivalence, Rank of a Matrix

A matrix A is said to be *row equivalent* to a matrix B , written

$$A \sim B$$

if B can be obtained from A by a sequence of elementary row operations. In the case that B is also an echelon matrix, B is called an *echelon form* of A .

The following are two basic results on row equivalence.

THEOREM 3.7: Suppose $A = [a_{ij}]$ and $B = [b_{ij}]$ are row equivalent echelon matrices with respective pivot entries

$$a_{1j_1}, a_{2j_2}, \dots, a_{rj_r} \quad \text{and} \quad b_{1k_1}, b_{2k_2}, \dots, b_{sk_s}$$

Then A and B have the same number of nonzero rows—that is, $r = s$ —and the pivot entries are in the same positions—that is, $j_1 = k_1, j_2 = k_2, \dots, j_r = k_r$.

THEOREM 3.8: Every matrix A is row equivalent to a unique matrix in row canonical form.

The proofs of the above theorems will be postponed to [Chapter 4](#). The unique matrix in [Theorem 3.8](#) is called the *row canonical form* of A .

Using the above theorems, we can now give our first definition of the rank of a matrix.

DEFINITION: The *rank* of a matrix A , written $\text{rank}(A)$, is equal to the number of pivots in an echelon form of A .

The rank is a very important property of a matrix and, depending on the context in which the matrix is used, it

will be defined in many different ways. Of course, all the definitions lead to the same number.

The next section gives the matrix format of Gaussian elimination, which finds an echelon form of any matrix A (and hence the rank of A), and also finds the row canonical form of A .

One can show that row equivalence is an *equivalence relation*. That is,

- (1) $A \sim A$ for any matrix A .
- (2) If $A \sim B$, then $B \sim A$.
- (3) If $A \sim B$ and $B \sim C$, then $A \sim C$.

Property (2) comes from the fact that each elementary row operation has an inverse operation of the same type.

Namely,

- (i) “Interchange R_i and R_j ” is its own inverse.
- (ii) “Replace R_i by kR_i ” and “Replace R_i by $(1/k) R_i$ ” are inverses.
- (iii) “Replace R_j by $kR_i + R_j$ ” and “Replace R_j by $kR_i + R_j$ ” are inverses.

There is a similar result for operation $[E]$ ([Problem 3.73](#)).

3.8 Gaussian Elimination, Matrix Formulation

This section gives two matrix algorithms that accomplish the following:

- (1) Algorithm 3.3 transforms any matrix A into an echelon form.

(2) Algorithm 3.4 transforms the echelon matrix into its row canonical form.

These algorithms, which use the elementary row operations, are simply restatements of *Gaussian elimination* as applied to matrices rather than to linear equations. (The term “row reduce” or simply “reduce” will mean to transform a matrix by the elementary row operations.)

ALGORITHM 3.3 (Forward Elimination): The input is any matrix A . (The algorithm puts 0's below each pivot, working from the “top-down.”) The output is an echelon form of A .

Step 1. Find the first column with a nonzero entry. Let j_1 denote this column.

(a) Arrange so that $a_{1j_1} \neq 0$. That is, if necessary, interchange rows so that a nonzero entry appears in the first row in column j_1 .

(b) Use a_{1j_1} as a pivot to obtain 0's below a_{1j_1} .

Specifically, for $i > 1$:

(1) Set $m = -a_{ij_1}/a_{1j_1}$; (2) Replace R_i by $mR_1 + R_i$

[That is, apply the operation $-(a_{ij_1}/a_{1j_1})R_1 + R_i \rightarrow R_i$.]

Step 2. Repeat Step 1 with the submatrix formed by all the rows excluding the first row. Here we let j_2 denote the first column in the subsystem with a nonzero entry. Hence, at the end of Step 2, we have $a_{2j_2} \neq 0$.

Steps 3 to r . Continue the above process until a submatrix has only zero rows.

We emphasize that at the end of the algorithm, the pivots will be

$$a_{1j_1}, a_{2j_2}, \dots, a_{rj_r}$$

where r denotes the number of nonzero rows in the final echelon matrix.

Remark 1: The following number m in Step 1(b) is called the *multiplier*:

$$m = -\frac{a_{ij_1}}{a_{1j_1}} = -\frac{\text{entry to be deleted}}{\text{pivot}}$$

Remark 2: One could replace the operation in Step 1(b) by the following which would avoid fractions if all the scalars were originally integers.

$$\text{Replace } R_i \text{ by } -a_{ij_1}R_1 + a_{1j_1}R_i.$$

ALGORITHM 3.4 (Backward Elimination): The input is a matrix $A = [a_{ij}]$ in echelon form with pivot entries

$$a_{1j_1}, a_{2j_2}, \dots, a_{rj_r}$$

The output is the row canonical form of A .

Step 1.

- (a) (Use row scaling so the last pivot equals 1.)
Multiply the last nonzero row R_r by $1/a_{rj_r}$.
- (b) (Use $a_{rj_r} = 1$ to obtain 0's above the pivot.) For $i = r-1, r-2, \dots, 2, 1$:
 - (1) Set $m = -a_{ij_r}$; (2) Replace R_i by $mR_r + R_i$

(That is, apply the operations $a_{ijr} R_r + R_i \rightarrow R_i$.)

Steps 2 to r 1. Repeat Step 1 for rows $R_{r1}, R_{r2}, \dots, R_2$.

Step r. (Use row scaling so the first pivot equals 1.) Multiply R_1 by $1 = a_{1j1}$.

There is an alternative form of Algorithm 3.4, which we describe here in words. The formal description of this algorithm is left to the reader as a supplementary problem.

ALTERNATIVE ALGORITHM 3.4 Puts 0's above the pivots row by row from the bottom up (rather than column by column from right to left).

The alternative algorithm, when applied to an augmented matrix M of a system of linear equations, is essentially the same as solving for the pivot unknowns one after the other from the bottom up.

Remark: We emphasize that Gaussian elimination is a two-stage process. Specifically,

Stage A (Algorithm 3.3). Puts 0's below each pivot, working from the top row R_1 down.

Stage B (Algorithm 3.4). Puts 0's above each pivot, working from the bottom row R_r up.

There is another algorithm, called *Gauss – Jordan*, that also row reduces a matrix to its row canonical form. The difference is that Gauss–Jordan puts 0's both below and above each pivot as it works its way from the top row R_1 down. Although Gauss–Jordan may be easier to state and understand, it is much less efficient than the two-stage Gaussian elimination algorithm.

EXAMPLE 3.11 Consider the matrix $A = \begin{bmatrix} 1 & 2 & -3 & 1 & 2 \\ 2 & 4 & -4 & 6 & 10 \\ 3 & 6 & -6 & 9 & 13 \end{bmatrix}$

- (a) Use Algorithm 3.3 to reduce A to an echelon form.
 - (b) Use Algorithm 3.4 to further reduce A to its row canonical form.
- (a) First use $a_{11} = 1$ as a pivot to obtain 0's below a_{11} ; that is, apply the operations "Replace R_2 by $2 R_1 + R_2$ " and "Replace R_3 by $3 R_1 + R_3$." Then use $a_{23} = 2$ as a pivot to obtain 0 below a_{23} ; that is, apply the operation "Replace R_3 by $3 R_2 + R_3$." This yields

$$A \sim \begin{bmatrix} 1 & 2 & -3 & 1 & 2 \\ 0 & 0 & 2 & 4 & 6 \\ 0 & 0 & 3 & 6 & 7 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & -3 & 1 & 2 \\ 0 & 0 & 2 & 4 & 6 \\ 0 & 0 & 0 & 0 & -2 \end{bmatrix}$$

The matrix is now in echelon form.

- (b) Multiply R_3 by $-\frac{1}{2}$ so the pivot entry $a_{35} = 1$, and then use $a_{35} = 1$ as a pivot to obtain 0's above it by the operations "Replace R_2 by $6 R_3 + R_2$ " and then "Replace R_1 by $2 R_3 + R_1$." This yields

$$A \sim \begin{bmatrix} 1 & 2 & -3 & 1 & 2 \\ 0 & 0 & 2 & 4 & 6 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & -3 & 1 & 0 \\ 0 & 0 & 2 & 4 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

Multiply R_2 by $\frac{1}{2}$ so the pivot entry $a_{23} = 1$, and then use $a_{23} = 1$ as a pivot to obtain 0's above it by the operation "Replace R_1 by $3 R_2 + R_1$." This yields

$$A \sim \begin{bmatrix} 1 & 2 & -3 & 1 & 0 \\ 0 & 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 0 & 7 & 0 \\ 0 & 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

The last matrix is the row canonical form of A .

Application to Systems of Linear Equations

One way to solve a system of linear equations is by working with its augmented matrix M rather than the equations themselves. Specifically, we reduce M to echelon form (which tells us whether the system has a solution), and then further reduce M to its row canonical form (which essentially gives the solution of the original system of linear equations). The justification for this process comes from the following facts:

- (1) Any elementary row operation on the augmented matrix M of the system is equivalent to applying the corresponding operation on the system itself.
- (2) The system has a solution if and only if the echelon form of the augmented matrix M does not have a row of the form $(0, 0, \dots, 0, b)$ with $b \neq 0$.
- (3) In the row canonical form of the augmented matrix M (excluding zero rows), the coefficient of each basic variable is a pivot entry equal to 1, and it is the only nonzero entry in its respective column; hence, the free-variable form of the solution of the system of linear equations is obtained by simply transferring the free variables to the other side.

This process is illustrated below.

EXAMPLE 3.12 Solve each of the following systems:

$$\begin{array}{rcl} x_1 + x_2 - 2x_3 + 4x_4 & = & 5 \\ 2x_1 + 2x_2 - 3x_3 + x_4 & = & 3 \\ 3x_1 + 3x_2 - 4x_3 - 2x_4 & = & 1 \end{array}$$

(a)

$$\begin{array}{rcl} x_1 + x_2 - 2x_3 + 3x_4 & = & 4 \\ 2x_1 + 3x_2 + 3x_3 - x_4 & = & 3 \\ 5x_1 + 7x_2 + 4x_3 + x_4 & = & 5 \end{array}$$

(b)

$$\begin{array}{rcl} x + 2y + z & = & 3 \\ 2x + 5y - z & = & -4 \\ 3x - 2y - z & = & 5 \end{array}$$

(c)

(a) Reduce its augmented matrix M to echelon form and then to row canonical form as follows:

$$M = \left[\begin{array}{ccccc} 1 & 1 & -2 & 4 & 5 \\ 2 & 2 & -3 & 1 & 3 \\ 3 & 3 & -4 & -2 & 1 \end{array} \right] \sim \left[\begin{array}{ccccc} 1 & 1 & -2 & 4 & 5 \\ 0 & 0 & 1 & -7 & -7 \\ 0 & 0 & 2 & -14 & -14 \end{array} \right] \sim \left[\begin{array}{ccccc} 1 & 1 & 0 & -10 & -9 \\ 0 & 0 & 1 & -7 & -7 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

Rewrite the row canonical form in terms of a system of linear equations to obtain the free variable form of the solution. That is,

$$\begin{array}{rcl} x_1 + x_2 - 10x_4 & = & -9 \\ x_3 - 7x_4 & = & -7 \end{array} \quad \text{or} \quad \begin{array}{rcl} x_1 & = & -9 - x_2 + 10x_4 \\ x_3 & = & -7 + 7x_4 \end{array}$$

(The zero row is omitted in the solution.) Observe that x_1 and x_3 are the pivot variables, and x_2 and x_4 are the free variables.

(b) First reduce its augmented matrix M to echelon form as follows:

$$M = \left[\begin{array}{ccccc} 1 & 1 & -2 & 3 & 4 \\ 2 & 3 & 3 & -1 & 3 \\ 5 & 7 & 4 & 1 & 5 \end{array} \right] \sim \left[\begin{array}{ccccc} 1 & 1 & -2 & 3 & 4 \\ 0 & 1 & 7 & -7 & -5 \\ 0 & 2 & 14 & -14 & -15 \end{array} \right] \sim \left[\begin{array}{ccccc} 1 & 1 & -2 & 3 & 4 \\ 0 & 1 & 7 & -7 & -5 \\ 0 & 0 & 0 & 0 & -5 \end{array} \right]$$

There is no need to continue to find the row canonical form of M , because the echelon form already tells us that the system has no solution. Specifically, the third row of the echelon matrix corresponds to the degenerate equation

$$0x_1 + 0x_2 + 0x_3 + 0x_4 = -5$$

which has no solution. Thus, the system has no solution.

(c) Reduce its augmented matrix M to echelon form and then to row canonical form as follows:

$$\begin{aligned}
 M &= \begin{bmatrix} 1 & 2 & 1 & 3 \\ 2 & 5 & -1 & -4 \\ 3 & -2 & -1 & 5 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 1 & 3 \\ 0 & 1 & -3 & -10 \\ 0 & -8 & -4 & -4 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 1 & 3 \\ 0 & 1 & -3 & -10 \\ 0 & 0 & -28 & -84 \end{bmatrix} \\
 &\sim \begin{bmatrix} 1 & 2 & 1 & 3 \\ 0 & 1 & -3 & -10 \\ 0 & 0 & 1 & 3 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 0 & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 3 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 3 \end{bmatrix}
 \end{aligned}$$

Thus, the system has the unique solution $x = 2$, $y = 1$, $z = 3$, or, equivalently, the vector $u = (2, 1, 3)$. We note that the echelon form of M already indicated that the solution was unique, because it corresponded to a triangular system.

Application to Existence and Uniqueness Theorems

This subsection gives theoretical conditions for the existence and uniqueness of a solution of a system of linear equations using the notion of the rank of a matrix.

THEOREM 3.9: Consider a system of linear equations in n unknowns with augmented matrix $M = [A, B]$. Then,

- (a) The system has a solution if and only if $\text{rank}(A) = \text{rank}(M)$.
- (b) The solution is unique if and only if $\text{rank}(A) = \text{rank}(M) = n$.

Proof of (a). The system has a solution if and only if an echelon form of $M = [A, B]$ does not have a row of the form

$$(0, 0, \dots, 0, b), \quad \text{with } b \neq 0$$

If an echelon form of M does have such a row, then b is a pivot of M but not of A , and hence, $\text{rank}(M) > \text{rank}(A)$. Otherwise, the echelon forms of A and M have the same pivots, and hence, $\text{rank}(A) = \text{rank}(M)$. This proves (a).

Proof of (b). The system has a unique solution if and only if an echelon form has no free variable. This means there is a pivot for each unknown. Accordingly, $n = \text{rank}(A) = \text{rank}(M)$. This proves (b).

The above proof uses the fact ([Problem 3.74](#)) that an echelon form of the augmented matrix $M = [A, B]$ also automatically yields an echelon form of A .

3.9 Matrix Equation of a System of Linear Equations

The general system (3.2) of m linear equations in n unknowns is equivalent to the matrix equation

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \cdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \cdots \\ b_m \end{bmatrix} \quad \text{or} \quad AX = B$$

where $A = [a_{ij}]$ is the coefficient matrix, $X = [x_j]$ is the column vector of unknowns, and $B = [b_i]$ is the column vector of constants. (Some texts write $Ax = b$ rather than $AX = B$, in order to emphasize that x and b are simply column vectors.)

The statement that the system of linear equations and the matrix equation are equivalent means that any vector

solution of the system is a solution of the matrix equation, and vice versa.

EXAMPLE 3.13 The following system of linear equations and matrix equation are equivalent:

$$\begin{aligned} x_1 + 2x_2 - 4x_3 + 7x_4 &= 4 \\ 3x_1 - 5x_2 + 6x_3 - 8x_4 &= 8 \\ 4x_1 - 3x_2 - 2x_3 + 6x_4 &= 11 \end{aligned} \quad \text{and} \quad \begin{bmatrix} 1 & 2 & -4 & 7 \\ 3 & -5 & 6 & -8 \\ 4 & -3 & -2 & 6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 4 \\ 8 \\ 11 \end{bmatrix}$$

We note that $x_1 = 3$, $x_2 = 1$, $x_3 = 2$, $x_4 = 1$, or, in other words, the vector $u = [3, 1, 2, 1]$ is a solution of the system. Thus, the (column) vector u is also a solution of the matrix equation.

The matrix form $AX = B$ of a system of linear equations is notationally very convenient when discussing and proving properties of systems of linear equations. This is illustrated with our first theorem (described in [Fig. 3-1](#)), which we restate for easy reference.

THEOREM 3.10: Suppose the field K is infinite. Then the system $AX = B$ has: (a) a unique solution, (b) no solution, or (c) an infinite number of solutions.

Proof. It suffices to show that if $AX = B$ has more than one solution, then it has infinitely many. Suppose u and v are distinct solutions of $AX = B$; that is, $Au = B$ and $Av = B$. Then, for any $k \in K$,

$$A[u + k(u - v)] = Au + k(Au - Av) = B + k(B - B) = B$$

Thus, for each $k \in K$, the vector $u + k(u - v)$ is a solution of $AX = B$. Because all such solutions are distinct ([Problem 3.47](#)), $AX = B$ has an infinite number of solutions.

Observe that the above theorem is true when K is the real field $\mathbf{R}$ (or the complex field $\mathbf{C}$). Section 3.3 shows that the theorem has a geometrical description when the system consists of two equations in two unknowns, where each equation represents a line in $\mathbf{R}^2$. The theorem also has a geometrical description when the system consists of three nondegenerate equations in three unknowns, where the three equations correspond to planes H_1, H_2, H_3 in $\mathbf{R}^3$. That is,

- (a) *Unique solution*: Here the three planes intersect in exactly one point.
- (b) *No solution*: Here the planes may intersect pairwise but with no common point of intersection, or two of the planes may be parallel.
- (c) *Infinite number of solutions*: Here the three planes may intersect in a line (one free variable), or they may coincide (two free variables).

These three cases are pictured in [Fig. 3-3](#).

Matrix Equation of a Square System of Linear Equations

A system $AX = B$ of linear equations is square if and only if the matrix A of coefficients is square. In such a case, we have the following important result.

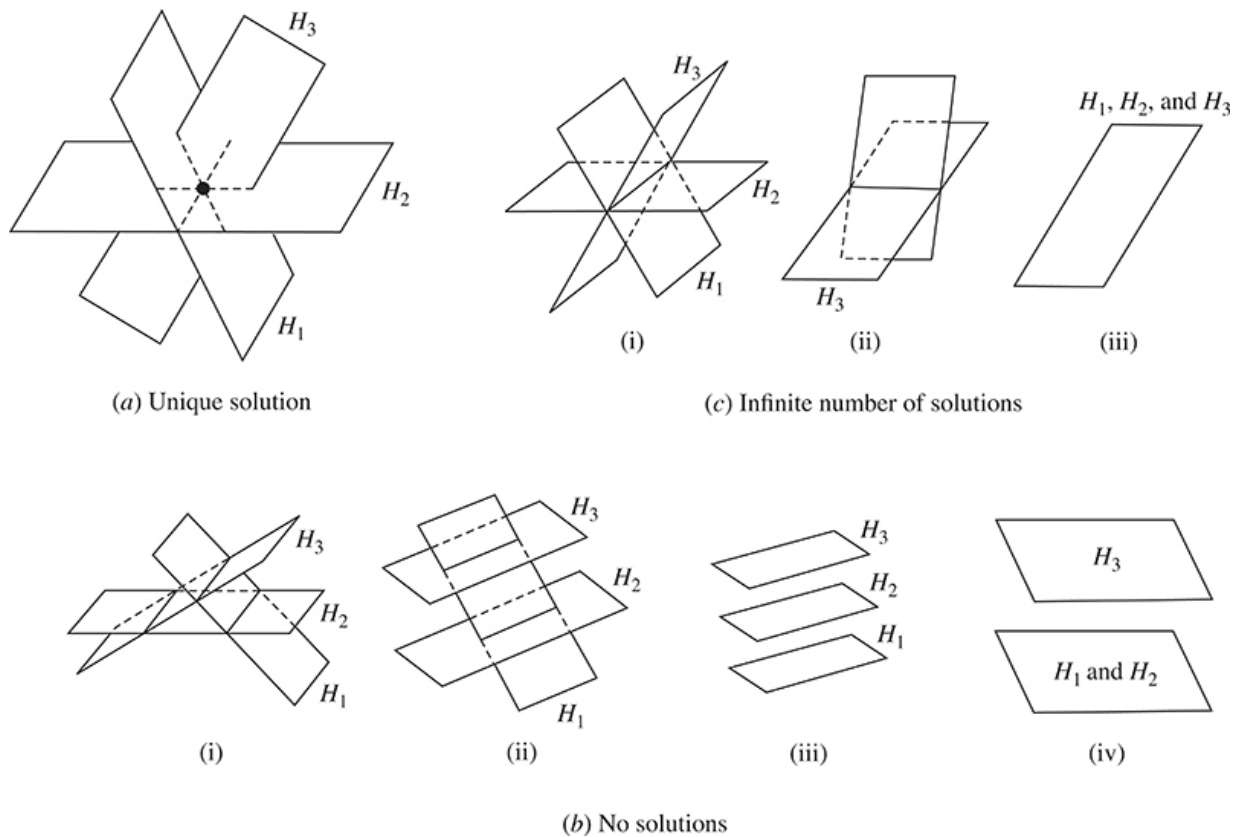


Figure 3-3

THEOREM 3.10: A square system $AX = B$ of linear equations has a unique solution if and only if the matrix A is invertible. In such a case, $A^{-1}B$ is the unique solution of the system.

We only prove here that if A is invertible, then $A^{-1}B$ is a unique solution. If A is invertible, then

$$A(A^{-1}B) = (AA^{-1})B = IB = B$$

and hence, $A^{-1}B$ is a solution. Now suppose v is any solution, so $Av = B$. Then

$$v = Iv = (A^{-1}A)v = A^{-1}(Av) = A^{-1}B$$

Thus, the solution $A^{-1}B$ is unique.

EXAMPLE 3.14 Consider the following system of linear equations, whose coefficient matrix A and inverse A^{-1} are also given:

$$\begin{array}{l} x + 2y + 3z = 1 \\ x + 3y + 6z = 3, \\ 2x + 6y + 13z = 5 \end{array} \quad A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 3 & 6 \\ 2 & 6 & 13 \end{bmatrix}, \quad A^{-1} = \begin{bmatrix} 3 & -8 & 3 \\ -1 & 7 & -3 \\ 0 & -2 & 1 \end{bmatrix}$$

By [Theorem 3.10](#), the unique solution of the system is

$$A^{-1}B = \begin{bmatrix} 3 & -8 & 3 \\ -1 & 7 & -3 \\ 0 & -2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix} = \begin{bmatrix} -6 \\ 5 \\ -1 \end{bmatrix}$$

That is, $x = 6$, $y = 5$, $z = 1$.

Remark: We emphasize that [Theorem 3.10](#) does not usually help us to find the solution of a square system. That is, finding the inverse of a coefficient matrix A is not usually any easier than solving the system directly. Thus, unless we are given the inverse of a coefficient matrix A , as in [Example 3.14](#), we usually solve a square system by Gaussian elimination (or some iterative method whose discussion lies beyond the scope of this text).

3.10 Systems of Linear Equations and Linear Combinations of Vectors

The general system (3.2) of linear equations may be rewritten as the following vector equation:

$$x_1 \begin{bmatrix} a_{11} \\ a_{21} \\ \dots \\ a_{m1} \end{bmatrix} + x_2 \begin{bmatrix} a_{12} \\ a_{22} \\ \dots \\ a_{m2} \end{bmatrix} + \dots + x_n \begin{bmatrix} a_{1n} \\ a_{2n} \\ \dots \\ a_{mn} \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \dots \\ b_m \end{bmatrix}$$

Recall that a vector v in K^n is said to be a *linear combination* of vectors $u_1, u_2, \dots, u_m$ in K^n if there exist scalars $a_1, a_2, \dots, a_m$ in K such that

$$v = a_1 u_1 + a_2 u_2 + \dots + a_m u_m$$

Accordingly, the general system (3.2) of linear equations and the above equivalent vector equation have a solution if and only if the column vector of constants is a linear combination of the columns of the coefficient matrix. We state this observation formally.

THEOREM 3.11: A system $AX = B$ of linear equations has a solution if and only if B is a linear combination of the columns of the coefficient matrix A .

Thus, the answer to the problem of expressing a given vector v in K^n as a linear combination of vectors $u_1, u_2, \dots, u_m$ in K^n reduces to solving a system of linear equations.

Linear Combination Example

Suppose we want to write the vector $v = (1, 2, 5)$ as a linear combination of the vectors

$$u_1 = (1, 1, 1), \quad u_2 = (1, 2, 3), \quad u_3 = (2, -1, 1)$$

First we write $v = xu_1 + yu_2 + zu_3$ with unknowns x, y, z , and then we find the equivalent system of linear equations which we solve. Specifically, we first write

$$\begin{bmatrix} 1 \\ -2 \\ 5 \end{bmatrix} = x \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + y \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + z \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix} \quad (*)$$

Then

$$\begin{bmatrix} 1 \\ -2 \\ 5 \end{bmatrix} = \begin{bmatrix} x \\ x \\ x \end{bmatrix} + \begin{bmatrix} y \\ 2y \\ 3y \end{bmatrix} + \begin{bmatrix} 2z \\ -z \\ z \end{bmatrix} = \begin{bmatrix} x + y + 2z \\ x + 2y - z \\ x + 3y + z \end{bmatrix}$$

Setting corresponding entries equal to each other yields the following equivalent system:

$$\begin{aligned} x + y + 2z &= 1 \\ x + 2y - z &= -2 \\ x + 3y + z &= 5 \end{aligned}$$

For notational convenience, we have written the vectors in $\mathbf{R}^n$ as columns, because it is then easier to find the equivalent system of linear equations. In fact, one can easily go from the vector equation (*) directly to the system (**).

Now we solve the equivalent system of linear equations by reducing the system to echelon form. This yields

$$\begin{aligned} x + y + 2z &= 1 \\ y - 3z &= -3 \\ 2y - z &= 4 \end{aligned} \quad \text{and then} \quad \begin{aligned} x + y + 2z &= 1 \\ y - 3z &= -3 \\ 5z &= 10 \end{aligned}$$

Back-substitution yields the solution $x = -6$, $y = 3$, $z = 2$. Thus, $u = -6 u_1 + 3 u_2 + 2 u_3$.

EXAMPLE 3.15

- (a) Write the vector $u = (4, 9, 19)$ as a linear combination of

$$u_1 = (1, -2, 3), \quad u_2 = (3, -7, 10), \quad u_3 = (2, 1, 9).$$

Find the equivalent system of linear equations by writing $v = xu_1 + yu_2 + zu_3$, and reduce the system to an echelon form. We have

$$\begin{array}{rcl} x + 3y + 2z = 4 & & x + 3y + 2z = 4 & & x + 3y + 2z = 4 \\ -2x - 7y + z = 9 & \text{or} & -y + 5z = 17 & \text{or} & -y + 5z = 17 \\ 3x + 10y + 9z = 19 & & y + 3z = 7 & & 8z = 24 \end{array}$$

Back-substitution yields the solution $x = 4, y = 2, z = 3$. Thus, v is a linear combination of u_1, u_2, u_3 .

Specifically, $v = 4u_1 + 2u_2 + 3u_3$.

(b) Write the vector $v = (2, 3, 5)$ as a linear combination of

$$u_1 = (1, 2, -3), \quad u_2 = (2, 3, -4), \quad u_3 = (1, 3, -5)$$

Find the equivalent system of linear equations by writing $v = xu_1 + yu_2 + zu_3$, and reduce the system to an echelon form. We have

$$\begin{array}{rcl} x + 2y + z = 2 & & x + 2y + z = 2 & & x + 2y + z = 2 \\ 2x + 3y + 3z = 3 & \text{or} & -y + z = -1 & \text{or} & -5y + 5z = -1 \\ -3x - 4y - 5z = -5 & & 2y - 2z = 1 & & 0 = 3 \end{array}$$

The system has no solution. Thus, it is impossible to write v as a linear combination of u_1, u_2, u_3 .

Linear Combinations of Orthogonal Vectors, Fourier Coefficients

Recall first (Section 1.4) that the dot (inner) product $u \cdot v$ of vectors $u = (a_1, \dots, a_n)$ and $v = (b_1, \dots, b_n)$ in $\mathbf{R}^n$ is defined by

$$u \cdot v = a_1b_1 + a_2b_2 + \cdots + a_nb_n$$

Furthermore, vectors u and v are said to be *orthogonal* if their dot product $u \cdot v = 0$.

Suppose that $u_1, u_2, \dots, u_n$ in $\mathbf{R}^n$ are n nonzero pairwise orthogonal vectors. This means

$$(i) \quad u_i \cdot u_j = 0 \quad \text{for } i \neq j \quad \text{and} \quad (ii) \quad u_i \cdot u_i \neq 0 \quad \text{for each } i$$

Then, for any vector v in $\mathbf{R}^n$, there is an easy way to write v as a linear combination of $u_1, u_2, \dots, u_n$, which is illustrated in the next example.

EXAMPLE 3.16 Consider the following three vectors in $\mathbf{R}^3$:

$$u_1 = (1, 1, 1), \quad u_2 = (1, -3, 2), \quad u_3 = (5, -1, -4)$$

These vectors are pairwise orthogonal; that is,

$$u_1 \cdot u_2 = 1 - 3 + 2 = 0, \quad u_1 \cdot u_3 = 5 - 1 - 4 = 0, \quad u_2 \cdot u_3 = 5 + 3 - 8 = 0$$

Suppose we want to write $v = (4, 14, 9)$ as a linear combination of u_1, u_2, u_3 .

Method 1. Find the equivalent system of linear equations as in [Example 3.14](#) and then solve, obtaining $v = 3u_1 + 4u_2 + u_3$.

Method 2. (This method uses the fact that the vectors u_1, u_2, u_3 are mutually orthogonal, and hence, the arithmetic is much simpler.) Set v as a linear combination of u_1, u_2, u_3 using unknown scalars x, y, z as follows:

$$(4, 14, -9) = x(1, 1, 1) + y(1, -3, 2) + z(5, -1, -4) \tag{*}$$

Take the dot product of (*) with respect to u_1 to get

$$(4, 14, -9) \cdot (1, 1, 1) = x(1, 1, 1) \cdot (1, 1, 1) \quad \text{or} \quad 9 = 3x \quad \text{or} \quad x = 3$$

(The last two terms drop out, because u_1 is orthogonal to u_2 and to u_3 .) Next take the dot product of (*) with respect to u_2 to obtain

$$(4, 14, -9) \cdot (1, -3, 2) = y(1, -3, 2) \cdot (1, -3, 2) \quad \text{or} \quad -56 = 14y \quad \text{or} \quad y = -4$$

Finally, take the dot product of (*) with respect to u_3 to get

$$(4, 14, -9) \cdot (5, -1, -4) = z(5, -1, -4) \cdot (5, -1, -4) \quad \text{or} \quad 42 = 42z \quad \text{or} \quad z = 1$$

Thus, $v = 3u_1 - 4u_2 + u_3$.

The procedure in Method 2 in [Example 3.16](#) is valid in general. Namely,

THEOREM 3.12: Suppose $u_1, u_2, \dots, u_n$ are nonzero mutually orthogonal vectors in $\mathbf{R}^n$. Then, for any vector v in $\mathbf{R}^n$,

$$v = \frac{v \cdot u_1}{u_1 \cdot u_1} u_1 + \frac{v \cdot u_2}{u_2 \cdot u_2} u_2 + \cdots + \frac{v \cdot u_n}{u_n \cdot u_n} u_n$$

We emphasize that there must be n such orthogonal vectors u_i in $\mathbf{R}^n$ for the formula to be used. Note also that each $u_i \neq 0$, because each u_i is a nonzero vector.

Remark: The following scalar k_i (appearing in [Theorem 3.12](#)) is called the *Fourier coefficient* of v with respect to u_i :

$$k_i = \frac{v \cdot u_i}{u_i \cdot u_i} = \frac{v \cdot u_i}{\|u_i\|^2}$$

It is analogous to a coefficient in the celebrated Fourier series of a function.

3.11 Homogeneous Systems of Linear Equations

A system of linear equations is said to be *homogeneous* if all the constant terms are zero. Thus, a homogeneous system has the form $AX = 0$. Clearly, such a system always has the zero vector $0 = (0, 0, \dots, 0)$ as a solution, called the *zero* or *trivial* solution. Accordingly, we are usually interested in whether or not the system has a nonzero solution.

Because a homogeneous system $AX = 0$ has at least the zero solution, it can always be put in an echelon form, say

$$\begin{array}{rcl} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + a_{14}x_4 + \cdots + a_{1n}x_n & = & 0 \\ a_{2j_2}x_{j_2} + a_{2,j_2+1}x_{j_2+1} + \cdots + a_{2n}x_n & = & 0 \\ & \cdots & \\ a_{rj_r}x_{j_r} + \cdots + a_{rn}x_n & = & 0 \end{array}$$

Here r denotes the number of equations in echelon form and n denotes the number of unknowns. Thus, the echelon system has $n - r$ free variables.

The question of nonzero solutions reduces to the following two cases:

- (i) $r = n$. The system has only the zero solution.
(ii) $r < n$. The system has a nonzero solution.

Accordingly, if we begin with fewer equations than unknowns, then, in echelon form, $r < n$, and the system has a nonzero solution. This proves the following important result.

THEOREM 3.13: A homogeneous system $AX = 0$ with more unknowns than equations has a nonzero solution.

EXAMPLE 3.17 Determine whether or not each of the following homogeneous systems has a nonzero solution: (a)

$$\begin{array}{r} x + y - z = 0 \\ 2x - 3y + z = 0 \\ x - 4y + 2z = 0 \end{array}$$

(a)

$$\begin{array}{r} x + y - z = 0 \\ 2x + 4y - z = 0 \\ 3x + 2y + 2z = 0 \end{array}$$

(b)

$$\begin{array}{r} x_1 + 2x_2 - 3x_3 + 4x_4 = 0 \\ 2x_1 - 3x_2 + 5x_3 - 7x_4 = 0 \\ 5x_1 + 6x_2 - 9x_3 + 8x_4 = 0 \end{array}$$

(c)

(a) Reduce the system to echelon form as follows:

$$\begin{array}{r} x + y - z = 0 \\ -5y + 3z = 0 \\ -5y + 3z = 0 \end{array} \quad \text{and then} \quad \begin{array}{r} x + y - z = 0 \\ -5y + 3z = 0 \end{array}$$

The system has a nonzero solution, because there are only two equations in the three unknowns in echelon form. Here z is a free variable. Let us, say, set $z = 5$. Then, by back-substitution, $y = 3$ and $x = 2$. Thus, the vector $u = (2, 3, 5)$ is a particular nonzero solution.

(b) Reduce the system to echelon form as follows:

$$\begin{array}{r} x + y - z = 0 \\ 2y + z = 0 \\ -y + 5z = 0 \end{array} \quad \text{and then} \quad \begin{array}{r} x + y - z = 0 \\ 2y + z = 0 \\ 11z = 0 \end{array}$$

In echelon form, there are three equations in three unknowns. Thus, the system has only the zero solution.

(c) The system must have a nonzero solution ([Theorem 3.13](#)), because there are four unknowns but only three equations. (Here we do not need to reduce the system to echelon form.)

Basis for the General Solution of a Homogeneous System

Let W denote the general solution of a homogeneous system $AX = 0$. A list of nonzero solution vectors $u_1, u_2, \dots, u_s$ of the system is said to be a *basis* for W if each solution vector $w \in W$ can be expressed uniquely as a linear combination of the vectors $u_1, u_2, \dots, u_s$; that is, there exist unique scalars $a_1, a_2, \dots, a_s$ such that

$$w = a_1u_1 + a_2u_2 + \cdots + a_su_s$$

The number s of such basis vectors is equal to the number of free variables. This number s is called the *dimension* of W , written as $\dim W = s$. When $W = \{0\}$ —that is, the system has only the zero solution—we define $\dim W = 0$.

The following theorem, proved in [Chapter 5](#), page 171, tells us how to find such a basis.

THEOREM 3.14: Let W be the general solution of a homogeneous system $AX = 0$, and suppose that the echelon form of the homogeneous system has s free variables. Let $u_1, u_2, \dots, u_s$ be the solutions obtained by setting one of the free variables equal to 1 (or any nonzero constant) and the remaining free variables equal to 0. Then $\dim W = s$, and the vectors $u_1, u_2, \dots, u_s$ form a basis of W .

We emphasize that the general solution W may have many bases, and that [Theorem 3.12](#) only gives us one such basis.

EXAMPLE 3.18 Find the dimension and a basis for the general solution W of the homogeneous system

$$\begin{aligned}x_1 + 2x_2 - 3x_3 + 2x_4 - 4x_5 &= 0 \\2x_1 + 4x_2 - 5x_3 + x_4 - 6x_5 &= 0 \\5x_1 + 10x_2 - 13x_3 + 4x_4 - 16x_5 &= 0\end{aligned}$$

First reduce the system to echelon form. Apply the following operations:

“Replace L_2 by $-2L_1 + L_2$ ” and “Replace L_3 by $-5L_1 + L_3$ ” and then “Replace L_3 by $-2L_2 + L_3$ ”

These operations yield

$$\begin{array}{lcl}x_1 + 2x_2 - 3x_3 + 2x_4 - 4x_5 = 0 & & x_1 + 2x_2 - 3x_3 + 2x_4 - 4x_5 = 0 \\& x_3 - 3x_4 + 2x_5 = 0 & \text{and} \\& 2x_3 - 6x_4 + 4x_5 = 0 & x_3 - 3x_4 + 2x_5 = 0\end{array}$$

The system in echelon form has three free variables, x_2 , x_4 , x_5 ; hence, $\dim W = 3$. Three solution vectors that form a basis for W are obtained as follows:

- (1) Set $x_2 = 1$, $x_4 = 0$, $x_5 = 0$. Back-substitution yields the solution $u_1 = (-2, 1, 0, 0, 0)$.
- (2) Set $x_2 = 0$, $x_4 = 1$, $x_5 = 0$. Back-substitution yields the solution $u_2 = (7, 0, 3, 1, 0)$.
- (3) Set $x_2 = 0$, $x_4 = 0$, $x_5 = 1$. Back-substitution yields the solution $u_3 = (-2, 0, -2, 0, 1)$.

The vectors

$u_1 = (-2, 1, 0, 0, 0)$, $u_2 = (7, 0, 3, 1, 0)$, $u_3 = (-2, 0, -2, 0, 1)$ form a basis for W .

Remark: Any solution of the system in [Example 3.18](#) can be written in the form

$$\begin{aligned}au_1 + bu_2 + cu_3 &= a(-2, 1, 0, 0, 0) + b(7, 0, 3, 1, 0) + c(-2, 0, -2, 0, 1) \\&= (-2a + 7b - 2c, \quad a, \quad 3b - 2c, \quad b, \quad c)\end{aligned}$$

or

$$x_1 = -2a + 7b - 2c, \quad x_2 = a, \quad x_3 = 3b - 2c, \quad x_4 = b, \quad x_5 = c$$

where a, b, c are arbitrary constants. Observe that this representation is nothing more than the parametric form of the general solution under the choice of parameters $x_2 = a, x_4 = b, x_5 = c$.

Nonhomogeneous and Associated Homogeneous Systems

Let $AX = B$ be a nonhomogeneous system of linear equations. Then $AX = 0$ is called the *associated homogeneous system*. For example,

$$\begin{array}{rcl} x + 2y - 4z = 7 & & x + 2y - 4z = 0 \\ 3x - 5y + 6z = 8 & \text{and} & 3x - 5y + 6z = 0 \end{array}$$

show a nonhomogeneous system and its associated homogeneous system.

The relationship between the solution U of a nonhomogeneous system $AX = B$ and the solution W of its associated homogeneous system $AX = 0$ is contained in the following theorem.

THEOREM 3.15: Let v_0 be a particular solution of $AX = B$ and let W be the general solution of $AX = 0$. Then the following is the general solution of $AX = B$:

$$U = v_0 + W = \{v_0 + w : w \in W\}$$

That is, $U = v_0 + W$ is obtained by adding v_0 to each element in W . We note that this theorem has a geometrical interpretation in $\mathbf{R}^3$. Specifically, suppose W is a line through the origin O . Then, as pictured in [Fig. 3-4](#), $U = v_0 + W$ is the line parallel to W obtained by adding v_0 to each element of

W . Similarly, whenever W is a plane through the origin O , then $U = v_0 + W$ is a plane parallel to W .

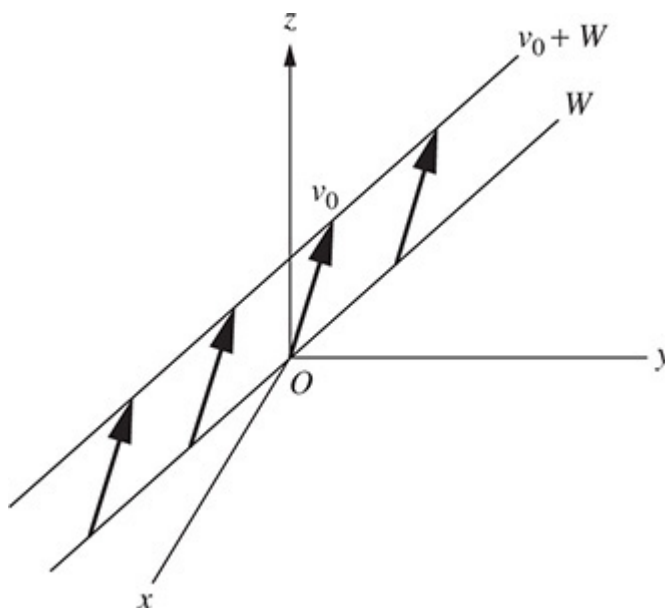


Figure 3-4

3.12 Elementary Matrices

Let e denote an elementary row operation and let $e(A)$ denote the results of applying the operation e to a matrix A . Now let E be the matrix obtained by applying e to the identity matrix I ; that is,

$$E = e(I)$$

Then E is called the *elementary matrix* corresponding to the elementary row operation e . Note that E is always a square matrix.

EXAMPLE 3.19 Consider the following three elementary row operations:

- (1) Interchange R_2 and R_3 . (2) Replace R_2 by $-6R_2$. (3) Replace R_3 by $-4R_1 + R_3$.

The 3×3 elementary matrices corresponding to the above elementary row operations are as follows:

$$E_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \quad E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -6 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad E_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -4 & 0 & 1 \end{bmatrix}$$

The following theorem, proved in [Problem 3.34](#), holds.

THEOREM 3.16: Let e be an elementary row operation and let E be the corresponding $m \times m$ elementary matrix. Then

$$e(A) = EA$$

where A is any $m \times n$ matrix.

In other words, the result of applying an elementary row operation e to a matrix A can be obtained by premultiplying A by the corresponding elementary matrix E .

Now suppose e' is the inverse of an elementary row operation e , and let E' and E be the corresponding matrices. We note ([Problem 3.33](#)) that E is invertible and E' is its inverse. This means, in particular, that any product

$$P = E_k \dots E_2 E_1$$

of elementary matrices is invertible.

Applications of Elementary Matrices

Using [Theorem 3.16](#), we are able to prove ([Problem 3.35](#)) the following important properties of matrices.

THEOREM 3.17: Let A be a square matrix. Then the following are equivalent:

(a) A is invertible (nonsingular).

(b) A is row equivalent to the identity matrix I .

(c) A is a product of elementary matrices.

Recall that square matrices A and B are inverses if $AB = BA = I$. The next theorem (proved in [Problem 3.36](#)) demonstrates that we need only show that one of the products is true, say $AB = I$, to prove that matrices are inverses.

THEOREM 3.18: Suppose $AB = I$. Then $BA = I$, and hence, $B = A^{-1}$.

Row equivalence can also be defined in terms of matrix multiplication. Specifically, we will prove ([Problem 3.37](#)) the following.

THEOREM 3.19: B is row equivalent to A if and only if there exists a nonsingular matrix P such that $B = PA$.

Application to Finding the Inverse of an $n \times n$ Matrix

The following algorithm finds the inverse of a matrix.

ALGORITHM 3.5: The input is a square matrix A . The output is the inverse of A or that the inverse does not exist.

Step 1. Form the $n - 2n$ (block) matrix $M = [A, I]$, where A is the left half of M and the identity matrix I is the right half of M .

Step 2. Row reduce M to echelon form. If the process generates a zero row in the A half of M , then STOP

STOP

A has no inverse. (Otherwise A is in triangular form.)

Step 3. Further row reduce M to its row canonical form

$$M \sim [I, B]$$

where the identity matrix I has replaced A in the left half of M .

Step 4. Set $A^{-1} = B$, the matrix that is now in the right half of M .

The justification for the above algorithm is as follows. Suppose A is invertible and, say, the sequence of elementary row operations $e_1, e_2, \dots, e_q$ applied to $M = [A, I]$ reduces the left half of M , which is A , to the identity matrix I . Let E_j be the elementary matrix corresponding to the operation e_j . Then, by applying [Theorem 3.16](#), we get

$$E_q \dots E_2 E_1 A = I \quad \text{or} \quad (E_q \dots E_2 E_1 I) A = I, \quad \text{so} \quad A^{-1} = E_q \dots E_2 E_1 I$$

That is, A^{-1} can be obtained by applying the elementary row operations $e_1, e_2, \dots, e_q$ to the identity matrix I , which appears in the right half of M . Thus, $B = A^{-1}$, as claimed.

EXAMPLE 3.20

Find the inverse of the matrix $A = \begin{bmatrix} 1 & 0 & 2 \\ 2 & -1 & 3 \\ 4 & 1 & 8 \end{bmatrix}$.

First form the (block) matrix $M = [A, I]$ and row reduce M to an echelon form:

$$M = \left[\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & 0 & 0 \\ 2 & -1 & 3 & 0 & 1 & 0 \\ 4 & 1 & 8 & 0 & 0 & 1 \end{array} \right] \sim \left[\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & -1 & -1 & -2 & 1 & 0 \\ 0 & 1 & 0 & -4 & 0 & 1 \end{array} \right] \sim \left[\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & -1 & -1 & -2 & 1 & 0 \\ 0 & 0 & -1 & -6 & 1 & 1 \end{array} \right]$$

In echelon form, the left half of M is in triangular form; hence, A has an inverse. Next we further row reduce M to its row canonical form:

$$M \sim \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -11 & 2 & 2 \\ 0 & -1 & 0 & 4 & 0 & -1 \\ 0 & 0 & 1 & 6 & -1 & -1 \end{array} \right] \sim \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -11 & 2 & 2 \\ 0 & 1 & 0 & -4 & 0 & 1 \\ 0 & 0 & 1 & 6 & -1 & -1 \end{array} \right]$$

The identity matrix is now in the left half of the final matrix; hence, the right half is A^{-1} . In other words,

$$A^{-1} = \begin{bmatrix} -11 & 2 & 2 \\ -4 & 0 & 1 \\ 6 & -1 & -1 \end{bmatrix}$$

Elementary Column Operations

Now let A be a matrix with columns $C_1, C_2, \dots, C_n$. The following operations on A , analogous to the elementary row operations, are called *elementary column operations*:

[F₁] (Column Interchange): Interchange columns C_i and C_j .

[F₂] (Column Scaling): Replace C_i by kC_i (where $k \neq 0$).

[F₃] (Column Addition): Replace C_j by $kC_i + C_j$.

We may indicate each of the column operations by writing, respectively,

$$(1) C_i \leftrightarrow C_j, \quad (2) kC_i \rightarrow C_i, \quad (3) (kC_i + C_j) \rightarrow C_j$$

Moreover, each column operation has an inverse operation of the same type, just like the corresponding row operation.

Now let f denote an elementary column operation, and let F be the matrix obtained by applying f to the identity matrix I ; that is,

$$F = f(I)$$

Then F is called the *elementary matrix* corresponding to the elementary column operation f . Note that F is always a square matrix.

EXAMPLE 3.21

Consider the following elementary column operations:

- (1) Interchange C_1 and C_3 ; (2) Replace C_3 by $-2C_3$; (3) Replace C_3 by $-3C_2 + C_3$

The corresponding three 3×3 elementary matrices are as follows:

$$F_1 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}, \quad F_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{bmatrix}, \quad F_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{bmatrix}$$

The following theorem is analogous to [Theorem 3.16](#) for the elementary row operations.

THEOREM 3.20: For any matrix A , $f(A) = AF$.

That is, the result of applying an elementary column operation f on a matrix A can be obtained by postmultiplying A by the corresponding elementary matrix F .

Matrix Equivalence

A matrix B is *equivalent* to a matrix A if B can be obtained from A by a sequence of row and column operations. Alternatively, B is equivalent to A , if there exist nonsingular matrices P and Q such that $B = PAQ$. Just like row equivalence, equivalence of matrices is an equivalence relation.

The main result of this subsection (proved in [Problem 3.38](#)) is as follows.

THEOREM 3.21: Every $m \times n$ matrix A is equivalent to a unique block matrix of the form

$$\begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$$

where I_r is the r -square identity matrix.

The following definition applies.

DEFINITION: The nonnegative integer r in [Theorem 3.21](#) is called the *rank* of A , written $\text{rank}(A)$.

Note that this definition agrees with the previous definition of the rank of a matrix.

3.13 LU DECOMPOSITION

Suppose A is a nonsingular matrix that can be brought into (upper) triangular form U using only row-addition operations; that is, suppose A can be triangularized by the following algorithm, which we write using computer notation.

ALGORITHM 3.6: The input is a matrix A and the output is a triangular matrix U .

Step 1. Repeat for $i = 1, 2, \dots, n - 1$:

Step 2. Repeat for $j = i + 1, i + 2, \dots, n$

(a) Set $m_{ij} := -a_{ij}/a_{ii}$.

(b) Set $R_j := m_{ij}R_i + R_j$

[End of Step 2 inner loop.]

[End of Step 1 outer loop.]

The numbers m_{ij} are called *multipliers*. Sometimes we keep track of these multipliers by means of the following lower triangular matrix L :

$$L = \begin{bmatrix} 1 & 0 & 0 & \dots & 0 & 0 \\ -m_{21} & 1 & 0 & \dots & 0 & 0 \\ -m_{31} & -m_{32} & 1 & \dots & 0 & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots \\ -m_{n1} & -m_{n2} & -m_{n3} & \dots & -m_{n,n-1} & 1 \end{bmatrix} \dots$$

That is, L has 1's on the diagonal, 0's above the diagonal, and the negative of the multiplier m_{ij} as its ij -entry below the diagonal.

The above matrix L and the triangular matrix U obtained in Algorithm 3.6 give us the classical LU factorization of such a matrix A . Namely,

THEOREM 3.22: Let A be a nonsingular matrix that can be brought into triangular form U using only row-addition operations. Then $A = LU$, where L is the above lower triangular matrix with 1's on the diagonal, and U is an upper triangular matrix with no 0's on the diagonal.

EXAMPLE 3.22 Suppose $A = \begin{bmatrix} 1 & 2 & -3 \\ -3 & -4 & 13 \\ 2 & 1 & -5 \end{bmatrix}$. We note that A may be reduced to triangular form by the operations

“Replace R_2 by $3R_1 + R_2$ ”; “Replace R_3 by $-2R_1 + R_3$ ”; and then “Replace R_3 by $\frac{3}{2}R_2 + R_3$ ”

That is,

$$A \sim \begin{bmatrix} 1 & 2 & -3 \\ 0 & 2 & 4 \\ 0 & -3 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & -3 \\ 0 & 2 & 4 \\ 0 & 0 & 7 \end{bmatrix}$$

This gives us the classical factorization $A = LU$, where

$$L = \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 2 & -\frac{3}{2} & 1 \end{bmatrix} \quad \text{and} \quad U = \begin{bmatrix} 1 & 2 & -3 \\ 0 & 2 & 4 \\ 0 & 0 & 7 \end{bmatrix}$$

We emphasize:

- (1) The entries $-3, 2, -\frac{3}{2}$ in L are the negatives of the multipliers in the above elementary row operations.
- (2) U is the triangular form of A .

Application to Systems of Linear Equations

Consider a computer algorithm M . Let $C(n)$ denote the running time of the algorithm as a function of the size n of the input data. [The function $C(n)$ is sometimes called the *time complexity* or simply the *complexity* of the algorithm M .] Frequently, $C(n)$ simply counts the number of multiplications and divisions executed by M , but does not count the number of additions and subtractions because they take much less time to execute.

Now consider a square system of linear equations $AX = B$, where

$$A = [a_{ij}], \quad X = [x_1, \dots, x_n]^T, \quad B = [b_1, \dots, b_n]^T$$

and suppose A has an LU factorization. Then the system can be brought into triangular form (in order to apply back-substitution) by applying Algorithm 3.6 to the augmented

matrix $M = [A, B]$ of the system. The time complexity of Algorithm 3.6 and back-substitution are, respectively,

$$C(n) \approx \frac{1}{2}n^3 \quad \text{and} \quad C(n) \approx \frac{1}{2}n^2$$

where n is the number of equations.

On the other hand, suppose we already have the factorization $A = LU$. Then, to triangularize the system, we need only apply the row operations in the algorithm (retained by the matrix L) to the column vector B . In this case, the time complexity is

$$C(n) \approx \frac{1}{2}n^2$$

Of course, to obtain the factorization $A = LU$ requires the original algorithm where $C(n) \approx \frac{1}{2}n^3$. Thus, nothing may be gained by first finding the LU factorization when a single system is involved. However, there are situations, illustrated below, where the LU factorization is useful.

Suppose, for a given matrix A , we need to solve the system

$$AX = B$$

repeatedly for a sequence of different constant vectors, say $B_1, B_2, \dots, B_k$. Also, suppose some of the B_i depend upon the solution of the system obtained while using preceding vectors B_j . In such a case, it is more efficient to first find the LU factorization of A , and then to use this factorization to solve the system for each new B .

EXAMPLE 3.23 Consider the following system of linear equations:

$$\begin{array}{l} x + 2y + z = k_1 \\ 2x + 3y + 3z = k_2 \\ -3x + 10y + 2z = k_3 \end{array} \quad \text{or} \quad AX = B, \quad \text{where} \quad A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 3 & 3 \\ -3 & 10 & 2 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} k_1 \\ k_2 \\ k_3 \end{bmatrix}$$

Suppose we want to solve the system three times where B is equal, say, to B_1, B_2, B_3 . Furthermore, suppose $B_1 = [1, 1, 1]^T$, and suppose

$$B_{j+1} = B_j + X_j \quad (\text{for } j = 1, 2)$$

where X_j is the solution of $AX = B_j$. Here it is more efficient to first obtain the LU factorization of A and then use the LU factorization to solve the system for each of the B 's. (This is done in [Problem 3.42](#).)

SOLVED PROBLEMS

Linear Equations, Solutions, 2×2 Systems

3.1. Determine whether each of the following equations is linear:

(a) $5x + 7y - 8yz = 16$, (b) $x + \pi y + ez = \log 5$, (c) $3x + ky - 8z = 16$

(a) No, because the product yz of two unknowns is of second degree.

(b) Yes, because π , e , and $\log 5$ are constants.

(c) As it stands, there are four unknowns: x, y, z, k . Because of the term ky it is not a linear equation. However, assuming k is a constant, the equation is linear in the unknowns x, y, z .

3.2. Determine whether the following vectors are solutions of $x_1 + 2x_2 - 4x_3 + 3x_4 = 15$: (a) $u = (3, 2, 1, 4)$ and (b) $v = (1, 2, 4, 5)$.

(a) Substitute to obtain $3 + 2(2) - 4(1) + 3(4) = 15$, or $15 = 15$; yes, it is a solution.

(b) Substitute to obtain $1 + 2(2) - 4(4) = 3(5) = 15$, or $4 = 15$; no, it is not a solution.

3.3. Solve

(a) $ex = \pi$, (b) $3x - 4 - x = 2x + 3$, (c) $7 + 2x - 4 = 3x + 3 - x$

(a) Because $e \neq 0$, multiply by $1/e$ to obtain $x = \pi/e$.

(b) Rewrite in standard form, obtaining $0x = 7$. The equation has no solution.

(c) Rewrite in standard form, obtaining $0x = 0$. Every scalar k is a solution.

3.4. Prove [Theorem 3.4](#): Consider the equation $ax = b$.

(i) If $a \neq 0$, then $x = b/a$ is a unique solution of $ax = b$.

(ii) If $a = 0$ but $b \neq 0$, then $ax = b$ has no solution.

(iii) If $a = 0$ and $b = 0$, then every scalar k is a solution of $ax = b$.

Suppose $a \neq 0$. Then the scalar b/a exists. Substituting b/a in $ax = b$ yields $a(b/a) = b$, or $b = b$; hence, b/a is a solution. On the other hand, suppose x_0 is a solution to $ax = b$, so that $ax_0 = b$. Multiplying both sides by $1/a$ yields $x_0 = b/a$. Hence, b/a is the unique solution of $ax = b$. Thus, (i) is proved.

On the other hand, suppose $a = 0$. Then, for any scalar k , we have $ak = 0k = 0$. If $b \neq 0$, then $ak \neq b$. Accordingly, k is not a solution of $ax = b$, and so (ii) is proved. If $b = 0$, then $ak = b$. That is, any scalar k is a solution of $ax = b$, and so (iii) is proved.

3.5. Solve each of the following systems:

(a)
$$\begin{aligned} 2x - 5y &= 11 \\ 3x + 4y &= 5 \end{aligned}$$

(b)
$$\begin{aligned} 2x - 3y &= 8 \\ -6x + 9y &= 6 \end{aligned}$$

(c)
$$\begin{aligned} 2x - 3y &= 8 \\ -4x + 6y &= -16 \end{aligned}$$

(a) Eliminate x from the equations by forming the new equation $L = 3L_1 + 2L_2$. This yields the equation

$$23y = -23, \quad \text{and so} \quad y = -1$$

Substitute $y = -1$ in one of the original equations, say L_1 , to get

$$2x - 5(-1) = 11 \quad \text{or} \quad 2x + 5 = 11 \quad \text{or} \quad 2x = 6 \quad \text{or} \quad x = 3$$

Thus, $x = 3$, $y = -1$ or the pair $u = (3, -1)$ is the unique solution of the system.

(b) Eliminate x from the equations by forming the new equation $L = 3L_1 + L_2$. This yields the equation

$$0x + 0y = 30$$

This is a degenerate equation with a nonzero constant; hence, this equation and the system have no solution. (Geometrically, the lines corresponding to the equations are parallel.)

(c) Eliminate x from the equations by forming the new equation $L = 2L_1 + L_2$. This yields the equation

$$0x + 0y = 0$$

This is a degenerate equation where the constant term is also zero. Thus, the system has an infinite number of solutions, which correspond to the solution of either equation. (Geometrically, the lines corresponding to the equations coincide.)

To find the general solution, set $y = a$ and substitute in L_1 to obtain

$$2x - 3a = 8 \quad \text{or} \quad 2x = 3a + 8 \quad \text{or} \quad x = \frac{3}{2}a + 4$$

Thus, the general solution is

$$x = \frac{3}{2}a + 4, \quad y = a \quad \text{or} \quad u = \left(\frac{3}{2}a + 4, a\right)$$

where a is any scalar.

3.6. Consider the system

$$\begin{aligned}x + ay &= 4 \\ax + 9y &= b\end{aligned}$$

- (a) For which values of a does the system have a unique solution?
- (b) Find those pairs of values (a, b) for which the system has more than one solution.
- (a) Eliminate x from the equations by forming the new equation $L = -aL_1 + L_2$. This yields the equation

$$(9 - a^2)y = b - 4a \tag{1}$$

The system has a unique solution if and only if the coefficient of y in (1) is not zero—that is, if $9 - a^2 \neq 0$ or if $a \neq \pm 3$.

- (b) The system has more than one solution if both sides of (1) are zero. The left-hand side is zero when $a = \pm 3$. When $a = 3$, the right-hand side is zero when $b - 12 = 0$ or $b = 12$. When $a = -3$, the right-hand side is zero when $b + 12 = 0$ or $b = -12$. Thus, $(3, 12)$ and $(-3, -12)$ are the pairs for which the system has more than one solution.

Systems in Triangular and Echelon Form

3.7. Determine the pivot and free variables in each of the following systems:

$$\begin{array}{rcl} 2x_1 - 3x_2 - 6x_3 - 5x_4 + 2x_5 & = & 7 \\ x_3 + 3x_4 - 7x_5 & = & 6 \\ x_4 - 2x_5 & = & 1 \end{array}$$

(a)

$$\begin{array}{rcl} 2x - 6y + 7z & = & 1 \\ 4y + 3z & = & 8 \\ 2z & = & 4 \end{array}$$

(b)

$$\begin{array}{rcl} x + 2y - 3z & = & 2 \\ 2x + 3y + z & = & 4 \\ 3x + 4y + 5z & = & 8 \end{array}$$

(c)

(a) In echelon form, the leading unknowns are the pivot variables, and the others are the free variables. Here x_1 , x_3 , x_4 are the pivot variables, and x_2 and x_5 are the free variables.

(b) The leading unknowns are x , y , z , so they are the pivot variables. There are no free variables (as in any triangular system).

(c) The notion of pivot and free variables applies only to a system in echelon form.

3.8. Solve the triangular system in [Problem 3.7\(b\)](#).

Because it is a triangular system, solve by back-substitution.

(i) The last equation gives $z = 2$.

(ii) Substitute $z = 2$ in the second equation to get $4y + 6 = 8$ or $y = \frac{1}{2}$.

(iii) Substitute $z = 2$ and $y = \frac{1}{2}$ in the first equation to get

$$2x - 6\left(\frac{1}{2}\right) + 7(2) = 1 \quad \text{or} \quad 2x + 11 = 1 \quad \text{or} \quad x = -5$$

Thus, $x = -5$, $y = \frac{1}{2}$, $z = 2$ or $u = (-5, \frac{1}{2}, 2)$ is the unique solution to the system.

3.9. Solve the echelon system in [Problem 3.7\(a\)](#).

Assign parameters to the free variables, say $x_2 = a$ and $x_5 = b$, and solve for the pivot variables by back-substitution.

(i) Substitute $x_5 = b$ in the last equation to get $x_4 - 2b = 1$ or $x_4 = 2b + 1$.

(ii) Substitute $x_5 = b$ and $x_4 = 2b + 1$ in the second equation to get

$$x_3 + 3(2b + 1) - 7b = 6 \quad \text{or} \quad x_3 - b + 3 = 6 \quad \text{or} \quad x_3 = b + 3$$

(iii) Substitute $z = 2$ and

$x_5 = b, x_4 = 2b + 1, x_3 = b + 3, x_2 = a$ in the first equation to get

$$2x_1 - 3a - 6(b + 3) - 5(2b + 1) + 2b = 7 \quad \text{or} \quad 2x_1 - 3a - 14b - 23 = 7$$
$$\text{or} \quad x_1 = \frac{3}{2}a + 7b + 15$$

Thus,

$$x_1 = \frac{3}{2}a + 7b + 15, \quad x_2 = a, \quad x_3 = b + 3, \quad x_4 = 2b + 1, \quad x_5 = b$$
$$\text{or} \quad u = \left(\frac{3}{2}a + 7b + 15, \quad a, \quad b + 3, \quad 2b + 1, \quad b \right)$$

is the parametric form of the general solution.

Alternatively, solving for the pivot variable x_1, x_3, x_4 in terms of the free variables x_2 and x_5 yields the following free-variable form of the general solution:

$$x_1 = \frac{3}{2}x_2 + 7x_5 + 15, \quad x_3 = x_5 + 3, \quad x_4 = 2x_5 + 1$$

3.10. Prove [Theorem 3.6](#). Consider the system (3.4) of linear equations in echelon form with r equations and n unknowns.

- (i) If $r = n$, then the system has a unique solution.
- (ii) If $r < n$, then we can arbitrarily assign values to the $n - r$ free variable and solve uniquely for the r pivot variables, obtaining a solution of the system.
- (i) Suppose $r = n$. Then we have a square system $AX = B$ where the matrix A of coefficients is (upper) triangular with nonzero diagonal elements. Thus, A is invertible. By [Theorem 3.10](#), the system has a unique solution.
- (ii) Assigning values to the $n - r$ free variables yields a triangular system in the pivot variables, which, by (i), has a unique solution.

Gaussian Elimination

3.11. Solve each of the following systems:

$$\begin{array}{rcl} x + 2y - 4z & = & -4 \\ 2x + 5y - 9z & = & -10 \\ 3x - 2y + 3z & = & 11 \end{array}$$

(a)

$$\begin{array}{rcl} x + 2y - 3z & = & -1 \\ -3x + y - 2z & = & -7 \\ 5x + 3y - 4z & = & 2 \end{array}$$

(b)

$$\begin{array}{rcl} x + 2y - 3z & = & 1 \\ 2x + 5y - 8z & = & 4 \\ 3x + 8y - 13z & = & 7 \end{array}$$

(c)

Reduce each system to triangular or echelon form using Gaussian elimination:

- (a) Apply “Replace L_2 by $-2L_1 + L_2$ ” and “Replace L_3 by $-3L_1 + L_3$ ” to eliminate x from the second and third equations, and then apply “Replace L_3 by $8L_2 + L_3$ ” to eliminate y from the third equation. These operations yield

$$\begin{array}{rcl}
 x + 2y - 4z = -4 & & x + 2y - 4z = -4 \\
 y - z = -2 & \text{and then} & y - z = -2 \\
 -8y + 15z = 23 & & 7z = 7
 \end{array}$$

The system is in triangular form. Solve by back-substitution to obtain the unique solution $u = (2, -1, 1)$.

- (b) Eliminate x from the second and third equations by the operations “Replace L_2 by $3L_1 + L_2$ ” and “Replace L_3 by $-5L_1 + L_3$.” This gives the equivalent system

$$\begin{array}{rcl}
 x + 2y - 3z & = & -1 \\
 7y - 11z & = & -10 \\
 -7y + 11z & = & 7
 \end{array}$$

The operation “Replace L_3 by $L_2 + L_3$ ” yields the following degenerate equation with a nonzero constant:

$$0x + 0y + 0z = -3$$

This equation and hence the system have no solution.

- (c) Eliminate x from the second and third equations by the operations “Replace L_2 by $-2L_1 + L_2$ ” and “Replace L_3 by $-3L_1 + L_3$.” This yields the new system $x +$

$$\begin{array}{rcl}
 x + 2y - 3z = 1 & & x + 2y - 3z = 1 \\
 y - 2z = 2 & \text{or} & y - 2z = 2 \\
 2y - 4z = 4 & &
 \end{array}$$

(The third equation is deleted, because it is a multiple of the second equation.) The system is in echelon form

with pivot variables x and y and free variable z .

To find the parametric form of the general solution, set $z = a$ and solve for x and y by back-substitution. Substitute $z = a$ in the second equation to get $y = 2 + 2a$. Then substitute $z = a$ and $y = 2 + 2a$ in the first equation to get

$$x + 2(2 + 2a) - 3a = 1 \quad \text{or} \quad x + 4 + a = 1 \quad \text{or} \quad x = -3 - a$$

Thus, the general solution is

$$x = -3 - a, \quad y = 2 + 2a, \quad z = a \quad \text{or} \quad u = (-3 - a, \quad 2 + 2a, \quad a)$$

where a is a parameter.

3.12. Solve each of the following systems:

$$\begin{array}{rcl} x_1 - 3x_2 + 2x_3 - x_4 + 2x_5 & = & 2 \\ 3x_1 - 9x_2 + 7x_3 - x_4 + 3x_5 & = & 7 \\ 2x_1 - 6x_2 + 7x_3 + 4x_4 - 5x_5 & = & 7 \end{array}$$

(a)

$$\begin{array}{rcl} x_1 + 2x_2 - 3x_3 + 4x_4 & = & 2 \\ 2x_1 + 5x_2 - 2x_3 + x_4 & = & 1 \\ 5x_1 + 12x_2 - 7x_3 + 6x_4 & = & 3 \end{array}$$

(b)

Reduce each system to echelon form using Gaussian elimination:

(a) Apply “Replace L_2 by $-3L_1 + L_2$ ” and “Replace L_3 by $-2L_1 + L_3$ ” to eliminate x from the second and third equations. This yields

$$\begin{array}{rcl} x_1 - 3x_2 + 2x_3 - x_4 + 2x_5 & = & 2 \\ x_3 + 2x_4 - 3x_5 & = & 1 \\ 3x_3 + 6x_4 - 9x_5 & = & 3 \end{array} \quad \text{or} \quad \begin{array}{rcl} x_1 - 3x_2 + 2x_3 - x_4 + 2x_5 & = & 2 \\ x_3 + 2x_4 - 3x_5 & = & 1 \end{array}$$

(We delete L_3 , because it is a multiple of L_2 .) The system is in echelon form with pivot variables x_1 and x_3 and free variables x_2, x_4, x_5 .

To find the parametric form of the general solution, set $x_2 = a$, $x_4 = b$, $x_5 = c$, where a , b , c are parameters. Back-substitution yields $x_3 = 1 - 2b + 3c$ and $x_1 = 3a + 5b - 8c$. The general solution is

$$x_1 = 3a + 5b - 8c, \quad x_2 = a, \quad x_3 = 1 - 2b + 3c, \quad x_4 = b, \quad x_5 = c$$

or, equivalently, $u = (3a + 5b - 8c, a, 1 - 2b + 3c, b, c)$.

- (b) Eliminate x_1 from the second and third equations by the operations “Replace L_2 by $-2L_1 + L_2$ ” and “Replace L_3 by $-5L_1 + L_3$.” This yields the system

$$\begin{aligned} x_1 + 2x_2 - 3x_3 + 4x_4 &= 2 \\ x_2 + 4x_3 - 7x_4 &= -3 \\ 2x_2 + 8x_3 - 14x_4 &= -7 \end{aligned}$$

The operation “Replace L_3 by $-2L_2 + L_3$ ” yields the degenerate equation $0 = -1$. Thus, the system has no solution (even though the system has more unknowns than equations).

3.13. Solve using the condensed format:

$$\begin{aligned} 2y + 3z &= 3 \\ x + y + z &= 4 \\ 4x + 8y - 3z &= 35 \end{aligned}$$

The condensed format follows:

	Number	Equation	Operation
(2)	(1)	$2y + 3z = 3$	$L_1 \leftrightarrow L_2$
(1)	(2)	$x + y + z = 4$	$L_1 \leftrightarrow L_2$
	(3)	$4x + 8y - 3z = 35$	
	(3')	$4y - 7z = 19$	Replace L_3 by $-4L_1 + L_3$
	(3'')	$-13z = 13$	Replace L_3 by $-2L_2 + L_3$

Here (1), (2), and (3'') form a triangular system. (We emphasize that the interchange of L_1 and L_2 is accomplished by simply renumbering L_1 and L_2 as above.)

Using back-substitution with the triangular system yields $z = -1$ from L_3 , $y = 3$ from L_2 , and $x = 2$ from L_1 . Thus, the unique solution of the system is $x = 2$, $y = 3$, $z = -1$ or the triple $u = (2, 3, -1)$.

3.14. Consider the system

$$\begin{array}{rcl} x + 2y + z & = & 3 \\ ay + 5z & = & 10 \\ 2x + 7y + az & = & b \end{array}$$

- Find those values of a for which the system has a unique solution.
- Find those pairs of values (a, b) for which the system has more than one solution.

Reduce the system to echelon form. That is, eliminate x from the third equation by the operation "Replace L_3 by $-2L_1 + L_3$ " and then eliminate y from the third equation by the operation "Replace L_3 by $-3L_2 + aL_3$." This yields

$$\begin{array}{rcl}
 x + 2y & + & z = 3 \\
 ay & + & 5z = 10 \\
 3y + (a - 2)z & = & b - 6
 \end{array}
 \quad \text{and then} \quad
 \begin{array}{rcl}
 x + 2y + z & = & 3 \\
 ay + 5z & = & 10 \\
 (a^2 - 2a - 15)z & = & ab - 6a - 30
 \end{array}$$

Examine the last equation $(a^2 - 2a - 15)z = ab - 6a - 30$.

- (a) The system has a unique solution if and only if the coefficient of z is not zero; that is, if

$$a^2 - 2a - 15 = (a - 5)(a + 3) \neq 0 \quad \text{or} \quad a \neq 5 \quad \text{and} \quad a \neq -3.$$

- (b) The system has more than one solution if both sides are zero. The left-hand side is zero when $a = 5$ or $a = -3$. When $a = 5$, the right-hand side is zero when $5b - 60 = 0$, or $b = 12$. When $a = -3$, the right-hand side is zero when $-3b - 12 = 0$, or $b = -4$. Thus, $(5, 12)$ and $(-3, -4)$ are the pairs for which the system has more than one solution.

Echelon Matrices, Row Equivalence, Row Canonical Form

3.15. Row reduce each of the following matrices to echelon form:

$$(a) \quad A = \begin{bmatrix} 1 & 2 & -3 & 0 \\ 2 & 4 & -2 & 2 \\ 3 & 6 & -4 & 3 \end{bmatrix}, \quad (b) \quad B = \begin{bmatrix} -4 & 1 & -6 \\ 1 & 2 & -5 \\ 6 & 3 & -4 \end{bmatrix}$$

- (a) Use $a_{11} = 1$ as a pivot to obtain 0's below a_{11} ; that is, apply the row operations "Replace R_2 by $-2R_1 + R_2$ " and "Replace R_3 by $-3R_1 + R_3$." Then use $a_{23} = 4$ as a pivot to obtain a 0 below a_{23} ; that is, apply the row operation "Replace R_3 by $-5R_2 + 4R_3$." These operations yield

$$A \sim \begin{bmatrix} 1 & 2 & -3 & 0 \\ 0 & 0 & 4 & 2 \\ 0 & 0 & 5 & 3 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & -3 & 0 \\ 0 & 0 & 4 & 2 \\ 0 & 0 & 0 & 2 \end{bmatrix}$$

The matrix is now in echelon form.

- (b) Hand calculations are usually simpler if the pivot element equals 1. Therefore, first interchange R_1 and R_2 . Next apply the operations “Replace R_2 by $4R_1 + R_2$ ” and “Replace R_3 by $-6R_1 + R_3$ ”; and then apply the operation “Replace R_3 by $R_2 + R_3$.” These operations yield

$$B \sim \begin{bmatrix} 1 & 2 & -5 \\ -4 & 1 & -6 \\ 6 & 3 & -4 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & -5 \\ 0 & 9 & -26 \\ 0 & -9 & 26 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & -5 \\ 0 & 9 & -26 \\ 0 & 0 & 0 \end{bmatrix}$$

The matrix is now in echelon form.

- 3.16.** Describe the *pivoting* row-reduction algorithm. Also describe the advantages, if any, of using this pivoting algorithm.

The row-reduction algorithm becomes a pivoting algorithm if the entry in column j of greatest absolute value is chosen as the pivot a_{1j_1} and if one uses the row operation

$$(-a_{ij_1}/a_{1j_1})R_1 + R_i \rightarrow R_i$$

The main advantage of the pivoting algorithm is that the above row operation involves division by the (current) pivot a_{1j_1} , and, on the computer, roundoff errors may be substantially reduced when one divides by a number as large in absolute value as possible.

3.17. Let $A = \begin{bmatrix} 2 & -2 & 2 & 1 \\ -3 & 6 & 0 & -1 \\ 1 & -7 & 10 & 2 \end{bmatrix}$. Reduce A to echelon form using the pivoting algorithm.

First interchange R_1 and R_2 so that -3 can be used as the pivot, and then apply the operations “Replace R_2 by $\frac{2}{3} R_1 + R_2$ ” and “Replace R_3 by $\frac{1}{3} R_1 + R_3$.” These operations yield

$$A \sim \begin{bmatrix} -3 & 6 & 0 & -1 \\ 2 & -2 & 2 & 1 \\ 1 & -7 & 10 & 2 \end{bmatrix} \sim \begin{bmatrix} -3 & 6 & 0 & -1 \\ 0 & 2 & 2 & \frac{1}{3} \\ 0 & -5 & 10 & \frac{5}{3} \end{bmatrix}$$

Now interchange R_2 and R_3 so that -5 can be used as the pivot, and then apply the operation “Replace R_3 by $\frac{2}{5} R_2 + R_3$.” We obtain

$$A \sim \begin{bmatrix} -3 & 6 & 0 & -1 \\ 0 & -5 & 10 & \frac{5}{3} \\ 0 & 2 & 2 & \frac{1}{3} \end{bmatrix} \sim \begin{bmatrix} -3 & 6 & 0 & -1 \\ 0 & -5 & 10 & \frac{5}{3} \\ 0 & 0 & 6 & 1 \end{bmatrix}$$

The matrix has been brought to echelon form using partial pivoting.

3.18. Reduce each of the following matrices to row canonical form:

$$(a) \quad A = \begin{bmatrix} 2 & 2 & -1 & 6 & 4 \\ 4 & 4 & 1 & 10 & 13 \\ 8 & 8 & -1 & 26 & 23 \end{bmatrix}, \quad (b) \quad B = \begin{bmatrix} 5 & -9 & 6 \\ 0 & 2 & 3 \\ 0 & 0 & 7 \end{bmatrix}$$

(a) First reduce A to echelon form by applying the operations “Replace R_2 by $-2R_1 + R_2$ ” and “Replace

R_3 by $4R_1 + R_3$,” and then applying the operation “Replace R_3 by $R_2 + R_3$.” These operations yield

$$A \sim \begin{bmatrix} 2 & 2 & -1 & 6 & 4 \\ 0 & 0 & 3 & -2 & 5 \\ 0 & 0 & 3 & 2 & 7 \end{bmatrix} \sim \begin{bmatrix} 2 & 2 & -1 & 6 & 4 \\ 0 & 0 & 3 & -2 & 5 \\ 0 & 0 & 0 & 4 & 2 \end{bmatrix}$$

Now use back-substitution on the echelon matrix to obtain the row canonical form of A . Specifically, first multiply R_3 by $\frac{1}{4}$ to obtain the pivot $a_{34} = 1$, and then apply the operations “Replace R_2 by $2R_3 + R_2$ ” and “Replace R_1 by $-6R_3 + R_1$.” These operations yield

$$A \sim \begin{bmatrix} 2 & 2 & -1 & 6 & 4 \\ 0 & 0 & 3 & -2 & 5 \\ 0 & 0 & 0 & 1 & \frac{1}{2} \end{bmatrix} \sim \begin{bmatrix} 2 & 2 & -1 & 0 & 1 \\ 0 & 0 & 3 & 0 & 6 \\ 0 & 0 & 0 & 1 & \frac{1}{2} \end{bmatrix}$$

Now multiply R_2 by $\frac{1}{3}$, making the pivot $a_{23} = 1$, and then apply “Replace R_1 by $R_2 + R_1$,” yielding

$$A \sim \begin{bmatrix} 2 & 2 & -1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 1 & \frac{1}{2} \end{bmatrix} \sim \begin{bmatrix} 2 & 2 & 0 & 0 & 3 \\ 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 1 & \frac{1}{2} \end{bmatrix}$$

Finally, multiply R_1 by $\frac{1}{2}$, so the pivot $a_{11} = 1$. Thus, we obtain the following row canonical form of A :

$$A \sim \begin{bmatrix} 1 & 1 & 0 & 0 & \frac{3}{2} \\ 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 1 & \frac{1}{2} \end{bmatrix}$$

- (b) Because B is in echelon form, use back-substitution to obtain

$$B \sim \begin{bmatrix} 5 & -9 & 6 \\ 0 & 2 & 3 \\ 0 & 0 & 1 \end{bmatrix} \sim \begin{bmatrix} 5 & -9 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \sim \begin{bmatrix} 5 & -9 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \sim \begin{bmatrix} 5 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

The last matrix, which is the identity matrix I , is the row canonical form of B . (This is expected, because B is invertible, and so its row canonical form must be I .)

3.19. Describe the Gauss-Jordan elimination algorithm, which also row reduces an arbitrary matrix A to its row canonical form.

The Gauss-Jordan algorithm is similar in some ways to the Gaussian elimination algorithm, except that here each pivot is used to place 0's both below and above the pivot, not just below the pivot, before working with the next pivot. Also, one variation of the algorithm first *normalizes* each row—that is, obtains a unit pivot—before it is used to produce 0's in the other rows, rather than normalizing the rows at the end of the algorithm.

3.20. Let $A = \begin{bmatrix} 1 & -2 & 3 & 1 & 2 \\ 1 & 1 & 4 & -1 & 3 \\ 2 & 5 & 9 & -2 & 8 \end{bmatrix}$. Use Gauss-Jordan to find the row canonical form of A .

Use $a_{11} = 1$ as a pivot to obtain 0's below a_{11} by applying the operations “Replace R_2 by $-R_1 + R_2$ ” and “Replace R_3 by $-2R_1 + R_3$.” This yields

$$A \sim \begin{bmatrix} 1 & -2 & 3 & 1 & 2 \\ 0 & 3 & 1 & -2 & 1 \\ 0 & 9 & 3 & -4 & 4 \end{bmatrix}$$

Multiply R_2 by $\frac{1}{3}$ to make the pivot $a_{22} = 1$, and then produce 0's below and above a_{22} by applying the operations "Replace R_3 by $-9R_2 + R_3$ " and "Replace R_1 by $2R_2 + R_1$." These operations yield

$$A \sim \begin{bmatrix} 1 & -2 & 3 & 1 & 2 \\ 0 & 1 & \frac{1}{3} & -\frac{2}{3} & \frac{1}{3} \\ 0 & 9 & 3 & -4 & 4 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & \frac{11}{3} & -\frac{1}{3} & \frac{8}{3} \\ 0 & 1 & \frac{1}{3} & -\frac{2}{3} & \frac{1}{3} \\ 0 & 0 & 0 & 2 & 1 \end{bmatrix}$$

Finally, multiply R_3 by $\frac{1}{2}$ to make the pivot $a_{34} = 1$, and then produce 0's above a_{34} by applying the operations "Replace R_2 by $\frac{2}{3}R_3 + R_2$ " and "Replace R_1 by $\frac{1}{3}R_3 + R_1$." These operations yield

$$A \sim \begin{bmatrix} 1 & 0 & \frac{11}{3} & -\frac{1}{3} & \frac{8}{3} \\ 0 & 1 & \frac{1}{3} & -\frac{2}{3} & \frac{1}{3} \\ 0 & 0 & 0 & 1 & \frac{1}{2} \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & \frac{11}{3} & 0 & \frac{17}{6} \\ 0 & 1 & \frac{1}{3} & 0 & \frac{2}{3} \\ 0 & 0 & 0 & 1 & \frac{1}{2} \end{bmatrix}$$

which is the row canonical form of A .

Systems of Linear Equations in Matrix Form

3.21. Find the augmented matrix M and the coefficient matrix A of the following system:

$$\begin{aligned} x + 2y - 3z &= 4 \\ 3y - 4z + 7x &= 5 \\ 6z + 8x - 9y &= 1 \end{aligned}$$

First align the unknowns in the system, and then use the aligned system to obtain M and A . We have

$$\begin{array}{l} x + 2y - 3z = 4 \\ 7x + 3y - 4z = 5; \\ 8x - 9y + 6z = 1 \end{array} \quad \text{then} \quad M = \begin{bmatrix} 1 & 2 & -3 & 4 \\ 7 & 3 & -4 & 5 \\ 8 & -9 & 6 & 1 \end{bmatrix} \quad \text{and} \quad A = \begin{bmatrix} 1 & 2 & -3 \\ 7 & 3 & -4 \\ 8 & -9 & 6 \end{bmatrix}$$

3.22. Solve each of the following systems using its augmented matrix M :

$$\begin{array}{rcl} x + 2y - z & = & 3 \\ x + 3y + z & = & 5 \\ 3x + 8y + 4z & = & 17 \end{array} \qquad \begin{array}{rcl} x - 2y + 4z & = & 2 \\ 2x - 3y + 5z & = & 3 \\ 3x - 4y + 6z & = & 7 \end{array} \qquad \begin{array}{rcl} x + y + 3z & = & 1 \\ 2x + 3y - z & = & 3 \\ 5x + 7y + z & = & 7 \end{array}$$

(a) Reduce the augmented matrix M to echelon form as follows:

$$M = \begin{bmatrix} 1 & 2 & -1 & 3 \\ 1 & 3 & 1 & 5 \\ 3 & 8 & 4 & 17 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & 1 & 2 & 2 \\ 0 & 2 & 7 & 8 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & 1 & 2 & 2 \\ 0 & 0 & 3 & 4 \end{bmatrix}$$

Now write down the corresponding triangular system

$$\begin{array}{rcl} x + 2y - z & = & 3 \\ y + 2z & = & 2 \\ 3z & = & 4 \end{array}$$

and solve by back-substitution to obtain the unique solution

$$x = \frac{17}{3}, y = -\frac{2}{3}, z = \frac{4}{3} \quad \text{or} \quad u = \left(\frac{17}{3}, -\frac{2}{3}, \frac{4}{3}\right)$$

Alternately, reduce the echelon form of M to row canonical form, obtaining

$$M \sim \begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & 1 & 2 & 2 \\ 0 & 0 & 1 & \frac{4}{3} \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 0 & \frac{13}{3} \\ 0 & 1 & 0 & -\frac{2}{3} \\ 0 & 0 & 1 & \frac{4}{3} \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & \frac{17}{3} \\ 0 & 1 & 0 & -\frac{2}{3} \\ 0 & 0 & 1 & \frac{4}{3} \end{bmatrix}$$

This also corresponds to the above solution.

- (b) First reduce the augmented matrix M to echelon form as follows:

$$M = \begin{bmatrix} 1 & -2 & 4 & 2 \\ 2 & -3 & 5 & 3 \\ 3 & -4 & 6 & 7 \end{bmatrix} \sim \begin{bmatrix} 1 & -2 & 4 & 2 \\ 0 & 1 & -3 & -1 \\ 0 & 2 & -6 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & -2 & 4 & 2 \\ 0 & 1 & -3 & -1 \\ 0 & 0 & 0 & 3 \end{bmatrix}$$

The third row corresponds to the degenerate equation $0x + 0y + 0z = 3$, which has no solution. Thus, "DO NOT CONTINUE." The original system also has no solution. (Note that the echelon form indicates whether or not the system has a solution.)

- (c) Reduce the augmented matrix M to echelon form and then to row canonical form:

$$M = \begin{bmatrix} 1 & 1 & 3 & 1 \\ 2 & 3 & -1 & 3 \\ 5 & 7 & 1 & 7 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 3 & 1 \\ 0 & 1 & -7 & 1 \\ 0 & 2 & -14 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 10 & 0 \\ 0 & 1 & -7 & 1 \end{bmatrix}$$

(The third row of the second matrix is deleted, because it is a multiple of the second row and will result in a zero row.) Write down the system corresponding to the row canonical form of M and then transfer the free variables to the other side to obtain the free-variable form of the solution:

$$\begin{array}{lcl} x + 10z = 0 & & x = -10z \\ y - 7z = 1 & \text{and} & y = 1 + 7z \end{array}$$

Here z is the only free variable. The parametric solution, using $z = a$, is as follows:

$$x = -10a, \quad y = 1 + 7a, \quad z = a \quad \text{or} \quad u = (-10a, 1 + 7a, a)$$

3.23. Solve the following system using its augmented matrix M :

$$\begin{aligned}x_1 + 2x_2 - 3x_3 - 2x_4 + 4x_5 &= 1 \\2x_1 + 5x_2 - 8x_3 - x_4 + 6x_5 &= 4 \\x_1 + 4x_2 - 7x_3 + 5x_4 + 2x_5 &= 8\end{aligned}$$

Reduce the augmented matrix M to echelon form and then to row canonical form:

$$\begin{aligned}M &= \begin{bmatrix} 1 & 2 & -3 & -2 & 4 & 1 \\ 2 & 5 & -8 & -1 & 6 & 4 \\ 1 & 4 & -7 & 5 & 2 & 8 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & -3 & -2 & 4 & 1 \\ 0 & 1 & -2 & 3 & -2 & 2 \\ 0 & 2 & -4 & 7 & -2 & 7 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & -3 & -2 & 4 & 1 \\ 0 & 1 & -2 & 3 & -2 & 2 \\ 0 & 0 & 0 & 1 & 2 & 3 \end{bmatrix} \\ &\sim \begin{bmatrix} 1 & 2 & -3 & 0 & 8 & 7 \\ 0 & 1 & -2 & 0 & -8 & -7 \\ 0 & 0 & 0 & 1 & 2 & 3 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 & 0 & 24 & 21 \\ 0 & 1 & -2 & 0 & -8 & -7 \\ 0 & 0 & 0 & 1 & 2 & 3 \end{bmatrix}\end{aligned}$$

Write down the system corresponding to the row canonical form of M and then transfer the free variables to the other side to obtain the free-variable form of the solution:

$$\begin{aligned}x_1 + x_3 + 24x_5 &= 21 \\x_2 - 2x_3 - 8x_5 &= -7 \\x_4 + 2x_5 &= 3\end{aligned} \quad \text{and} \quad \begin{aligned}x_1 &= 21 - x_3 - 24x_5 \\x_2 &= -7 + 2x_3 + 8x_5 \\x_4 &= 3 - 2x_5\end{aligned}$$

Here x_1, x_2, x_4 are the pivot variables and x_3 and x_5 are the free variables. Recall that the parametric form of the solution can be obtained from the free-variable form of the solution by simply setting the free variables equal to parameters, say $x_3 = a$, $x_5 = b$. This process yields

$$x_1 = 21 - a - 24b, \quad x_2 = -7 + 2a + 8b, \quad x_3 = a, \quad x_4 = 3 - 2b, \quad x_5 = b$$

or

$$u = (21 - a - 24b, -7 + 2a + 8b, a, 3 - 2b, b)$$

which is another form of the solution.

Linear Combinations, Homogeneous Systems

3.24. Write v as a linear combination of u_1, u_2, u_3 , where

(a) $v = (3, 10, 7)$ and $u_1 = (1, 3, -2)$, $u_2 = (1, 4, 2)$, $u_3 = (2, 8, 1)$;

(b) $v = (2, 7, 10)$ and $u_1 = (1, 2, 3)$, $u_2 = (1, 3, 5)$, $u_3 = (1, 5, 9)$;

(c) $v = (1, 5, 4)$ and $u_1 = (1, 3, -2)$, $u_2 = (2, 7, -1)$, $u_3 = (1, 6, 7)$.

Find the equivalent system of linear equations by writing $v = xu_1 + yu_2 + zu_3$. Alternatively, use the augmented matrix M of the equivalent system, where $M = [u_1, u_2, u_3, v]$. (Here u_1, u_2, u_3, v are the columns of M .)

(a) The vector equation $v = xu_1 + yu_2 + zu_3$ for the given vectors is as follows:

$$\begin{bmatrix} 3 \\ 10 \\ 7 \end{bmatrix} = x \begin{bmatrix} 1 \\ 3 \\ -2 \end{bmatrix} + y \begin{bmatrix} 1 \\ 4 \\ 2 \end{bmatrix} + z \begin{bmatrix} 2 \\ 8 \\ 1 \end{bmatrix} = \begin{bmatrix} x + y + 2z \\ 3x + 4y + 8z \\ -2x + 2y + z \end{bmatrix}$$

Form the equivalent system of linear equations by setting corresponding entries equal to each other, and then reduce the system to echelon form:

$$\begin{array}{lll} x + y + 2z = 3 & & x + y + 2z = 3 \\ 3x + 4y + 8z = 10 & \text{or} & y + 2z = 1 \\ -2x + 2y + z = 7 & & 4y + 5z = 13 \end{array} \quad \begin{array}{lll} x + y + 2z = 3 & & x + y + 2z = 3 \\ y + 2z = 1 & \text{or} & y + 2z = 1 \\ -3z = 9 & & -3z = 9 \end{array}$$

The system is in triangular form. Back-substitution yields the unique solution $x = 2$, $y = 7$, $z = -3$. Thus, $v = 2u_1 + 7u_2 - 3u_3$.

Alternatively, form the augmented matrix $M = [u_1, u_2, u_3, v]$ of the equivalent system, and reduce M to echelon form:

$$M = \begin{bmatrix} 1 & 1 & 2 & 3 \\ 3 & 4 & 8 & 10 \\ -2 & 2 & 1 & 7 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 2 & 3 \\ 0 & 1 & 2 & 1 \\ 0 & 4 & 5 & 13 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 2 & 3 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & -3 & 9 \end{bmatrix}$$

The last matrix corresponds to a triangular system that has a unique solution. Back-substitution yields the solution $x = 2$, $y = 7$, $z = -3$. Thus, $v = 2u_1 + 7u_2 - 3u_3$.

- (b) Form the augmented matrix $M = [u_1, u_2, u_3, v]$ of the equivalent system, and reduce M to the echelon form:

$$M = \begin{bmatrix} 1 & 1 & 1 & 2 \\ 2 & 3 & 5 & 7 \\ 3 & 5 & 9 & 10 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 & 2 \\ 0 & 1 & 3 & 3 \\ 0 & 2 & 6 & 4 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 & 2 \\ 0 & 1 & 3 & 3 \\ 0 & 0 & 0 & -2 \end{bmatrix}$$

The third row corresponds to the degenerate equation $0x + 0y + 0z = -2$, which has no solution. Thus, the system also has no solution, and v cannot be written as a linear combination of u_1, u_2, u_3 .

- (c) Form the augmented matrix $M = [u_1, u_2, u_3, v]$ of the equivalent system, and reduce M to echelon form:

$$M = \begin{bmatrix} 1 & 2 & 1 & 1 \\ 3 & 7 & 6 & 5 \\ -2 & -1 & 7 & 4 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 1 & 1 \\ 0 & 1 & 3 & 2 \\ 0 & 3 & 9 & 6 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 1 & 1 \\ 0 & 1 & 3 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

The last matrix corresponds to the following system with free variable z :

$$\begin{aligned}x + 2y + z &= 1 \\ y + 3z &= 2\end{aligned}$$

Thus, v can be written as a linear combination of u_1 , u_2 , u_3 in many ways. For example, let the free variable $z = 1$, and, by back-substitution, we get $y = -2$ and $x = 2$. Thus, $v = 2u_1 - 2u_2 + u_3$.

3.25. Let $u_1 = (1, 2, 4)$, $u_2 = (2, -3, 1)$, $u_3 = (2, 1, -1)$ in $\mathbf{R}^3$. Show that u_1 , u_2 , u_3 are orthogonal, and write v as a linear combination of u_1 , u_2 , u_3 , where (a) $v = (7, 16, 6)$, (b) $v = (3, 5, 2)$.

Take the dot product of pairs of vectors to get

$$u_1 \cdot u_2 = 2 - 6 + 4 = 0, \quad u_1 \cdot u_3 = 2 + 2 - 4 = 0, \quad u_2 \cdot u_3 = 4 - 3 - 1 = 0$$

Thus, the three vectors in $\mathbf{R}^3$ are orthogonal, and hence Fourier coefficients can be used. That is, $v = xu_1 + yu_2 + zu_3$, where

$$x = \frac{v \cdot u_1}{u_1 \cdot u_1}, \quad y = \frac{v \cdot u_2}{u_2 \cdot u_2}, \quad z = \frac{v \cdot u_3}{u_3 \cdot u_3}$$

(a) We have

$$x = \frac{7 + 32 + 24}{1 + 4 + 16} = \frac{63}{21} = 3, \quad y = \frac{14 - 48 + 6}{4 + 9 + 1} = \frac{-28}{14} = -2, \quad z = \frac{14 + 16 - 6}{4 + 1 + 1} = \frac{24}{6} = 4$$

Thus, $v = 3u_1 - 2u_2 + 4u_3$.

(b) We have

$$x = \frac{3 + 10 + 8}{1 + 4 + 16} = \frac{21}{21} = 1, \quad y = \frac{6 - 15 + 2}{4 + 9 + 1} = \frac{-7}{14} = -\frac{1}{2}, \quad z = \frac{6 + 5 - 2}{4 + 1 + 1} = \frac{9}{6} = \frac{3}{2}$$

Thus, $v = 3u_1 - 2u_2 + 4u_3$.

3.26. Find the dimension and a basis for the general solution W of each of the following homogeneous systems:

$$2x_1 + 4x_2 - 5x_3 + 3x_4 = 0$$

$$3x_1 + 6x_2 - 7x_3 + 4x_4 = 0$$

$$5x_1 + 10x_2 - 11x_3 + 6x_4 = 0$$

(a)

$$x - 2y - 3z = 0$$

$$2x + y + 3z = 0$$

$$3x - 4y - 2z = 0$$

(b)

(a) Reduce the system to echelon form using the operations "Replace L_2 by $-3L_1 + 2L_2$," "Replace L_3 by $-5L_1 + 2L_3$," and then "Replace L_3 by $-2L_2 + L_3$." These operations yield

$$2x_1 + 4x_2 - 5x_3 + 3x_4 = 0$$

$$x_3 - x_4 = 0$$

$$3x_3 - 3x_4 = 0$$

and

$$2x_1 + 4x_2 - 5x_3 + 3x_4 = 0$$

$$x_3 - x_4 = 0$$

The system in echelon form has two free variables, x_2 and x_4 , so $\dim W = 2$. A basis $[u_1, u_2]$ for W may be obtained as follows:

(1) Set $x_2 = 1$, $x_4 = 0$. Back-substitution yields $x_3 = 0$, and then $x_1 = -2$. Thus, $u_1 = (2, 1, 0, 0)$.

(2) Set $x_2 = 0$, $x_4 = 1$. Back-substitution yields $x_3 = 1$, and then $x_1 = 1$. Thus, $u_2 = (1, 0, 1, 1)$.

(b) Reduce the system to echelon form, obtaining

$$x - 2y - 3z = 0$$

$$5y + 9z = 0$$

$$2y + 7z = 0$$

and

$$x - 2y - 3z = 0$$

$$5y + 9z = 0$$

$$17z = 0$$

There are no free variables (the system is in triangular form). Hence, $\dim W = 0$, and W has no basis. Specifically, W consists only of the zero solution; that is, $W = \{0\}$.

3.27. Find the dimension and a basis for the general solution W of the following homogeneous system using matrix notation:

$$\begin{aligned}x_1 + 2x_2 + 3x_3 - 2x_4 + 4x_5 &= 0 \\2x_1 + 4x_2 + 8x_3 + x_4 + 9x_5 &= 0 \\3x_1 + 6x_2 + 13x_3 + 4x_4 + 14x_5 &= 0\end{aligned}$$

Show how the basis gives the parametric form of the general solution of the system.

When a system is homogeneous, we represent the system by its coefficient matrix A rather than by its augmented matrix M , because the last column of the augmented matrix M is a zero column, and it will remain a zero column during any row-reduction process.

Reduce the coefficient matrix A to echelon form, obtaining

$$A = \begin{bmatrix} 1 & 2 & 3 & -2 & 4 \\ 2 & 4 & 8 & 1 & 9 \\ 3 & 6 & 13 & 4 & 14 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 3 & -2 & 4 \\ 0 & 0 & 2 & 5 & 1 \\ 0 & 0 & 4 & 10 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 3 & -2 & 4 \\ 0 & 0 & 2 & 5 & 1 \end{bmatrix}$$

(The third row of the second matrix is deleted, because it is a multiple of the second row and will result in a zero row.) We can now proceed in one of two ways.

(a) Write down the corresponding homogeneous system in echelon form:

$$\begin{aligned}x_1 + 2x_2 + 3x_3 - 2x_4 + 4x_5 &= 0 \\2x_3 + 5x_4 + x_5 &= 0\end{aligned}$$

The system in echelon form has three free variables, x_2, x_4, x_5 , so $\dim W = 3$. A basis $[u_1, u_2, u_3]$ for W may be obtained as follows:

(1) Set $x_2 = 1, x_4 = 0, x_5 = 0$. Back-substitution yields $x_3 = 0$, and then $x_1 = -2$. Thus,

$$u_1 = (-2, 1, 0, 0, 0)$$

(2) Set $x_2 = 0, x_4 = 1, x_5 = 0$. Back-substitution yields $x_3 = -\frac{5}{2}$, and then $x_1 = \frac{19}{2}$. Thus,

$$u_2 = (\frac{19}{2}, 0, -\frac{5}{2}, 1, 0)$$

(3) Set $x_2 = 0, x_4 = 0, x_5 = 1$. Back-substitution yields $x_3 = -\frac{1}{2}$, and then $x_1 = -\frac{5}{2}$. Thus,

$$u_3 = (-\frac{5}{2}, 0, -\frac{1}{2}, 0, 1)$$

[One could avoid fractions in the basis by choosing $x_4 = 2$ in (2) and $x_5 = 2$ in (3), which yields multiples of u_2 and u_3 .] The parametric form of the general solution is obtained from the following linear combination of the basis vectors using parameters a, b, c :

$$au_1 + bu_2 + cu_3 = (-2a + \frac{19}{2}b - \frac{5}{2}c, a, -\frac{5}{2}b - \frac{1}{2}c, b, c)$$

(b) Reduce the echelon form of A to row canonical form:

$$A \sim \begin{bmatrix} 1 & 2 & 3 & -2 & 4 \\ 0 & 0 & 1 & \frac{5}{2} & \frac{1}{2} \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 3 & -\frac{19}{2} & \frac{5}{2} \\ 0 & 0 & 1 & \frac{5}{2} & \frac{1}{2} \end{bmatrix}$$

Write down the corresponding free-variable solution:

$$x_1 = -2x_2 + \frac{19}{2}x_4 - \frac{5}{2}x_5$$

$$x_3 = -\frac{5}{2}x_4 - \frac{1}{2}x_5$$

Using these equations for the pivot variables x_1 and x_3 , repeat the above process to obtain a basis $[u_1, u_2, u_3]$ for W . That is, set $x_2 = 1, x_4 = 0, x_5 = 0$ to get u_1 ; set $x_2 = 0, x_4 = 1, x_5 = 0$ to get u_2 ; and set $x_2 = 0, x_4 = 0, x_5 = 1$ to get u_3 .

3.28. Prove [Theorem 3.15](#). Let v_0 be a particular solution of $AX = B$, and let W be the general solution of $AX = 0$. Then $U = v_0 + W = \{v_0 + w : w \in W \text{ is the general solution of } AX = B\}$.

Let w be a solution of $AX = 0$. Then

$$A(v_0 + w) = Av_0 + Aw = B + 0 = B$$

Thus, the sum $v_0 + w$ is a solution of $AX = B$. On the other hand, suppose v is also a solution of $AX = B$. Then

$$A(v - v_0) = Av - Av_0 = B - B = 0$$

Therefore, $v - v_0$ belongs to W . Because $v = v_0 + (v - v_0)$, we find that any solution of $AX = B$ can be obtained by adding a solution of $AX = 0$ to a solution of $AX = B$. Thus, the theorem is proved.

Elementary Matrices, Applications

3.29. Let e_1, e_2, e_3 denote, respectively, the elementary row operations

“Interchange rows R_1 and R_2 ,” “Replace R_3 by $7R_3$,” “Replace R_2 by $-3R_1 + R_2$ ”

Find the corresponding three-square elementary matrices E_1, E_2, E_3 . Apply each operation to the 3 - 3 identity matrix I_3 to obtain

$$E_1 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 7 \end{bmatrix}, \quad E_3 = \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

3.30. Consider the elementary row operations in [Problem 3.29](#).

- Describe the inverse operations $e_1^{-1}, e_2^{-1}, e_3^{-1}$
- Find the corresponding three-square elementary matrices E'_1, E'_2, E'_3
- What is the relationship between the matrices E'_1, E'_2, E'_3 and the matrices E_1, E_2, E_3 ?

(a) The inverses of e_1, e_2, e_3 are, respectively,

“Interchange rows R_1 and R_2 ,” “Replace R_3 by $\frac{1}{7}R_3$,” “Replace R_2 by $3R_1 + R_2$.”

(b) Apply each inverse operation to the 3 - 3 identity matrix I_3 to obtain

$$E'_1 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad E'_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \frac{1}{7} \end{bmatrix}, \quad E'_3 = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(c) The matrices E'_1, E'_2, E'_3 are, respectively, the inverses of the matrices E_1, E_2, E_3 .

3.31. Write each of the following matrices as a product of elementary matrices:

$$(a) \ A = \begin{bmatrix} 1 & -3 \\ -2 & 4 \end{bmatrix}, \quad (b) \ B = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix}, \quad (c) \ C = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 3 & 8 \\ -3 & -1 & 2 \end{bmatrix}$$

The following three steps write a matrix M as a product of elementary matrices:

Step 1. Row reduce M to the identity matrix I , keeping track of the elementary row operations.

Step 2. Write down the inverse row operations.

Step 3. Write M as the product of the elementary matrices corresponding to the inverse operations. This gives the desired result.

If a zero row appears in Step 1, then M is not row equivalent to the identity matrix I , and M cannot be written as a product of elementary matrices.

(a) (1) We have

$$A = \begin{bmatrix} 1 & -3 \\ -2 & 4 \end{bmatrix} \sim \begin{bmatrix} 1 & -3 \\ 0 & -2 \end{bmatrix} \sim \begin{bmatrix} 1 & -3 \\ 0 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

where the row operations are, respectively,

“Replace R_2 by $-4R_3 + R_2$,” “Replace R_1 by $-3R_3 + R_1$,” “Replace R_1 by $-2R_2 + R_1$ ”

(2) Inverse operations:

“Replace R_2 by $-2R_1 + R_2$,” “Replace R_2 by $-2R_2$,” “Replace R_1 by $-3R_2 + R_1$ ”

$$(3) \ A = \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -2 \end{bmatrix} \begin{bmatrix} 1 & -3 \\ 0 & 1 \end{bmatrix}$$

(b) (1) We have

$$B = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$$

where the row operations are, respectively,

“Replace R_2 by $-2R_1 + R_2$,” “Replace R_2 by $-2R_2$,” “Replace R_1 by $-3R_2 + R_1$ ”

(2) Inverse operations:

“Replace R_2 by $4R_3 + R_2$,” “Replace R_1 by $3R_3 + R_1$,” “Replace R_1 by $2R_2 + R_1$ ”

$$(3) B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(c) (1) First row reduce C to echelon form. We have

$$C = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 3 & 8 \\ -3 & -1 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 4 \\ 0 & 2 & 8 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 4 \\ 0 & 0 & 0 \end{bmatrix}$$

In echelon form, C has a zero row. “STOP.” The matrix C cannot be row reduced to the identity matrix I , and C cannot be written as a product of elementary matrices. (We note, in particular, that C has no inverse.)

3.32. Find the inverse of

$$A = \begin{bmatrix} 1 & 2 & -4 \\ -1 & -1 & 5 \\ 2 & 7 & -3 \end{bmatrix}, \text{ (b) } B = \begin{bmatrix} 1 & 3 & -4 \\ 1 & 5 & -1 \\ 3 & 13 & -6 \end{bmatrix}.$$

(a) Form the matrix $M = [A, I]$ and row reduce M to echelon form:

$$M = \left[\begin{array}{ccc|ccc} 1 & 2 & -4 & 1 & 0 & 0 \\ -1 & -1 & 5 & 0 & 1 & 0 \\ 2 & 7 & -3 & 0 & 0 & 1 \end{array} \right] \sim \left[\begin{array}{ccc|ccc} 1 & 2 & -4 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 0 \\ 0 & 3 & 5 & -2 & 0 & 1 \end{array} \right]$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & 2 & -4 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 2 & -5 & -3 & 1 \end{array} \right]$$

In echelon form, the left half of M is in triangular form; hence, A has an inverse. Further reduce M to row canonical form:

$$M \sim \left[\begin{array}{ccc|ccc} 1 & 2 & 0 & -9 & -6 & 2 \\ 0 & 1 & 0 & \frac{7}{2} & \frac{5}{2} & -\frac{1}{2} \\ 0 & 0 & 1 & -\frac{5}{2} & -\frac{3}{2} & \frac{1}{2} \end{array} \right] \sim \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -16 & -11 & 3 \\ 0 & 1 & 0 & \frac{7}{2} & \frac{5}{2} & -\frac{1}{2} \\ 0 & 0 & 1 & -\frac{5}{2} & -\frac{3}{2} & \frac{1}{2} \end{array} \right]$$

The final matrix has the form $[I, A^{-1}]$; that is, A^{-1} is the right half of the last matrix. Thus,

$$A^{-1} = \begin{bmatrix} -16 & -11 & 3 \\ \frac{7}{2} & \frac{5}{2} & -\frac{1}{2} \\ -\frac{5}{2} & -\frac{3}{2} & \frac{1}{2} \end{bmatrix}$$

(b) Form the matrix $M = [B, I]$ and row reduce M to echelon form:

$$M = \left[\begin{array}{ccc|ccc} 1 & 3 & -4 & 1 & 0 & 0 \\ 1 & 5 & -1 & 0 & 1 & 0 \\ 3 & 13 & -6 & 0 & 0 & 1 \end{array} \right] \sim \left[\begin{array}{ccc|ccc} 1 & 3 & -4 & 1 & 0 & 0 \\ 0 & 2 & 3 & -1 & 1 & 0 \\ 0 & 4 & 6 & -3 & 0 & 1 \end{array} \right] \sim \left[\begin{array}{ccc|ccc} 1 & 3 & -4 & 1 & 0 & 0 \\ 0 & 2 & 3 & -1 & 1 & 0 \\ 0 & 0 & 0 & -1 & -2 & 1 \end{array} \right]$$

In echelon form, M has a zero row in its left half; that is, B is not row reducible to triangular form. Accordingly, B has no inverse.

3.33. Show that every elementary matrix E is invertible, and its inverse is an elementary matrix.

Let E be the elementary matrix corresponding to the elementary operation e ; that is, $e(I) = E$. Let e' be the inverse operation of e and let E' be the corresponding elementary matrix; that is, $e'(I) = E'$. Then

$$I = e'(e(I)) = e'(E) = E'E \quad \text{and} \quad I = e(e'(I)) = e(E') = EE'$$

Therefore, E' is the inverse of E .

3.34. Prove [Theorem 3.16](#): Let e be an elementary row operation and let E be the corresponding m -square elementary matrix; that is, $E = e(I)$. Then $e(A) = EA$, where A is any $m \times n$ matrix.

Let R_i be the row i of A ; we denote this by writing $A = [R_1, \dots, R_m]$. If B is a matrix for which AB is defined then $AB = [R_1B, \dots, R_mB]$. We also let

$$e_i = (0, \dots, 0, \hat{1}, 0, \dots, 0), \quad \hat{} = i$$

Here $\hat{} = i$ means 1 is the i th entry. One can show (Problem 2.45) that $e_i A = R_i$. We also note that $I = [e_1, e_2, \dots, e_m]$ is the identity matrix.

(i) Let e be the elementary row operation “Interchange rows R_i and R_j .” Then, for $\hat{} = i$ and $\hat{} = j$,

$$E = e(I) = [e_1, \dots, \hat{e}_j, \dots, \hat{e}_i, \dots, e_m]$$

and

$$e(A) = [R_1, \dots, \hat{R}_j, \dots, \hat{R}_i, \dots, R_m]$$

Thus,

$$EA = [e_1A, \dots, \widehat{e_jA}, \dots, \widehat{e_iA}, \dots, e_mA] = [R_1, \dots, \widehat{R_j}, \dots, \widehat{R_i}, \dots, R_m] = e(A)$$

(ii) Let e be the elementary row operation “Replace R_i by kR_i ($k \neq 0$).” Then, for $i = i$,

$$E = e(I) = [e_1, \dots, \widehat{ke_i}, \dots, e_m]$$

and

$$e(A) = [R_1, \dots, \widehat{kR_i}, \dots, R_m]$$

Thus,

$$EA = [e_1A, \dots, \widehat{ke_iA}, \dots, e_mA] = [R_1, \dots, \widehat{kR_i}, \dots, R_m] = e(A)$$

(iii) Let e be the elementary row operation “Replace R_j by $kR_j + R_i$.” Then, for $i = i$,

$$E = e(I) = [e_1, \dots, \widehat{ke_j + e_i}, \dots, e_m]$$

and

$$e(A) = [R_1, \dots, \widehat{kR_j + R_i}, \dots, R_m]$$

Using $(ke_j + e_i)A = k(e_jA) + e_iA = kR_j + R_i$, we have

$$\begin{aligned} EA &= [e_1A, \dots, (ke_j + e_i)A, \dots, e_mA] \\ &= [R_1, \dots, \widehat{kR_j + R_i}, \dots, R_m] = e(A) \end{aligned}$$

3.35. Prove [Theorem 3.17](#): Let A be a square matrix. Then the following are equivalent:

(a) A is invertible (nonsingular).

(b) A is row equivalent to the identity matrix I .

(c) A is a product of elementary matrices.

Suppose A is invertible and suppose A is row equivalent to matrix B in row canonical form. Then there exist elementary matrices $E_1, E_2, \dots, E_s$ such that $E_s \dots E_2 E_1 A = B$. Because A is invertible and each elementary matrix is invertible, B is also invertible. But if $B \neq I$, then B has a zero row; whence B is not invertible. Thus, $B = I$, and (a) implies (b).

If (b) holds, then there exist elementary matrices $E_1, E_2, \dots, E_s$ such that $E_s \dots E_2 E_1 A = I$. Hence, $A = (E_s \dots E_2 E_1)^{-1} = E_1^{-1} E_2^{-1} \dots E_s^{-1}$. But the E_i^{-1} are also elementary matrices. Thus (b) implies (c).

If (c) holds, then $A = E_1 E_2 \dots E_s$. The E_i are invertible matrices; hence, their product A is also invertible. Thus, (c) implies (a). Accordingly, the theorem is proved.

3.36. Prove [Theorem 3.18](#): If $AB = I$, then $BA = I$, and hence $B = A^{-1}$.

Suppose A is not invertible. Then A is not row equivalent to the identity matrix I , and so A is row equivalent to a matrix with a zero row. In other words, there exist elementary matrices $E_1, \dots, E_s$ such that $E_s \dots E_2 E_1 A$ has a zero row. Hence, $E_s \dots E_2 E_1 AB = E_s \dots E_2 E_1$, an invertible matrix, also has a zero row. But invertible matrices cannot have zero rows; hence A is invertible, with inverse A^{-1} . Then also,

$$B = IB = (A^{-1}A)B = A^{-1}(AB) = A^{-1}I = A^{-1}$$

3.37. Prove [Theorem 3.19](#): B is row equivalent to A (written $B \sim A$) if and only if there exists a nonsingular matrix P such that $B = PA$.

If $B \sim A$, then $B = e_s(\dots(e_2(e_1(A))) \dots) = E_s \dots E_2 E_1 A = PA$ where $P = E_s \dots E_2 E_1$ is nonsingular. Conversely, suppose $B = PA$, where P is nonsingular. By [Theorem 3.17](#), P is a product of elementary matrices, and so B can be obtained from A by a sequence of elementary row operations; that is, $B \sim A$. Thus, the theorem is proved.

3.38. Prove [Theorem 3.21](#): Every $m \times n$ matrix A is equivalent to a unique block matrix of the form $\begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$, where I_r is the $r \times r$ identity matrix.

The proof is constructive, in the form of an algorithm.

Step 1. Row reduce A to row canonical form, with leading nonzero entries $a_{1j_1}, a_{2j_2}, \dots, a_{rj_r}$.

Step 2. Interchange C_1 and C_{1j_1} , interchange C_2 and C_{2j_2} , $\dots$, and interchange C_r and C_{rj_r} . This gives a matrix in the form $\left[\begin{array}{c|c} I_r & B \\ \hline 0 & 0 \end{array} \right]$, with leading nonzero entries $a_{11}, a_{22}, \dots, a_{rr}$.

Step 3. Use column operations, with the a_{ij} as pivots, to replace each entry in B with a zero; that is, for $i = 1, 2, \dots, r$ and $j = r + 1, r + 2, \dots, n$, apply the operation $-b_{ij}C_i + C_j \rightarrow C_j$.

The final matrix has the desired form $\left[\begin{array}{c|c} I_r & 0 \\ \hline 0 & 0 \end{array} \right]$.

Lu Factorization

3.39. Find the LU factorization of

$$(a) \quad A = \begin{bmatrix} 1 & -3 & 5 \\ 2 & -4 & 7 \\ -1 & -2 & 1 \end{bmatrix}, \quad (b) \quad B = \begin{bmatrix} 1 & 4 & -3 \\ 2 & 8 & 1 \\ -5 & -9 & 7 \end{bmatrix}.$$

(a) Reduce A to triangular form by the following operations:

“Replace R_2 by $-2R_1 + R_2$,” “Replace R_3 by $R_1 + R_3$,” and then
“Replace R_3 by $\frac{5}{2}R_2 + R_3$ ”

These operations yield the following, where the triangular form is U :

$$A \sim \begin{bmatrix} 1 & -3 & 5 \\ 0 & 2 & -3 \\ 0 & -5 & 6 \end{bmatrix} \sim \begin{bmatrix} 1 & -3 & 5 \\ 0 & 2 & -3 \\ 0 & 0 & -\frac{3}{2} \end{bmatrix} = U \quad \text{and} \quad L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & -\frac{5}{2} & 1 \end{bmatrix}$$

The entries $2, -1, -\frac{5}{2}$ in L are the negatives of the multipliers $-2, 1, \frac{5}{2}$ in the above row operations. (As a check, multiply L and U to verify $A = LU$.)

(b) Reduce B to triangular form by first applying the operations “Replace R_2 by $-2R_1 + R_2$ ” and “Replace R_3 by $5R_1 + R_3$.” These operations yield

$$B \sim \begin{bmatrix} 1 & 4 & -3 \\ 0 & 0 & 7 \\ 0 & 11 & -8 \end{bmatrix}.$$

Observe that the second diagonal entry is 0. Thus, B cannot be brought into triangular form without row interchange operations. Accordingly, B is not LU -factorable. (There does exist a PLU factorization of such a matrix B , where P is a permutation matrix,

but such a factorization lies beyond the scope of this text.)

3.40. Find the LDU factorization of the matrix A in [Problem 3.39](#).

The $A = LDU$ factorization refers to the situation where L is a lower triangular matrix with 1's on the diagonal (as in the LU factorization of A), D is a diagonal matrix, and U is an upper triangular matrix with 1's on the diagonal. Thus, simply factor out the diagonal entries in the matrix U in the above LU factorization of A to obtain D and L . That is,

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & -\frac{5}{2} & 1 \end{bmatrix}, \quad D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & -\frac{3}{2} \end{bmatrix}, \quad U = \begin{bmatrix} 1 & -3 & 5 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{bmatrix}$$

3.41. Find the LU factorization of the matrix

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 3 & 3 \\ -3 & -10 & 2 \end{bmatrix}.$$

Reduce A to triangular form by the following operations:

(1) "Replace R_2 by $-2R_1 + R_2$," (2) "Replace R_3 by $3R_1 + R_3$," (3) "Replace R_3 by $-4R_2 + R_3$ "

These operations yield the following, where the triangular form is U :

$$A \sim \begin{bmatrix} 1 & 2 & 1 \\ 0 & -1 & 1 \\ 0 & -4 & 5 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 1 \\ 0 & -1 & 1 \\ 0 & 0 & 1 \end{bmatrix} = U \quad \text{and} \quad L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -3 & 4 & 1 \end{bmatrix}$$

The entries 2, -3, 4 in L are the negatives of the multipliers -2, 3, -4 in the above row operations. (As a check, multiply L and U to verify $A = LU$.)

3.42. Let A be the matrix in [Problem 3.41](#). Find X_1, X_2, X_3 , where X_i is the solution of $AX = B_i$ for (a) $B_1 = (1, 1, 1)$, (b) $B_2 = B_1 + X_1$, (c) $B_3 = B_2 + X_2$.

(a) Find $L^{-1}B_1$ by applying the row operations (1), (2), and then (3) in [Problem 3.41](#) to B_1 :

$$B_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \xrightarrow{(1) \text{ and } (2)} \begin{bmatrix} 1 \\ -1 \\ 4 \end{bmatrix} \xrightarrow{(3)} \begin{bmatrix} 1 \\ -1 \\ 8 \end{bmatrix}$$

Solve $UX = B$ for $B = (1, -1, 8)$ by back-substitution to obtain $X_1 = (-25, 9, 8)$.

(b) First find $B_2 = B_1 + X_1 = (1, 1, 1) + (-25, 9, 8) = (-24, 10, 9)$. Then as above

$$B_2 = [-24, 10, 9]^T \xrightarrow{(1) \text{ and } (2)} [-24, 58, -63]^T \xrightarrow{(3)} [-24, 58, -295]^T$$

Solve $UX = B$ for $B = (-24, 58, -295)$ by back-substitution to obtain $X_2 = (943, -353, -295)$.

(c) First find $B_3 = B_2 + X_2 = (-24, 10, 9) + (943, -353, -295) = (919, -343, -286)$. Then, as above

$$B_3 = [919, -343, -286]^T \xrightarrow{(1) \text{ and } (2)} [919, -2181, 2671]^T \xrightarrow{(3)} [919, -2181, 11\,395]^T$$

Solve $UX = B$ for $B = (919, -2181, 11\,395)$ by back-substitution to obtain

$$X_3 = (-37\,628, 13\,576, 11\,395).$$

Miscellaneous Problems

3.43. Let L be a linear combination of the m equations in n unknowns in the system (3.2). Say L is the equation

$$(c_1a_{11} + \cdots + c_ma_{m1})x_1 + \cdots + (c_1a_{1n} + \cdots + c_ma_{mn})x_n = c_1b_1 + \cdots + c_mb_m \quad (1)$$

Show that any solution of the system (3.2) is also a solution of L .

Let $u = (k_1, \dots, k_n)$ be a solution of (3.2). Then

$$a_{i1}k_1 + a_{i2}k_2 + \cdots + a_{in}k_n = b_i \quad (i = 1, 2, \dots, m) \quad (2)$$

Substituting u in the left-hand side of (1) and using (2), we get

$$\begin{aligned} & (c_1a_{11} + \cdots + c_ma_{m1})k_1 + \cdots + (c_1a_{1n} + \cdots + c_ma_{mn})k_n \\ &= c_1(a_{11}k_1 + \cdots + a_{1n}k_n) + \cdots + c_m(a_{m1}k_1 + \cdots + a_{mn}k_n) \\ &= c_1b_1 + \cdots + c_mb_m \end{aligned}$$

This is the right-hand side of (1); hence, u is a solution of (1).

3.44. Suppose a system $\mathcal{M}$ of linear equations is obtained from a system $\mathcal{L}$ by applying an elementary operation (page 64). Show that $\mathcal{M}$ and $\mathcal{L}$ have the same solutions.

Each equation L in $\mathcal{M}$ is a linear combination of equations in $\mathcal{L}$. Hence, by [Problem 3.43](#), any solution of $\mathcal{L}$ will also be a solution of $\mathcal{M}$. On the other hand, each elementary operation has an inverse elementary operation, so $\mathcal{L}$ can be obtained from $\mathcal{M}$ by an elementary operation. This means that any solution of $\mathcal{M}$ is a solution of $\mathcal{L}$. Thus, $\mathcal{L}$ and $\mathcal{M}$ have the same solutions.

3.45. Prove [Theorem 3.4](#): Suppose a system $\mathcal{M}$ of linear equations is obtained from a system $\mathcal{L}$ by a sequence of elementary operations. Then $\mathcal{M}$ and $\mathcal{L}$ have the same solutions.

Each step of the sequence does not change the solution set ([Problem 3.44](#)). Thus, the original system $\mathcal{L}$ and the final system $\mathcal{M}$ (and any system in between) have the same solutions.

3.46. A system $\mathcal{L}$ of linear equations is said to be *consistent* if no linear combination of its equations is a degenerate equation L with a nonzero constant. Show that $\mathcal{L}$ is consistent if and only if $\mathcal{L}$ is reducible to echelon form.

Suppose $\mathcal{L}$ is reducible to echelon form. Then $\mathcal{L}$ has a solution, which must also be a solution of every linear combination of its equations. Thus, L , which has no solution, cannot be a linear combination of the equations in $\mathcal{L}$. Thus, $\mathcal{L}$ is consistent.

On the other hand, suppose $\mathcal{L}$ is not reducible to echelon form. Then, in the reduction process, it must yield a degenerate equation L with a nonzero constant, which is a linear combination of the equations in $\mathcal{L}$. Therefore, $\mathcal{L}$ is not consistent; that is, $\mathcal{L}$ is inconsistent.

3.47. Suppose u and v are distinct vectors. Show that, for distinct scalars k , the vectors $u + k(u - v)$ are distinct.

Suppose $u + k_1(u - v) = u + k_2(u - v)$. We need only show that $k_1 = k_2$. We have

$$k_1(u - v) = k_2(u - v), \quad \text{and so} \quad (k_1 - k_2)(u - v) = 0$$

Because u and v are distinct, $u - v \neq 0$. Hence, $k_1 - k_2 = 0$, and so $k_1 = k_2$.

3.48. Suppose AB is defined. Prove

(a) Suppose A has a zero row. Then AB has a zero row.

(b) Suppose B has a zero column. Then AB has a zero column.

(a) Let R_i be the zero row of A , and $C_1, \dots, C_n$ the columns of B . Then the i th row of AB is

$$(R_i C_1, R_i C_2, \dots, R_i C_n) = (0, 0, 0, \dots, 0)$$

(b) B^T has a zero row, and so $B^T A^T = (AB)^T$ has a zero row. Hence, AB has a zero column.

SUPPLEMENTARY PROBLEMS

Linear Equations, 2×2 Systems

3.49. Determine whether each of the following systems is linear:

(a) $3x - 4y + 2yz = 8$, (b) $ex + 3y = \pi$, (c) $2x - 3y + kz = 4$

3.50. Solve (a) $\pi x = 2$, (b) $3x + 2 = 5x + 7 - 2x$, (c) $6x + 2 - 4x = 5 + 2x - 3$

3.51. Solve each of the following systems:

(a) $2x + 3y = 1$ $5x + 7y = 3$	(b) $4x - 2y = 5$ $-6x + 3y = 1$	(c) $2x - 4 = 3y$ $5y - x = 5$	(d) $2x - 4y = 10$ $3x - 6y = 15$
------------------------------------	-------------------------------------	-----------------------------------	--------------------------------------

3.52. Consider each of the following systems in unknowns x and y :

(a) $x - ay = 1$ $ax - 4y = b$	(b) $ax + 3y = 2$ $12x + ay = b$	(c) $x + ay = 3$ $2x + 5y = b$
-----------------------------------	-------------------------------------	-----------------------------------

For which values of a does each system have a unique solution, and for which pairs of values (a, b) does each system have more than one solution?

General Systems of Linear Equations

3.53. Solve

$$\begin{aligned} \text{(a)} \quad x + y + 2z &= 4 \\ 2x + 3y + 6z &= 10 \\ 3x + 6y + 10z &= 17 \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad x - 2y + 3z &= 2 \\ 2x - 3y + 8z &= 7 \\ 3x - 4y + 13z &= 8 \end{aligned}$$

$$\begin{aligned} \text{(c)} \quad x + 2y + 3z &= 3 \\ 2x + 3y + 8z &= 4 \\ 5x + 8y + 19z &= 11 \end{aligned}$$

3.54. Solve

$$\begin{aligned} \text{(a)} \quad x - 2y &= 5 \\ 2x + 3y &= 3 \\ 3x + 2y &= 7 \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad x + 2y - 3z + 2t &= 2 \\ 2x + 5y - 8z + 6t &= 5 \\ 3x + 4y - 5z + 2t &= 4 \end{aligned}$$

$$\begin{aligned} \text{(c)} \quad x + 2y + 4z - 5t &= 3 \\ 3x - y + 5z + 2t &= 4 \\ 5x - 4y + 6z + 9t &= 2 \end{aligned}$$

3.55. Solve

$$\begin{aligned} \text{(a)} \quad 2x - y - 4z &= 2 \\ 4x - 2y - 6z &= 5 \\ 6x - 3y - 8z &= 8 \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad x + 2y - z + 3t &= 3 \\ 2x + 4y + 4z + 3t &= 9 \\ 3x + 6y - z + 8t &= 10 \end{aligned}$$

3.56. Consider each of the following systems in unknowns x, y, z :

$$\begin{aligned} \text{(a)} \quad x - 2y &= 1 \\ x - y + az &= 2 \\ ay + 9z &= b \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad x + 2y + 2z &= 1 \\ x + ay + 3z &= 3 \\ x + 11y + az &= b \end{aligned}$$

$$\begin{aligned} \text{(c)} \quad x + y + az &= 1 \\ x + ay + z &= 4 \\ ax + y + z &= b \end{aligned}$$

For which values of a does the system have a unique solution, and for which pairs of values (a, b) does the system have more than one solution? The value of b does not have any effect on whether the system has a unique solution. Why?

Linear Combinations, Homogeneous Systems

3.57. Write v as a linear combination of u_1, u_2, u_3 , where

- (a) $v = (4, -9, 2)$, $u_1 = (1, 2, -1)$, $u_2 = (1, 4, 2)$, $u_3 = (1, -3, 2)$;
 (b) $v = (1, 3, 2)$, $u_1 = (1, 2, 1)$, $u_2 = (2, 6, 5)$, $u_3 = (1, 7, 8)$;
 (c) $v = (1, 4, 6)$, $u_1 = (1, 1, 2)$, $u_2 = (2, 3, 5)$, $u_3 = (3, 5, 8)$.

3.58. Let $u_1 = (1, 1, 2)$, $u_2 = (1, 3, -2)$, $u_3 = (4, -2, -1)$ in $\mathbf{R}^3$. Show that u_1, u_2, u_3 are orthogonal, and write v as a linear combination of u_1, u_2, u_3 , where (a) $v = (5, -5, 9)$, (b) $v = (1, -3, 3)$, (c) $v = (1, 1, 1)$. (*Hint: Use Fourier coefficients.*)

3.59. Find the dimension and a basis of the general solution W of each of the following homogeneous systems:

- | | | |
|----------------------|-----------------------|---------------------------|
| (a) $x - y + 2z = 0$ | (b) $x + 2y - 3z = 0$ | (c) $x + 2y + 3z + t = 0$ |
| $2x + y + z = 0$ | $2x + 5y + 2z = 0$ | $2x + 4y + 7z + 4t = 0$ |
| $5x + y + 4z = 0$ | $3x - y - 4z = 0$ | $3x + 6y + 10z + 5t = 0$ |

3.360. Find the dimension and a basis of the general solution W of each of the following systems:

Echelon Matrices, Row Canonical Form

3.61. Reduce each of the following matrices to echelon form and then to row canonical form: (a)

- | | |
|---|---|
| (a) $x_1 + 3x_2 + 2x_3 - x_4 - x_5 = 0$ | (b) $2x_1 - 4x_2 + 3x_3 - x_4 + 2x_5 = 0$ |
| $2x_1 + 6x_2 + 5x_3 + x_4 - x_5 = 0$ | $3x_1 - 6x_2 + 5x_3 - 2x_4 + 4x_5 = 0$ |
| $5x_1 + 15x_2 + 12x_3 + x_4 - 3x_5 = 0$ | $5x_1 - 10x_2 + 7x_3 - 3x_4 + 18x_5 = 0$ |

3.62. Reduce each of the following matrices to echelon form and then to row canonical form:

- | | | |
|--|--|---|
| (a) $\begin{bmatrix} 1 & 1 & 2 \\ 2 & 4 & 9 \\ 1 & 5 & 12 \end{bmatrix}$, | (b) $\begin{bmatrix} 1 & 2 & -1 & 2 & 1 \\ 2 & 4 & 1 & -2 & 5 \\ 3 & 6 & 3 & -7 & 7 \end{bmatrix}$, | (c) $\begin{bmatrix} 2 & 4 & 2 & -2 & 5 & 1 \\ 3 & 6 & 2 & 2 & 0 & 4 \\ 4 & 8 & 2 & 6 & -5 & 7 \end{bmatrix}$ |
|--|--|---|

3.63. Using only 0's and 1's, list all possible 2×2 matrices in row canonical form.

3.64. Using only 0's and 1's, find the number n of possible 3×3 matrices in row canonical form.

Elementary Matrices, Applications

3.65. Let e_1, e_2, e_3 denote, respectively, the following elementary row operations:

“Interchange R_2 and R_3 ,” “Replace R_2 by $3R_2$,” “Replace R_1 by $2R_3 + R_1$ ”

(a) Find the corresponding elementary matrices E_1, E_2, E_3 .

(b) Find the inverse operations $e_1^{-1}, e_2^{-1}, e_3^{-1}$; their corresponding elementary matrices E'_1, E'_2, E'_3 ; and the relationship between them and E_1, E_2, E_3 .

(c) Describe the corresponding elementary column operations f_1, f_2, f_3 .

(d) Find elementary matrices F_1, F_2, F_3 corresponding to f_1, f_2, f_3 , and the relationship between them and E_1, E_2, E_3 .

3.66. Express each of the following matrices as a product of elementary matrices:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 3 & -6 \\ -2 & 4 \end{bmatrix}, \quad C = \begin{bmatrix} 2 & 6 \\ -3 & -7 \end{bmatrix}, \quad D = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 3 \\ 3 & 8 & 7 \end{bmatrix}$$

3.67. Find the inverse of each of the following matrices (if it exists):

$$A = \begin{bmatrix} 1 & -2 & -1 \\ 2 & -3 & 1 \\ 3 & -4 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 6 & 1 \\ 3 & 10 & -1 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 3 & -2 \\ 2 & 8 & -3 \\ 1 & 7 & 1 \end{bmatrix}, \quad D = \begin{bmatrix} 2 & 1 & -1 \\ 5 & 2 & -3 \\ 0 & 2 & 1 \end{bmatrix}$$

3.68. Find the inverse of each of the following $n \times n$ matrices:

(a) A has 1's on the diagonal and *superdiagonal* (entries directly above the diagonal) and 0's elsewhere.

(b) B has 1's on and above the diagonal, and 0's below the diagonal.

Lu Factorization

3.69. Find the LU factorization of each of the following matrices:

(a) $\begin{bmatrix} 1 & -1 & -1 \\ 3 & -4 & -2 \\ 2 & -3 & -2 \end{bmatrix}$, (b) $\begin{bmatrix} 1 & 3 & -1 \\ 2 & 5 & 1 \\ 3 & 4 & 2 \end{bmatrix}$, (c) $\begin{bmatrix} 2 & 3 & 6 \\ 4 & 7 & 9 \\ 3 & 5 & 4 \end{bmatrix}$, (d) $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 7 \\ 3 & 7 & 10 \end{bmatrix}$

3.70. Let A be the matrix in [Problem 3.69\(a\)](#). Find X_1, X_2, X_3, X_4 , where

(a) X_1 is the solution of $AX = B_1$, where $B_1 = (1, 1, 1)^T$.

(b) For $k > 1$, X_k is the solution of $AX = B_k$, where $B_k = B_{k-1} + X_{k-1}$.

3.71. Let B be the matrix in [Problem 3.69\(b\)](#). Find the LDU factorization of B .

Miscellaneous Problems

3.72. Consider the following systems in unknowns x and y :

$$(a) \quad \begin{cases} ax + by = 1 \\ cx + dy = 0 \end{cases}$$

$$(b) \quad \begin{cases} ax + by = 0 \\ cx + dy = 1 \end{cases}$$

Suppose $D = ad - bc \neq 0$. Show that each system has the unique solution:

$$(a) \quad x = d/D, y = -c/D, \quad (b) \quad x = -b/D, y = a/D.$$

3.73. Find the inverse of the row operation “Replace R_i by $kR_i + lR_j$ ($k \neq 0$).”

3.74. Prove that deleting the last column of an echelon form (respectively, the row canonical form) of an augmented matrix $M = [A, B]$ yields an echelon form (respectively, the row canonical form) of A .

3.75. Let e be an elementary row operation and E its elementary matrix, and let f be the corresponding elementary column operation and F its elementary matrix. Prove

$$(a) \quad f(A) = (e(A^T))^T, \quad (b) \quad F = E^T, \quad (c) \quad f(A) = AF.$$

3.76. Matrix A is *equivalent* to matrix B , written $A \approx B$, if there exist nonsingular matrices P and Q such that $B = PAQ$. Prove that $\approx$ is an *equivalence* relation; that is,

$$(a) \quad A \approx A, \quad (b) \quad \text{If } A \approx B, \text{ then } B \approx A, \quad (c) \quad \text{If } A \approx B \text{ and } B \approx C, \text{ then } A \approx C.$$

ANSWERS TO SUPPLEMENTARY PROBLEMS

Notation: $A = [R_1; R_2; \dots]$ denotes the matrix A with rows $R_1, R_2, \dots$. The elements in each row are separated by commas (which may be omitted with single digits), the rows are

separated by semicolons, and 0 denotes a zero row. For example,

$$A = [1, 2, 3, 4; \quad 5, -6, 7, -8; \quad 0] = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & -6 & 7 & -8 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

3.49.

- (a) no, (b) yes, (c) linear in x, y, z , not linear in x, y, z, k

3.50.

- (a) $x = 2/\pi$, (b) no solution, (c) every scalar k is a solution

3.51.

- (a) $(2, -1)$, (b) no solution, (c) $(5, 2)$, (d) $(5 + 2a, a)$

3.52.

- (a) $a \neq \pm 2$, $(2, 2)$, $(-2, -2)$, (b) $a \neq \pm 6$, $(6, 4)$, $(-6, -4)$, (c) $a \neq \frac{5}{2}$, $(\frac{5}{2}, 6)$

3.53.

- (a) $(2, 1, \frac{1}{2})$, (b) no solution, (c) $u = (-7a - 1, 2a + 2, a)$.

3.54.

- (a) $(3, -1)$, (b) $u = (-a + 2b, 1 + 2a - 2b, a, b)$, (c) no solution

3.55.

- (a) $u = (\frac{1}{2}a + 2, a, \frac{1}{2})$, (b) $u = (\frac{1}{2}(7 - 5b - 4a), a, \frac{1}{2}(1 + b), b)$

3.56.

- (a) $a \neq \pm 3$, $(3, 3)$, $(-3, -3)$, (b) $a \neq 5$ and $a \neq -1$, $(5, 7)$, $(-1, -5)$,
(c) $a \neq 1$ and $a \neq -2$, $(-2, 5)$

3.57. (a) $2, -1, 3$, (b) $6, -3, 1$, (c) not possible

3.58. (a) $3, -2, 1$, (b) $\frac{2}{3}, -1, \frac{1}{3}$, (c) $\frac{2}{3}, \frac{1}{7}, \frac{1}{21}$

3.59.

- (a) $\dim W = 1$, $u_1 = (-1, 1, 1)$, (b) $\dim W = 0$, no basis,
(c) $\dim W = 2$, $u_1 = (-2, 1, 0, 0)$, $u_2 = (5, 0, -2, 1)$

3.60.

- (a) $\dim W = 3$, $u_1 = (-3, 1, 0, 0, 0)$, $u_2 = (7, 0, -3, 1, 0)$, $u_3 = (3, 0, -1, 0, 1)$,
 (b) $\dim W = 2$, $u_1 = (2, 1, 0, 0, 0)$, $u_2 = (5, 0, -5, -3, 1)$

3.61.

- (a) $[1, 0, -\frac{1}{2}; 0, 1, \frac{5}{2}; 0]$, (b) $[1, 2, 0, 0, 2; 0, 0, 1, 0, 5; 0, 0, 0, 1, 2]$,
 (c) $[1, 2, 0, 4, -5, 3; 0, 0, 1, -5, \frac{15}{2}, -\frac{5}{2}; 0]$

3.62.

- (a) $[1, 2, 0, 0, -4, -2; 0, 0, 1, 0, 1, 2; 0, 0, 0, 1, 2, 1; 0]$,
 (b) $[0, 1, 0, 0; 0, 0, 1, 0; 0, 0, 0, 1; 0]$, (c) $[1, 0, 0, 4; 0, 1, 0, -1; 0, 0, 1, 2; 0]$

3.63. 5: $[1, 0; 0, 1], [1, 1; 0, 0], [1, 0; 0, 0], [0, 1; 0, 0], 0$ **3.64.** 16**3.65.**

- (a) $[1, 0, 0; 0, 0, 1; 0, 1, 0], [1, 0, 0; 0, 3, 0; 0, 0, 1], [1, 0, 2; 0, 1, 0; 0, 0, 1]$,
 (b) $R_2 \leftrightarrow R_3; \frac{1}{3}R_2 \rightarrow R_2; -2R_3 + R_1 \rightarrow R_1; \text{each } E'_i = E_i^{-1}$,
 (c) $C_2 \leftrightarrow C_3, 3C_2 \rightarrow C_2, 2C_3 + C_1 \rightarrow C_1$, (d) $\text{each } F_i = E_i^T$.

3.66.

$$A = [1, 0; 3, 1][1, 0; 0, -2][1, 2; 0, 1], \quad B \text{ is not invertible,}$$

$$C = [1, 0; -\frac{3}{2}, 1][1, 0; 0, 2][1, 6; 0, 1][2, 0; 0, 1],$$

$$D = [100; 010; 301][100; 010; 021][100; 013; 001][120; 010; 001]$$

3.67.

$$A^{-1} = [-8, 12, -5; -5, 7, -3; 1, -2, 1], \quad B \text{ has no inverse,}$$

$$C^{-1} = [\frac{29}{2}, -\frac{17}{2}, \frac{7}{2}; -\frac{5}{2}, \frac{3}{2}, -\frac{1}{2}; 3, -2, 1], \quad D^{-1} = [8, -3, -1; -5, 2, 1; 10, -4, -1]$$

3.68.

$$A^{-1} = [1, -1, 1, -1, \dots; 0, 1, -1, 1, -1, \dots; 0, 0, 1, -1, 1, -1, \dots; \dots; \dots; 0, \dots, 0, 1]$$

B^{-1} has 1's on diagonal, -1's on superdiagonal, and 0's elsewhere.

- (a) $[100; 310; 211][1, -1, -1; 0, -1, 1; 0, 0, -1]$,
 (b) $[100; 210; 351][1, 3, -1; 0, -1, 3; 0, 0, -10]$,
 (c) $[100; 210; \frac{3}{2}, \frac{1}{2}, 1][2, 3, 6; 0, 1, -3; 0, 0, -\frac{7}{2}]$,

3.69. (d) There is no LU decomposition.**3.70.**

$$X_1 = [1, 1, -1]^T, \quad B_2 = [2, 2, 0]^T, \quad X_2 = [6, 4, 0]^T, \quad B_3 = [8, 6, 0]^T, \quad X_3 = [22, 16, -2]^T,$$

$$B_4 = [30, 22, -2]^T, \quad X_4 = [86, 62, -6]^T$$

3.71.

$$B = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 5 & 1 \end{bmatrix} \text{diag}(1, -1, -10) \begin{bmatrix} 1 & 3 & -1 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix}$$

3.73. Replace R_i by $-kR_j + (1/k)R_i$.

3.75. (c) $f(A) = (e(A^T))^T = (EA^T)^T = (A^T)^T E^T = AF$

3.76. (a) $A = IAI$. (b) If $A = PBQ$, then $B = P^{-1}AQ^{-1}$.
(c) If $A = PBQ$ and $B = P'CQ'$, then $A = (PP')C(Q'Q)$.

OceanofPDF.com

CHAPTER 4

Vector Spaces

4.1 Introduction

This chapter introduces the underlying structure of linear algebra, that of a finite-dimensional vector space. The definition of a vector space V , whose elements are called *vectors*, involves an arbitrary field K , whose elements are called *scalars*. The following notation will be used (unless otherwise stated or implied):

V	the given vector space
u, v, w	vectors in V
K	the given number field
a, b, c , or k	scalars in K

Almost nothing essential is lost if the reader assumes that K is the real field $\mathbf{R}$ or the complex field $\mathbf{C}$.

The reader might suspect that the real line $\mathbf{R}$ has “dimension” one, the cartesian plane $\mathbf{R}^2$ has “dimension” two, and the space $\mathbf{R}^3$ has “dimension” three. This chapter

formalizes the notion of “dimension,” and this definition will agree with the reader’s intuition.

Throughout this text, we will use the following set notation:

$a \in A$	Element a belongs to set A
$a, b \in A$	Elements a and b belong to A
$\forall x \in A$	For every x in A
$\exists x \in A$	There exists an x in A
$A \subseteq B$	A is a subset of B
$A \cap B$	Intersection of A and B
$A \cup B$	Union of A and B
$\emptyset$	Empty set

4.2 Vector Spaces

The following defines the notion of a vector space V where K is the field of scalars.

DEFINITION 4.1: Let V be a nonempty set with two operations:

- (i) **Vector Addition:** This assigns to any $u, v \in V$ a sum $u + v$ in V .
- (ii) **Scalar Multiplication:** This assigns to any $u \in V, k \in K$ a product $ku \in V$.

Then V is called a *vector space* (over the field K) if the following axioms hold for any vectors $u, v, w \in V$:

$$[A_1] \quad (u + v) + w = u + (v + w)$$

[A₂] There is a vector in V , denoted by 0 and called the *zero vector*, such that, for any $u \in V$,

$$u + 0 = 0 + u = u$$

[A₃] For each $u \in V$, there is a vector in V , denoted by $-u$, and called the *negative* of u , such that

$$u + (-u) = (-u) + u = 0.$$

$$[A_4] \quad u + v = v + u.$$

$$[M_1] \quad k(u + v) = ku + kv, \text{ for any scalar } k \in K.$$

$$[M_2] \quad (a + b)u = au + bu, \text{ for any scalars } a, b \in K.$$

$$[M_3] \quad (ab)u = a(bu), \text{ for any scalars } a, b \in K.$$

$$[M_4] \quad 1u = u, \text{ for the unit scalar } 1 \in K.$$

The above axioms naturally split into two sets (as indicated by the labeling of the axioms). The first four are concerned only with the additive structure of V and can be summarized by saying V is a *commutative group* under addition. This means

- (a) Any sum $v_1 + v_2 + \dots + v_m$ of vectors requires no parentheses and does not depend on the order of the summands.
- (b) The zero vector 0 is unique, and the negative $-u$ of a vector u is unique.
- (c) (Cancellation Law) If $u + w = v + w$, then $u = v$.

Also, *subtraction* in V is defined by $u - v = u + (-v)$, where $-v$ is the unique negative of v .

On the other hand, the remaining four axioms are concerned with the “action” of the field K of scalars on the vector space V . Using these additional axioms, we prove ([Problem 4.2](#)) the following simple properties of a vector space.

THEOREM 4.1: Let V be a vector space over a field K .

- (i) For any scalar $k \in K$ and $0 \in V$, $k0 = 0$.
- (ii) For $0 \in K$ and any vector $u \in V$; $0u = 0$.

- (iii) If $ku = 0$, where $k \in K$ and $u \in V$, then $k = 0$ or $u = 0$.
- (iv) For any $k \in K$ and any $u \in V$, $(-k)u = k(-u) = -ku$.

4.3 Examples of Vector Spaces

This section lists important examples of vector spaces that will be used throughout the text.

Space K^n

Let K be an arbitrary field. The notation K^n is frequently used to denote the set of all n -tuples of elements in K . Here K^n is a vector space over K using the following operations:

- (i) **Vector Addition:** $(a_1, a_2, \dots, a_n) + (b_1, b_2, \dots, b_n) = (a_1 + b_1, a_2 + b_2, \dots, a_n + b_n)$
- (ii) **Scalar Multiplication:** $k(a_1, a_2, \dots, a_n) = (ka_1, ka_2, \dots, ka_n)$

The zero vector in K^n is the n -tuple of zeros,

$$0 = (0, 0, \dots, 0)$$

and the negative of a vector is defined by

$$-(a_1, a_2, \dots, a_n) = (-a_1, -a_2, \dots, -a_n)$$

Observe that these are the same as the operations defined for $\mathbf{R}^n$ in [Chapter 1](#). The proof that K^n is a vector space is identical to the proof of [Theorem 1.1](#), which we now regard as stating that $\mathbf{R}^n$ with the operations defined there is a vector space over $\mathbf{R}$.

Polynomial Space $P(t)$

Let $\mathbf{P}(t)$ denote the set of all polynomials of the form

$$p(t) = a_0 + a_1t + a_2t^2 + \cdots + a_st^s \quad (s = 1, 2, \dots)$$

where the coefficients a_i belong to a field K . Then $\mathbf{P}(t)$ is a vector space over K using the following operations:

- (i) **Vector Addition:** Here $p(t) + q(t)$ in $\mathbf{P}(t)$ is the usual operation of addition of polynomials.
- (ii) **Scalar Multiplication:** Here $kp(t)$ in $\mathbf{P}(t)$ is the usual operation of the product of a scalar k and a polynomial $p(t)$.

The zero polynomial 0 is the zero vector in $\mathbf{P}(t)$.

Polynomial Space $\mathbf{P}_n(t)$

Let $\mathbf{P}_n(t)$ denote the set of all polynomials $p(t)$ over a field K , where the degree of $p(t)$ is less than or equal to n ; that is,

$$p(t) = a_0 + a_1t + a_2t^2 + \cdots + a_st^s$$

where $s \leq n$. Then $\mathbf{P}_n(t)$ is a vector space over K with respect to the usual operations of addition of polynomials and of multiplication of a polynomial by a constant (just like the vector space $\mathbf{P}(t)$ above). We include the zero polynomial 0 as an element of $\mathbf{P}_n(t)$, even though its degree is undefined.

Matrix Space $\mathbf{M}_{m,n}$

The notation $\mathbf{M}_{m,n}$, or simply $\mathbf{M}$, will be used to denote the set of all $m \times n$ matrices with entries in a field K . Then $\mathbf{M}_{m,n}$ is a vector space over K with respect to the usual operations of matrix addition and scalar multiplication of matrices, as indicated by [Theorem 2.1](#).

Function Space $F(X)$

Let X be a nonempty set and let K be an arbitrary field. Let $F(X)$ denote the set of all functions of X into K . [Note that $F(X)$ is nonempty, because X is nonempty.] Then $F(X)$ is a vector space over K with respect to the following operations:

- (i) **Vector Addition:** The sum of two functions f and g in $F(X)$ is the function $f + g$ in $F(X)$ defined by

$$(f + g)(x) = f(x) + g(x) \quad \forall x \in X$$

- (ii) **Scalar Multiplication:** The product of a scalar $k \in K$ and a function f in $F(X)$ is the function kf in $F(X)$ defined by

$$(kf)(x) = kf(x) \quad \forall x \in X$$

The zero vector in $F(X)$ is the zero function $\mathbf{0}$, which maps every $x \in X$ into the zero element $0 \in K$;

$$\mathbf{0}(x) = 0 \quad \forall x \in X$$

Also, for any function f in $F(X)$, negative of f is the function $-f$ in $F(X)$ defined by

$$(-f)(x) = -f(x) \quad \forall x \in X$$

Fields and Subfields

Suppose a field E is an extension of a field K ; that is, suppose E is a field that contains K as a subfield. Then E may be viewed as a vector space over K using the following operations:

- (i) **Vector Addition:** Here $u + v$ in E is the usual addition in E .

- (ii) **Scalar Multiplication:** Here ku in E , where $k \in K$ and $u \in E$, is the usual product of k and u as elements of E .

That is, the eight axioms of a vector space are satisfied by E and its subfield K with respect to the above two operations.

4.4 Linear Combinations, Spanning Sets

Let V be a vector space over a field K . A vector v in V is a *linear combination* of vectors $u_1, u_2, \dots, u_m$ in V if there exist scalars $a_1, a_2, \dots, a_m$ in K such that

$$v = a_1u_1 + a_2u_2 + \cdots + a_mu_m$$

Alternatively, v is a linear combination of $u_1, u_2, \dots, u_m$ if there is a solution to the vector equation

$$v = x_1u_1 + x_2u_2 + \cdots + x_mu_m$$

where $x_1, x_2, \dots, x_m$ are unknown scalars.

EXAMPLE 4.1 (Linear Combinations in $\mathbf{R}^n$) Suppose we want to express $v = (3, 7, -4)$ in $\mathbf{R}^3$ as a linear combination of the vectors

$$u_1 = (1, 2, 3), \quad u_2 = (2, 3, 7), \quad u_3 = (3, 5, 6)$$

We seek scalars x, y, z such that $v = xu_1 + yu_2 + zu_3$; that is,

$$\begin{bmatrix} 3 \\ 7 \\ -4 \end{bmatrix} = x \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + y \begin{bmatrix} 2 \\ 3 \\ 7 \end{bmatrix} + z \begin{bmatrix} 3 \\ 5 \\ 6 \end{bmatrix} \quad \text{or} \quad \begin{array}{rcl} x + 2y + 3z & = & 3 \\ 2x + 3y + 5z & = & 7 \\ 3x + 7y + 6z & = & -4 \end{array}$$

(For notational convenience, we have written the vectors in $\mathbf{R}^3$ as columns, because it is then easier to find the equivalent system of linear equations.) Reducing the system to echelon form yields

$$\begin{array}{rcl} x + 2y + 3z = & 3 \\ -y - z = & 1 \\ y - 3z = & -13 \end{array} \quad \text{and then} \quad \begin{array}{rcl} x + 2y + 3z = & 3 \\ -y - z = & 1 \\ -4z = & -12 \end{array}$$

Back-substitution yields the solution $x = 2$, $y = -4$, $z = 3$. Thus, $v = 2u_1 - 4u_2 + 3u_3$.

Remark: Generally speaking, the question of expressing a given vector v in K^n as a linear combination of vectors $u_1, u_2, \dots, u_m$ in K^n is equivalent to solving a system $AX = B$ of linear equations, where v is the column B of constants, and the u 's are the columns of the coefficient matrix A . Such a system may have a unique solution (as above), many solutions, or no solution. The last case—no solution—means that v cannot be written as a linear combination of the u 's.

EXAMPLE 4.2 (Linear combinations in $\mathbf{P}(t)$) Suppose we want to express the polynomial $v = 3t^2 + 5t - 5$ as a linear combination of the polynomials

$$p_1 = t^2 + 2t + 1, \quad p_2 = 2t^2 + 5t + 4, \quad p_3 = t^2 + 3t + 6$$

We seek scalars x, y, z such that $v = xp_1 + yp_2 + zp_3$; that is,

$$3t^2 + 5t - 5 = x(t^2 + 2t + 1) + y(2t^2 + 5t + 4) + z(t^2 + 3t + 6) \quad (*)$$

There are two ways to proceed from here.

(1) Expand the right-hand side of (*) obtaining:

$$\begin{aligned}
 3t^2 + 5t - 5 &= xt^2 + 2xt + x + 2yt^2 + 5yt + 4y + zt^2 + 3zt + 6z \\
 &= (x + 2y + z)t^2 + (2x + 5y + 3z)t + (x + 4y + 6z)
 \end{aligned}$$

Set coefficients of the same powers of t equal to each other, and reduce the system to echelon form:

$$\begin{array}{rcl}
 x + 2y + z = 3 & & x + 2y + z = 3 & & x + 2y + z = 3 \\
 2x + 5y + 3z = 5 & \text{or} & y + z = -1 & \text{or} & y + z = -1 \\
 x + 4y + 6z = -5 & & 2y + 5z = -8 & & 3z = -6
 \end{array}$$

The system is in triangular form and has a solution. Back-substitution yields the solution $x = 3, y = 1, z = -2$. Thus,

$$v = 3p_1 + p_2 - 2p_3$$

- (2) The equation (*) is actually an identity in the variable t ; that is, the equation holds for any value of t . We can obtain three equations in the unknowns x, y, z by setting t equal to any three values. For example,

$$\text{Set } t = 0 \text{ in (1) to obtain: } x + 4y + 6z = -5$$

$$\text{Set } t = 1 \text{ in (1) to obtain: } 4x + 11y + 10z = 3$$

$$\text{Set } t = -1 \text{ in (1) to obtain: } y + 4z = -7$$

Reducing this system to echelon form and solving by back-substitution again yields the solution $x = 3, y = 1, z = -2$. Thus (again), $v = 3p_1 + p_2 - 2p_3$.

Spanning Sets

Let V be a vector space over K . Vectors $u_1, u_2, \dots, u_m$ in V are said to *span* V or to form a *spanning set* of V if every v in

V is a linear combination of the vectors $u_1, u_2, \dots, u_m$ —that is, if there exist scalars $a_1, a_2, \dots, a_m$ in K such that

$$v = a_1u_1 + a_2u_2 + \cdots + a_mu_m$$

The following remarks follow directly from the definition.

Remark 1: Suppose $u_1, u_2, \dots, u_m$ span V . Then, for any vector w , the set $w, u_1, u_2, \dots, u_m$ also spans V .

Remark 2: Suppose $u_1, u_2, \dots, u_m$ span V and suppose u_k is a linear combination of some of the other u 's. Then the u 's without u_k also span V .

Remark 3: Suppose $u_1, u_2, \dots, u_m$ span V and suppose one of the u 's is the zero vector. Then the u 's without the zero vector also span V .

EXAMPLE 4.3 Consider the vector space $V = \mathbf{R}^3$.

(a) We claim that the following vectors form a spanning set of $\mathbf{R}^3$:

$$e_1 = (1, 0, 0), \quad e_2 = (0, 1, 0), \quad e_3 = (0, 0, 1)$$

Specifically, if $v = (a, b, c)$ is any vector in $\mathbf{R}^3$, then

$$v = ae_1 + be_2 + ce_3$$

For example, $v = (5, -6, 2) = 5e_1 - 6e_2 + 2e_3$.

(b) We claim that the following vectors also form a spanning set of $\mathbf{R}^3$:

$$w_1 = (1, 1, 1), \quad w_2 = (1, 1, 0), \quad w_3 = (1, 0, 0)$$

Specifically, if $v = (a, b, c)$ is any vector in $\mathbf{R}^3$, then
(Problem 4.62)

$$v = (a, b, c) = cw_1 + (b - c)w_2 + (a - b)w_3$$

For example, $v = (5, -6, 2) = 2w_1 - 8w_2 + 11w_3$.

(c) One can show (Problem 3.24) that $v = (2, 7, 8)$ cannot be written as a linear combination of the vectors

$$u_1 = (1, 2, 3), \quad u_2 = (1, 3, 5), \quad u_3 = (1, 5, 9)$$

Accordingly, u_1, u_2, u_3 do not span $\mathbf{R}^3$.

EXAMPLE 4.4 Consider the vector space $V = \mathbf{P}_n(t)$
consisting of all polynomials of degree $\leq n$.

(a) Clearly every polynomial in $\mathbf{P}_n(t)$ can be expressed as a linear combination of the $n + 1$ polynomials

$$1, \quad t, \quad t^2, \quad t^3, \quad \dots, \quad t^n$$

Thus, these powers of t (where $1 = t^0$) form a spanning set for $\mathbf{P}_n(t)$.

(b) One can also show that, for any scalar c , the following $n + 1$ powers of $t - c$,

$$1, \quad t - c, \quad (t - c)^2, \quad (t - c)^3, \quad \dots, \quad (t - c)^n$$

(where $(t - c)^0 = 1$), also form a spanning set for $\mathbf{P}_n(t)$.

EXAMPLE 4.5 Consider the vector space $\mathbf{M} = \mathbf{M}_{2,2}$
consisting of all 2×2 matrices, and consider
the following four matrices in $\mathbf{M}$:

$$E_{11} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad E_{12} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad E_{21} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \quad E_{22} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

Then clearly any matrix A in $\mathbf{M}$ can be written as a linear combination of the four matrices. For example,

$$A = \begin{bmatrix} 5 & -6 \\ 7 & 8 \end{bmatrix} = 5E_{11} - 6E_{12} + 7E_{21} + 8E_{22}$$

Accordingly, the four matrices $E_{11}, E_{12}, E_{21}, E_{22}$ span $\mathbf{M}$.

4.5 Subspaces

This section introduces the important notion of a subspace.

DEFINITION 4.2: Let V be a vector space over a field K and let W be a subset of V . Then W is a *subspace* of V if W is itself a vector space over K with respect to the operations of vector addition and scalar multiplication on V .

The way in which one shows that any set W is a vector space is to show that W satisfies the eight axioms of a vector space. However, if W is a subset of a vector space V , then some of the axioms automatically hold in W , because they already hold in V . Simple criteria for identifying subspaces follow.

THEOREM 4.2: Suppose W is a subset of a vector space V . Then W is a subspace of V if the following two conditions hold:

- (a) The zero vector 0 belongs to W .
- (b) For every $u, v \in W, k \in K$: (i) The sum $u + v \in W$. (ii) The multiple $ku \in W$.

Property (i) in (b) states that W is *closed under vector addition*, and property (ii) in (b) states that W is *closed under scalar multiplication*. Both properties may be combined into the following equivalent single statement:

(b') For every $u, v \in W$, $a, b \in K$, the linear combination $au + bv \in W$.

Now let V be any vector space. Then V automatically contains two subspaces: the set $\{0\}$ consisting of the zero vector alone and the whole space V itself. These are sometimes called the *trivial* subspaces of V . Examples of nontrivial subspaces follow.

EXAMPLE 4.6 Consider the vector space $V = \mathbf{R}^3$.

(a) Let U consist of all vectors in $\mathbf{R}^3$ whose entries are equal; that is,

$$U = \{(a, b, c) : a = b = c\}$$

For example, $(1, 1, 1)$, $(-3, -3, -3)$, $(7, 7, 7)$, $(-2, -2, -2)$ are vectors in U . Geometrically, U is the line through the origin O and the point $(1, 1, 1)$ as shown in Fig. 4-1(a). Clearly $0 = (0, 0, 0)$ belongs to U , because all entries in 0 are equal. Further, suppose u and v are arbitrary vectors in U , say, $u = (a, a, a)$ and $v = (b, b, b)$. Then, for any scalar $k \in \mathbf{R}$, the following are also vectors in U :

$$u + v = (a + b, a + b, a + b) \quad \text{and} \quad ku = (ka, ka, ka)$$

Thus, U is a subspace of $\mathbf{R}^3$.

(b) Let W be any plane in $\mathbf{R}^3$ passing through the origin, as pictured in Fig. 4-1(b). Then $0 = (0, 0, 0)$ belongs to W , because we assumed W passes through, the origin O . Further, suppose u and v are vectors in W . Then u and v

may be viewed as arrows in the plane W emanating from the origin O , as in Fig. 4-1(b). The sum $u + v$ and any multiple ku of u also lie in the plane W . Thus, W is a subspace of $\mathbf{R}^3$.

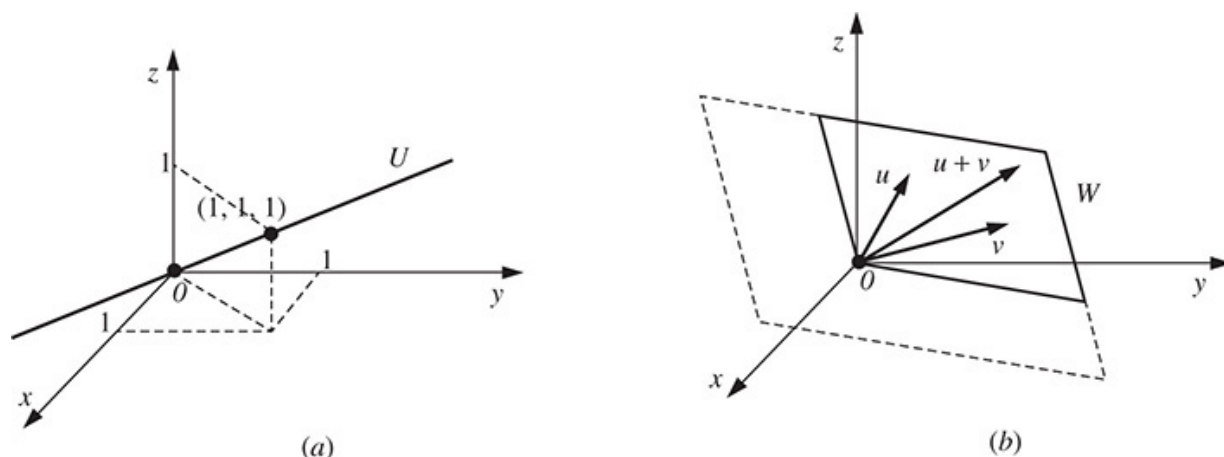


Figure 4-1

EXAMPLE 4.7

- (a) Let $V = \mathbf{M}_{n,n}$, the vector space of $n \times n$ matrices. Let W_1 be the subset of all (upper) triangular matrices and let W_2 be the subset of all symmetric matrices. Then W_1 is a subspace of V , because W_1 contains the zero matrix 0 and W_1 is closed under matrix addition and scalar multiplication; that is, the sum and scalar multiple of such triangular matrices are also triangular. Similarly, W_2 is a subspace of V .
- (b) Let $V = \mathbf{P}(t)$, the vector space $\mathbf{P}(t)$ of polynomials. Then the space $\mathbf{P}_n(t)$ of polynomials of degree at most n may be viewed as a subspace of $\mathbf{P}(t)$. Let $\mathbf{Q}(t)$ be the collection of polynomials with only even powers of t . For example, the following are polynomials in $\mathbf{Q}(t)$:

$$p_1 = 3 + 4t^2 - 5t^6 \quad \text{and} \quad p_2 = 6 - 7t^4 + 9t^6 + 3t^{12}$$

(We assume that any constant $k = kt^0$ is an even power of t .) Then $\mathbf{Q}(t)$ is a subspace of $\mathbf{P}(t)$.

(c) Let V be the vector space of real-valued functions. Then the collection W_1 of continuous functions and the collection W_2 of differentiable functions are subspaces of V .

Intersection of Subspaces

Let U and W be subspaces of a vector space V . We show that the intersection $U \cap W$ is also a subspace of V . Clearly, $0 \in U$ and $0 \in W$, because U and W are subspaces; whence $0 \in U \cap W$. Now suppose u and v belong to the intersection $U \cap W$. Then $u, v \in U$ and $u, v \in W$. Further, because U and W are subspaces, for any scalars $a, b \in K$,

$$au + bv \in U \quad \text{and} \quad au + bv \in W$$

Thus, $au + bv \in U \cap W$. Therefore, $U \cap W$ is a subspace of V . The above result generalizes as follows.

THEOREM 4.3: The intersection of any number of subspaces of a vector space V is a subspace of V .

Solution Space of a Homogeneous System

Consider a system $AX = B$ of linear equations in n unknowns. Then every solution u may be viewed as a vector in K^n . Thus, the solution set of such a system is a subset of K^n . Now suppose the system is homogeneous; that is, suppose the system has the form $AX = 0$. Let W be its solution set. Because $A0 = 0$, the zero vector $0 \in W$. Moreover, suppose u and v belong to W . Then u and v are solutions of $AX = 0$, or, in other words, $Au = 0$ and $Av = 0$. Therefore, for any scalars a and b , we have

$$A(au + bv) = aAu + bAv = a0 + b0 = 0 + 0 = 0$$

Thus, $au + bv$ belongs to W , because it is a solution of $AX = 0$. Accordingly, W is a subspace of K^n .

We state the above result formally.

THEOREM 4.4: The solution set W of a homogeneous system $AX = 0$ in n unknowns is a subspace of K^n .

We emphasize that the solution set of a nonhomogeneous system $AX = B$ is not a subspace of K^n . In fact, the zero vector 0 does not belong to its solution set.

4.6 Linear Spans, Row Space of a Matrix

Suppose $u_1, u_2, \dots, u_m$ are any vectors in a vector space V . Recall (Section 4.4) that any vector of the form $a_1u_1 + a_2u_2 + \dots + a_mu_m$, where the a_i are scalars, is called a *linear combination* of $u_1, u_2, \dots, u_m$. The collection of all such linear combinations, denoted by

$$\text{span}(u_1, u_2, \dots, u_m) \quad \text{or} \quad \text{span}(u_i)$$

is called the *linear span* of $u_1, u_2, \dots, u_m$.

Clearly the zero vector 0 belongs to $\text{span}(u_i)$, because

$$0 = 0u_1 + 0u_2 + \dots + 0u_m$$

Furthermore, suppose v and v' belong to $\text{span}(u_i)$, say,

$$v = a_1u_1 + a_2u_2 + \dots + a_mu_m \quad \text{and} \quad v' = b_1u_1 + b_2u_2 + \dots + b_mu_m$$

Then,

$$v + v' = (a_1 + b_1)u_1 + (a_2 + b_2)u_2 + \cdots + (a_m + b_m)u_m$$

and, for any scalar $k \in K$,

$$kv = ka_1u_1 + ka_2u_2 + \cdots + ka_mu_m$$

Thus, $v + v'$ and kv also belong to $\text{span}(u_i)$. Accordingly, $\text{span}(u_i)$ is a subspace of V .

More generally, for any subset S of V , $\text{span}(S)$ consists of all linear combinations of vectors in S or, when $S = \phi$, $\text{span}(S) = \{0\}$. Thus, in particular, S is a spanning set (Section 4.4) of $\text{span}(S)$.

The following theorem, which was partially proved above, holds.

THEOREM 4.5: Let S be a subset of a vector space V .

- (i) Then $\text{span}(S)$ is a subspace of V that contains S .
- (ii) If W is a subspace of V containing S , then $\text{span}(S) \subseteq W$.

Condition (ii) in theorem 4.5 may be interpreted as saying that $\text{span}(S)$ is the “smallest” subspace of V containing S .

EXAMPLE 4.8 Consider the vector space $V = \mathbf{R}^3$.

- (a) Let u be any nonzero vector in $\mathbf{R}^3$. Then $\text{span}(u)$ consists of all scalar multiples of u . Geometrically, $\text{span}(u)$ is the line through the origin O and the endpoint of u , as shown in Fig. 4-2(a).

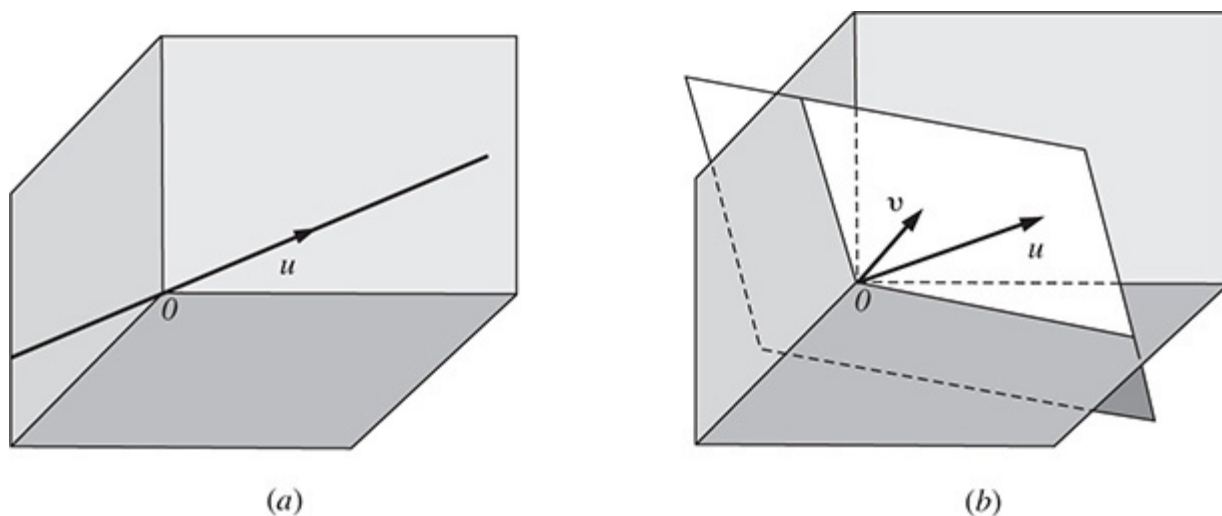


Figure 4-2

- (b) Let u and v be vectors in $\mathbf{R}^3$ that are not multiples of each other. Then $\text{span}(u, v)$ is the plane through the origin O and the endpoints of u and v as shown in Fig. 4-2(b).
- (c) Consider the vectors $e_1 = (1, 0, 0)$, $e_2 = (0, 1, 0)$, $e_3 = (0, 0, 1)$ in $\mathbf{R}^3$. Recall [Example 4.1(a)] that every vector in $\mathbf{R}^3$ is a linear combination of e_1, e_2, e_3 . That is, e_1, e_2, e_3 form a spanning set of $\mathbf{R}^3$. Accordingly, $\text{span}(e_1, e_2, e_3) = \mathbf{R}^3$.

Row Space of a Matrix

Let $A = [a_{ij}]$ be an arbitrary $m \times n$ matrix over a field K . The rows of A ,

$$R_1 = (a_{11}, a_{12}, \dots, a_{1n}), \quad R_2 = (a_{21}, a_{22}, \dots, a_{2n}), \quad \dots, \quad R_m = (a_{m1}, a_{m2}, \dots, a_{mn})$$

may be viewed as vectors in K^n ; hence, they span a subspace of K^n called the *row space* of A and denoted by $\text{rowsp}(A)$. That is,

$$\text{rowsp}(A) = \text{span}(R_1, R_2, \dots, R_m)$$

Analogously, the columns of A may be viewed as vectors in K^m called the *column space* of A and denoted by $\text{colsp}(A)$. Observe that $\text{colsp}(A) = \text{rowsp}(A^T)$.

Recall that matrices A and B are row equivalent, written $A \sim B$, if B can be obtained from A by a sequence of elementary row operations. Now suppose M is the matrix obtained by applying one of the following elementary row operations on a matrix A :

- (1) Interchange R_i and R_j , (2) Replace R_i by kR_i , (3) Replace R_j by $kR_i + R_j$

Then each row of M is a row of A or a linear combination of rows of A . Hence, the row space of M is contained in the row space of A . On the other hand, we can apply the inverse elementary row operation on M to obtain A ; hence, the row space of A is contained in the row space of M . Accordingly, A and M have the same row space. This will be true each time we apply an elementary row operation. Thus, we have proved the following theorem.

THEOREM 4.6: Row equivalent matrices have the same row space.

We are now able to prove (Problems 4.45–4.47) basic results on row equivalence (which first appeared as Theorems 3.7 and 3.8 in [Chapter 3](#)).

THEOREM 4.7: Suppose $A = [a_{ij}]$ and $B = [b_{ij}]$ are row equivalent echelon matrices with respective pivot entries

$$a_{1j_1}, a_{2j_2}, \dots, a_{rj_r} \quad \text{and} \quad b_{1k_1}, b_{2k_2}, \dots, b_{sk_s}$$

Then A and B have the same number of nonzero rows—that is, $r = s$ —and their pivot

entries are in the same positions—that is, $j_1 = k_1, j_2 = k_2, \dots, j_r = k_r$.

THEOREM 4.8: Suppose A and B are row canonical matrices. Then A and B have the same row space if and only if they have the same nonzero rows.

COROLLARY 4.9: Every matrix A is row equivalent to a unique matrix in row canonical form.

We apply the above results in the next example.

EXAMPLE 4.9 Consider the following two sets of vectors in $\mathbf{R}^4$:

$$\begin{aligned} u_1 &= (1, 2, -1, 3), & u_2 &= (2, 4, 1, -2), & u_3 &= (3, 6, 3, -7) \\ w_1 &= (1, 2, -4, 11), & w_2 &= (2, 4, -5, 14) \end{aligned}$$

Let $U = \text{span}(u_i)$ and $W = \text{span}(w_i)$. There are two ways to show that $U = W$.

- (a) Show that each u_i is a linear combination of w_1 and w_2 , and show that each w_i is a linear combination of u_1, u_2, u_3 . Observe that we have to show that six systems of linear equations are consistent.
- (b) Form the matrix A whose rows are u_1, u_2, u_3 and row reduce A to row canonical form, and form the matrix B whose rows are w_1 and w_2 and row reduce B to row canonical form:

$$A = \begin{bmatrix} 1 & 2 & -1 & 3 \\ 2 & 4 & 1 & -2 \\ 3 & 6 & 3 & -7 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & 0 & 3 & -8 \\ 0 & 0 & 6 & -16 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 0 & \frac{1}{3} \\ 0 & 0 & 1 & -\frac{8}{3} \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & 2 & -4 & 11 \\ 2 & 4 & -5 & 14 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & -4 & 11 \\ 0 & 0 & 3 & -8 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 0 & \frac{1}{3} \\ 0 & 0 & 1 & -\frac{8}{3} \end{bmatrix}$$

Because the nonzero rows of the matrices in row canonical form are identical, the row spaces of A and B are equal. Therefore, $U = W$.

Clearly, the method in (b) is more efficient than the method in (a).

4.7 Linear Dependence and Independence

Let V be a vector space over a field K . The following defines the notion of linear dependence and independence of vectors over K . (One usually suppresses mentioning K when the field is understood.) This concept plays an essential role in the theory of linear algebra and in mathematics in general.

DEFINITION 4.3: We say that the vectors $v_1, v_2, \dots, v_m$ in V are *linearly dependent* if there exist scalars $a_1, a_2, \dots, a_m$ in K , not all of them 0, such that

$$a_1 v_1 + a_2 v_2 + \cdots + a_m v_m = 0$$

Otherwise, we say that the vectors are *linearly independent*.

The above definition may be restated as follows.
Consider the vector equation

$$x_1 v_1 + x_2 v_2 + \cdots + x_m v_m = 0 \quad (*)$$

where the x 's are unknown scalars. This equation always has the *zero solution* $x_1 = 0, x_2 = 0, \dots, x_m = 0$. Suppose this is the only solution; that is, suppose we can show:

$$x_1 v_1 + x_2 v_2 + \cdots + x_m v_m = 0 \quad \text{implies} \quad x_1 = 0, \quad x_2 = 0, \quad \dots, \quad x_m = 0$$

Then the vectors $v_1, v_2, \dots, v_m$ are linearly independent. On the other hand, suppose the equation (*) has a nonzero solution; then the vectors are linearly dependent.

A set $S = \{v_1, v_2, \dots, v_m\}$ of vectors in V is linearly dependent or independent according to whether the vectors $v_1, v_2, \dots, v_m$ are linearly dependent or independent.

An infinite set S of vectors is linearly dependent or independent according to whether there do or do not exist vectors $v_1, v_2, \dots, v_k$ in S that are linearly dependent.

Warning: The set $S = \{v_1, v_2, \dots, v_m\}$ above represents a *list* or, in other words, a finite sequence of vectors where the vectors are ordered and repetition is permitted.

The following remarks follow directly from the above definition.

Remark 1: Suppose 0 is one of the vectors $v_1, v_2, \dots, v_m$, say $v_1 = 0$. Then the vectors must be linearly dependent, because we have the following linear combination where the coefficient of $v_1 \neq 0$:

$$1v_1 + 0v_2 + \cdots + 0v_m = 1 \cdot 0 + 0 + \cdots + 0 = 0$$

Remark 2: Suppose v is a nonzero vector. Then v , by itself, is linearly independent, because

$$kv = 0, \quad v \neq 0 \quad \text{implies} \quad k = 0$$

Remark 3: Suppose two of the vectors $v_1, v_2, \dots, v_m$ are equal or one is a scalar multiple of the other, say $v_1 = kv_2$. Then the vectors must be linearly dependent, because we have the following linear combination where the coefficient of $v_1 \neq 0$:

$$v_1 - kv_2 + 0v_3 + \cdots + 0v_m = 0$$

Remark 4: Two vectors v_1 and v_2 are linearly dependent if and only if one of them is a multiple of the other.

Remark 5: If the set $\{v_1, \dots, v_m\}$ is linearly independent, then any rearrangement of the vectors $\{v_{i_1}, v_{i_2}, \dots, v_{i_m}\}$ is also linearly independent.

Remark 6: If a set S of vectors is linearly independent, then any subset of S is linearly independent. Alternatively, if S contains a linearly dependent subset, then S is linearly dependent.

EXAMPLE 4.10

(a) Let $u = (1, 1, 0)$, $v = (1, 3, 2)$, $w = (4, 9, 5)$. Then u, v, w are linearly dependent, because

$$3u + 5v - 2w = 3(1, 1, 0) + 5(1, 3, 2) - 2(4, 9, 5) = (0, 0, 0) = 0$$

(b) We show that the vectors $u = (1, 2, 3)$, $v = (2, 5, 7)$, $w = (1, 3, 5)$ are linearly independent. We form the vector

equation $xu + yv + zw = 0$, where x, y, z are unknown scalars. This yields

$$x \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + y \begin{bmatrix} 2 \\ 5 \\ 7 \end{bmatrix} + z \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \text{or} \quad \begin{array}{l} x + 2y + z = 0 \\ 2x + 5y + 3z = 0 \\ 3x + 7y + 5z = 0 \end{array} \quad \text{or} \quad \begin{array}{l} x + 2y + z = 0 \\ y + z = 0 \\ 2z = 0 \end{array}$$

Back-substitution yields $x = 0, y = 0, z = 0$. We have shown that

$$xu + yv + zw = 0 \quad \text{implies} \quad x = 0, \quad y = 0, \quad z = 0$$

Accordingly, u, v, w are linearly independent.

(c) Let V be the vector space of functions from $\mathbf{R}$ into $\mathbf{R}$. We show that the functions $f(t) = \sin t, g(t) = e^t, h(t) = t^2$ are linearly independent. We form the vector (function) equation $xf + yg + zh = 0$, where x, y, z are unknown scalars. This function equation means that, for every value of t ,

$$x \sin t + ye^t + zt^2 = 0$$

Thus, in this equation, we choose appropriate values of t to easily get $x = 0, y = 0, z = 0$. For example,

- | | | | | |
|-------|------------------------|--|----|---------|
| (i) | Substitute $t = 0$ | to obtain $x(0) + y(1) + z(0) = 0$ | or | $y = 0$ |
| (ii) | Substitute $t = \pi$ | to obtain $x(0) + 0(e^\pi) + z(\pi^2) = 0$ | or | $z = 0$ |
| (iii) | Substitute $t = \pi/2$ | to obtain $x(1) + 0(e^{\pi/2}) + 0(\pi^2/4) = 0$ | or | $x = 0$ |

We have shown

$$xf + yg + zh = 0 \quad \text{implies} \quad x = 0, \quad y = 0, \quad z = 0$$

Accordingly, u, v, w are linearly independent.

Linear Dependence in $\mathbf{R}^3$

Linear dependence in the vector space $V = \mathbf{R}^3$ can be described geometrically as follows:

- (a) Any two vectors u and v in $\mathbf{R}^3$ are linearly dependent if and only if they lie on the same line through the origin O , as shown in Fig. 4-3(a).
- (b) Any three vectors u, v, w in $\mathbf{R}^3$ are linearly dependent if and only if they lie on the same plane through the origin O , as shown in Fig. 4-3(b).

Later, we will be able to show that any four or more vectors in $\mathbf{R}^3$ are automatically linearly dependent.

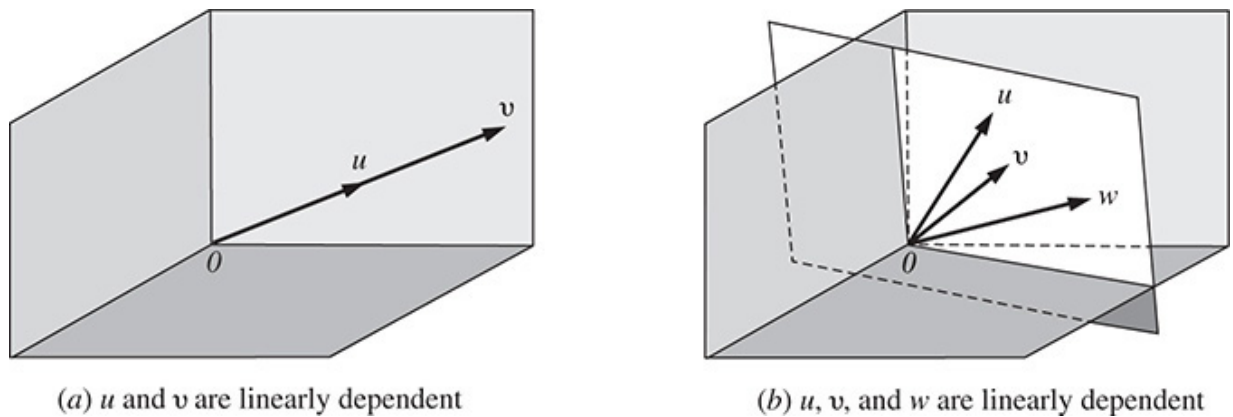


Figure 4-3

Linear Dependence and Linear Combinations

The notions of linear dependence and linear combinations are closely related. Specifically, for more than one vector, we show that the vectors $v_1, v_2, \dots, v_m$ are linearly dependent if and only if one of them is a linear combination of the others.

Suppose, say, v_i is a linear combination of the others,

$$v_i = a_1 v_1 + \cdots + a_{i-1} v_{i-1} + a_{i+1} v_{i+1} + \cdots + a_m v_m$$

Then by adding v_i to both sides, we obtain

$$a_1 v_1 + \cdots + a_{i-1} v_{i-1} - v_i + a_{i+1} v_{i+1} + \cdots + a_m v_m = 0$$

where the coefficient of v_i is not 0. Hence, the vectors are linearly dependent. Conversely, suppose the vectors are linearly dependent, say,

$$b_1 v_1 + \cdots + b_j v_j + \cdots + b_m v_m = 0, \quad \text{where} \quad b_j \neq 0$$

Then we can solve for v_j obtaining

$$v_j = b_j^{-1} b_1 v_1 - \cdots - b_j^{-1} b_{j-1} v_{j-1} - b_j^{-1} b_{j+1} v_{j+1} - \cdots - b_j^{-1} b_m v_m$$

and so v_j is a linear combination of the other vectors.

We now state a slightly stronger statement than the one above. This result has many important consequences.

LEMMA 4.10: Suppose two or more nonzero vectors $v_1, v_2, \dots, v_m$ are linearly dependent. Then one of the vectors is a linear combination of the preceding vectors; that is, there exists $k > 1$ such that

$$v_k = c_1 v_1 + c_2 v_2 + \cdots + c_{k-1} v_{k-1}$$

Linear Dependence and Echelon Matrices

Consider the following echelon matrix A , whose pivots have been circled:

$$A = \begin{bmatrix} 0 & \textcircled{2} & 3 & 4 & 5 & 6 & 7 \\ 0 & 0 & \textcircled{4} & 3 & 2 & 3 & 4 \\ 0 & 0 & 0 & 0 & \textcircled{7} & 8 & 9 \\ 0 & 0 & 0 & 0 & 0 & \textcircled{6} & 7 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Observe that the rows R_2, R_3, R_4 have 0's in the second column below the nonzero pivot in R_1 , and hence any linear combination of R_2, R_3, R_4 must have 0 as its second entry. Thus, R_1 cannot be a linear combination of the rows below it. Similarly, the rows R_3 and R_4 have 0's in the third column below the nonzero pivot in R_2 , and hence R_2 cannot be a linear combination of the rows below it. Finally, R_3 cannot be a multiple of R_4 , because R_4 has a 0 in the fifth column below the nonzero pivot in R_3 . Viewing the nonzero rows from the bottom up, R_4, R_3, R_2, R_1 , no row is a linear combination of the preceding rows. Thus, the rows are linearly independent by Lemma 4.10.

The argument used with the above echelon matrix A can be used for the nonzero rows of any echelon matrix. Thus, we have the following very useful result.

THEOREM 4.11: The nonzero rows of a matrix in echelon form are linearly independent.

4.8 Basis and Dimension

First we state two equivalent ways to define a basis of a vector space V . (The equivalence is proved in [Problem 4.28](#).)

DEFINITION 4.4 A: A set $S = \{u_1, u_2, \dots, u_n\}$ of vectors is a *basis* of V if it has the following two properties: (1) S is linearly independent. (2) S spans V .

DEFINITION 4.4 B: A set $S = \{u_1, u_2, \dots, u_n\}$ of vectors is a *basis* of V if every $v \in V$ can be written uniquely as a linear combination of the basis vectors.

The following is a fundamental result in linear algebra.

THEOREM 4.12: Let V be a vector space such that one basis has m elements and another basis has n elements. Then $m = n$.

A vector space V is said to be of *finite dimension* n or *n -dimensional*, written

$$\dim V = n$$

if V has a basis with n elements. [Theorem 4.12](#) tells us that all bases of V have the same number of elements, so this definition is well defined.

The vector space $\{0\}$ is defined to have dimension 0.

Suppose a vector space V does not have a finite basis. Then V is said to be of *infinite dimension* or to be *infinite-dimensional*.

The above fundamental [Theorem 4.12](#) is a consequence of the following “replacement lemma” (proved in [Problem 4.35](#)).

LEMMA 4.13: Suppose $\{v_1, v_2, \dots, v_n\}$ spans V , and suppose $\{w_1, w_2, \dots, w_m\}$ is linearly independent. Then $m \leq n$, and V is spanned by a set of the form

$$\{w_1, w_2, \dots, w_m, v_{i_1}, v_{i_2}, \dots, v_{i_{n-m}}\}$$

Thus, in particular, $n + 1$ or more vectors in V are linearly dependent.

Observe in the above lemma that we have replaced m of the vectors in the spanning set of V by the m independent vectors and still retained a spanning set.

Examples of Bases

This subsection presents important examples of bases of some of the main vector spaces appearing in this text.

(a) Vector space K^n : Consider the following n vectors in K^n :

$$e_1 = (1, 0, 0, 0, \dots, 0, 0), \quad e_2 = (0, 1, 0, 0, \dots, 0, 0), \quad \dots, \quad e_n = (0, 0, 0, 0, \dots, 0, 1)$$

These vectors are linearly independent. (For example, they form a matrix in echelon form.) Furthermore, any vector $u = (a_1, a_2, \dots, a_n)$ in K^n can be written as a linear combination of the above vectors. Specifically,

$$v = a_1 e_1 + a_2 e_2 + \dots + a_n e_n$$

Accordingly, the vectors form a basis of K^n called the *usual* or *standard* basis of K^n . Thus (as one might expect), K^n has dimension n . In particular, any other basis of K^n has n elements.

(b) Vector space $\mathbf{M} = \mathbf{M}_{r,s}$ of all $r \times s$ matrices: The following six matrices form a basis of the vector space $\mathbf{M}_{2,3}$ of all 2×3 matrices over K :

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

More generally, in the vector space $\mathbf{M} = \mathbf{M}_{r,s}$ of all $r \times s$ matrices, let E_{ij} be the matrix with ij -entry 1 and 0's elsewhere. Then all such matrices form a basis of $\mathbf{M}_{r,s}$ called the *usual* or *standard* basis of $\mathbf{M}_{r,s}$. Accordingly, $\dim \mathbf{M}_{r,s} = rs$.

(c) Vector space $\mathbf{P}_n(t)$ of all polynomials of degree $\leq n$
: The set $S = \{1, t, t^2, t^3, \dots, t^n\}$ of $n + 1$ polynomials is a

basis of $\mathbf{P}_n(t)$. Specifically, any polynomial $f(t)$ of degree $\leq n$ can be expressed as a linear combination of these powers of t , and one can show that these polynomials are linearly independent. Therefore, $\dim \mathbf{P}_n(t) = n + 1$.

(d) Vector space $\mathbf{P}(t)$ of all polynomials: Consider any finite set $S = \{f_1(t), f_2(t), \dots, f_m(t)\}$ of polynomials in $\mathbf{P}(t)$, and let m denote the largest of the degrees of the polynomials. Then any polynomial $g(t)$ of degree exceeding m cannot be expressed as a linear combination of the elements of S . Thus, S cannot be a basis of $\mathbf{P}(t)$. This means that the dimension of $\mathbf{P}(t)$ is infinite. We note that the infinite set $S' = \{1, t, t^2, t^3, \dots\}$, consisting of all the powers of t , spans $\mathbf{P}(t)$ and is linearly independent. Accordingly, S' is an infinite basis of $\mathbf{P}(t)$.

Theorems on Bases

The following three theorems (proved in Problems 4.37, 4.38, and 4.39) will be used frequently.

THEOREM 4.14: Let V be a vector space of finite dimension n . Then:

- (i) Any $n + 1$ or more vectors in V are linearly dependent.
- (ii) Any linearly independent set $S = \{u_1, u_2, \dots, u_n\}$ with n elements is a basis of V .
- (iii) Any spanning set $T = \{v_1, v_2, \dots, v_n\}$ of V with n elements is a basis of V .

THEOREM 4.15: Suppose S spans a vector space V . Then:

- (i) Any maximum number of linearly independent vectors in S form a basis of V .

- (ii) Suppose one deletes from S every vector that is a linear combination of preceding vectors in S . Then the remaining vectors form a basis of V .

THEOREM 4.16: Let V be a vector space of finite dimension and let $S = \{u_1, u_2, \dots, u_r\}$ be a set of linearly independent vectors in V . Then S is part of a basis of V ; that is, S may be extended to a basis of V .

EXAMPLE 4.11

- (a) The following four vectors in $\mathbf{R}^4$ form a matrix in echelon form:

$$(1, 1, 1, 1), (0, 1, 1, 1), (0, 0, 1, 1), (0, 0, 0, 1)$$

Thus, the vectors are linearly independent, and, because $\dim \mathbf{R}^4 = 4$, the four vectors form a basis of $\mathbf{R}^4$.

- (b) The following $n + 1$ polynomials in $\mathbf{P}_n(t)$ are of increasing degree:

$$1, t - 1, (t - 1)^2, \dots, (t - 1)^n$$

Therefore, no polynomial is a linear combination of preceding polynomials; hence, the polynomials are linear independent. Furthermore, they form a basis of $\mathbf{P}_n(t)$, because $\dim \mathbf{P}_n(t) = n + 1$.

- (c) Consider any four vectors in $\mathbf{R}^3$, say

$$(257, -132, 58), \quad (43, 0, -17), \quad (521, -317, 94), \quad (328, -512, -731)$$

By [Theorem 4.14\(i\)](#), the four vectors must be linearly dependent, because they come from the three-dimensional vector space $\mathbf{R}^3$.

Dimension and Subspaces

The following theorem (proved in [Problem 4.40](#)) gives the basic relationship between the dimension of a vector space and the dimension of a subspace.

THEOREM 4.17: Let W be a subspace of an n -dimensional vector space V . Then $\dim W \leq n$. In particular, if $\dim W = n$, then $W = V$.

EXAMPLE 4.12 Let W be a subspace of the real space $\mathbf{R}^3$. Note that $\dim \mathbf{R}^3 = 3$. [Theorem 4.17](#) tells us that the dimension of W can only be 0, 1, 2, or 3. The following cases apply:

- (a) If $\dim W = 0$, then $W = \{0\}$, a point.
- (b) If $\dim W = 1$, then W is a line through the origin 0.
- (c) If $\dim W = 2$, then W is a plane through the origin 0.
- (d) If $\dim W = 3$, then W is the entire space $\mathbf{R}^3$.

4.9 Application to Matrices, Rank of a Matrix

Let A be any $m \times n$ matrix over a field K . Recall that the rows of A may be viewed as vectors in K^n and that the row space of A , written $\text{rowsp}(A)$, is the subspace of K^n spanned by the rows of A . The following definition applies.

DEFINITION 4.5: The *rank* of a matrix A , written $\text{rank}(A)$, is equal to the maximum number of linearly

independent rows of A or, equivalently, the dimension of the row space of A .

Recall, on the other hand, that the columns of an $m \times n$ matrix A may be viewed as vectors in K^m and that the column space of A , written $\text{colsp}(A)$, is the subspace of K^m spanned by the columns of A . Although m may not be equal to n —that is, the rows and columns of A may belong to different vector spaces—we have the following fundamental result.

THEOREM 4.18: The maximum number of linearly independent rows of any matrix A is equal to the maximum number of linearly independent columns of A . Thus, the dimension of the row space of A is equal to the dimension of the column space of A .

Accordingly, one could restate the above definition of the rank of A using columns instead of rows.

Basis-Finding Problems

This subsection shows how an echelon form of any matrix A gives us the solution to certain problems about A itself. Specifically, let A and B be the following matrices, where the echelon matrix B (whose pivots are circled) is an echelon form of A :

$$A = \begin{bmatrix} 1 & 2 & 1 & 3 & 1 & 2 \\ 2 & 5 & 5 & 6 & 4 & 5 \\ 3 & 7 & 6 & 11 & 6 & 9 \\ 1 & 5 & 10 & 8 & 9 & 9 \\ 2 & 6 & 8 & 11 & 9 & 12 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} \textcircled{1} & 2 & 1 & 3 & 1 & 2 \\ 0 & \textcircled{1} & 3 & 1 & 2 & 1 \\ 0 & 0 & 0 & \textcircled{1} & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

We solve the following four problems about the matrix A , where $C_1, C_2, \dots, C_6$ denote its columns:

- (a) Find a basis of the row space of A .
- (b) Find each column C_k of A that is a linear combination of preceding columns of A .
- (c) Find a basis of the column space of A .
- (d) Find the rank of A .

(a) We are given that A and B are row equivalent, so they have the same row space. Moreover, B is in echelon form, so its nonzero rows are linearly independent and hence form a basis of the row space of B . Thus, they also form a basis of the row space of A . That is,

basis of $\text{rowsp}(A)$: $(1, 2, 1, 3, 1, 2), (0, 1, 3, 1, 2, 1), (0, 0, 0, 1, 1, 2)$

(b) Let $M_k = [C_1, C_2, \dots, C_k]$, the submatrix of A consisting of the first k columns of A . Then M_{k-1} and M_k are, respectively, the coefficient matrix and augmented matrix of the vector equation

$$x_1 C_1 + x_2 C_2 + \dots + x_{k-1} C_{k-1} = C_k$$

[Theorem 3.9](#) tells us that the system has a solution, or, equivalently, C_k is a linear combination of the preceding columns of A if and only if $\text{rank}(M_k) = \text{rank}(M_{k-1})$, where $\text{rank}(M_k)$ means the number of pivots in an echelon form of M_k . Now the first k column of the echelon matrix B is also an echelon form of M_k . Accordingly,

$$\text{rank}(M_2) = \text{rank}(M_3) = 2 \quad \text{and} \quad \text{rank}(M_4) = \text{rank}(M_5) = \text{rank}(M_6) = 3$$

Thus, C_3, C_5, C_6 are each a linear combination of the preceding columns of A .

(c) The fact that the remaining columns C_1, C_2, C_4 are not linear combinations of their respective preceding columns also tells us that they are linearly independent. Thus, they form a basis of the column space of A . That is,

basis of $\text{colsp}(A)$: $[1, 2, 3, 1, 2]^T, [2, 5, 7, 5, 6]^T, [3, 6, 11, 8, 11]^T$

Observe that C_1, C_2, C_4 may also be characterized as those columns of A that contain the pivots in any echelon form of A .

(d) Here we see that three possible definitions of the rank of A yield the same value.

- (i) There are three pivots in B , which is an echelon form of A .
- (ii) The three pivots in B correspond to the nonzero rows of B , which form a basis of the row space of A .
- (iii) The three pivots in B correspond to the columns of A , which form a basis of the column space of A .

Thus, $\text{rank}(A) = 3$.

Application to Finding a Basis for $W = \text{span}(u_1, u_2, \dots, u_r)$

Frequently, we are given a list $S = \{u_1, u_2, \dots, u_r\}$ of vectors in K^n and we want to find a basis for the subspace W of K^n spanned by the given vectors—that is, a basis of

$$W = \text{span}(S) = \text{span}(u_1, u_2, \dots, u_r)$$

The following two algorithms, which are essentially described in the above subsection, find such a basis (and hence the dimension) of W .

Algorithm 4.1 (Row space algorithm)

Step 1. Form the matrix M whose *rows* are the given vectors.

Step 2. Row reduce M to echelon form.

Step 3. Output the nonzero rows of the echelon matrix.

Sometimes we want to find a basis that only comes from the original given vectors. The next algorithm accomplishes this task.

Algorithm 4.2 (Casting-out algorithm)

Step 1. Form the matrix M whose *columns* are the given vectors.

Step 2. Row reduce M to echelon form.

Step 3. For each column C_k in the echelon matrix without a pivot, delete (cast out) the vector u_k from the list S of given vectors.

Step 4. Output the remaining vectors in S (which correspond to columns with pivots).

We emphasize that in the first algorithm we form a matrix whose rows are the given vectors, whereas in the second algorithm we form a matrix whose columns are the given vectors.

EXAMPLE 4.13 Let W be the subspace of $\mathbf{R}^5$ spanned by the following vectors:

$$\begin{aligned} u_1 &= (1, 2, 1, 3, 2), & u_2 &= (1, 3, 3, 5, 3), & u_3 &= (3, 8, 7, 13, 8) \\ u_4 &= (1, 4, 6, 9, 7), & u_5 &= (5, 13, 13, 25, 19) \end{aligned}$$

Find a basis of W consisting of the original given vectors, and find $\dim W$.

Form the matrix M whose columns are the given vectors, and reduce M to echelon form:

$$M = \begin{bmatrix} 1 & 1 & 3 & 1 & 5 \\ 2 & 3 & 8 & 4 & 13 \\ 1 & 3 & 7 & 6 & 13 \\ 3 & 5 & 13 & 9 & 25 \\ 2 & 3 & 8 & 7 & 19 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 3 & 1 & 5 \\ 0 & 1 & 2 & 2 & 3 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

The pivots in the echelon matrix appear in columns C_1 , C_2 , C_4 . Accordingly, we “cast out” the vectors u_3 and u_5 from the original five vectors. The remaining vectors u_1 , u_2 , u_4 , which correspond to the columns in the echelon matrix with pivots, form a basis of W . Thus, in particular, $\dim W = 3$.

Remark: The justification of the casting-out algorithm is essentially described above, but we repeat it again here for emphasis. The fact that column C_3 in the echelon matrix in [Example 4.13](#) does not have a pivot means that the vector equation

$$xu_1 + yu_2 = u_3$$

has a solution, and hence u_3 is a linear combination of u_1 and u_2 . Similarly, the fact that C_5 does not have a pivot means that u_5 is a linear combination of the preceding vectors. We have deleted each vector in the original spanning set that is a linear combination of preceding vectors. Thus, the remaining vectors are linearly independent and form a basis of W .

Application to Homogeneous Systems of Linear Equations

Consider again a homogeneous system $AX = 0$ of linear equations over K with n unknowns. By [Theorem 4.4](#), the solution set W of such a system is a subspace of K^n , and hence W has a dimension. The following theorem, whose proof is postponed until [Chapter 5](#), holds.

THEOREM 4.19: The dimension of the solution space W of a homogeneous system $AX = 0$ is $n - r$, where n is the number of unknowns and r is the rank of the coefficient matrix A .

In the case where the system $AX = 0$ is in echelon form, it has precisely $n - r$ free variables, say $x_{i_1}, x_{i_2}, \dots, x_{i_{n-r}}$. Let v_j be the solution obtained by setting $x_{i_j} = 1$ (or any nonzero constant) and the remaining free variables equal to 0. We show ([Problem 4.50](#)) that the solutions $v_1, v_2, \dots, v_{n-r}$ are linearly independent; hence, they form a basis of the solution space W .

We have already used the above process to find a basis of the solution space W of a homogeneous system $AX = 0$ in Section 3.11. [Problem 4.48](#) gives three other examples.

4.10 Sums and Direct Sums

Let U and W be subsets of a vector space V . The sum of U and W , written $U + W$, consists of all sums $u + w$ where $u \in U$ and $w \in W$. That is,

$$U + W = \{v : v = u + w, \text{ where } u \in U \text{ and } w \in W\}$$

Now suppose U and W are subspaces of V . Then one can easily show ([Problem 4.53](#)) that $U + W$ is a subspace of V .

Recall that $U \cap W$ is also a subspace of V . The following theorem (proved in [Problem 4.58](#)) relates the dimensions of these subspaces.

THEOREM 4.20: Suppose U and W are finite-dimensional subspaces of a vector space V . Then $U + W$ has finite dimension and

$$\dim(U + W) = \dim U + \dim W - \dim(U \cap W)$$

EXAMPLE 4.14 Let $V = \mathbf{M}_{2,2}$, the vector space of 2×2 matrices. Let U consist of those matrices whose second row is zero, and let W consist of those matrices whose second column is zero. Then

$$U = \left\{ \begin{bmatrix} a & b \\ 0 & 0 \end{bmatrix} \right\}, \quad W = \left\{ \begin{bmatrix} a & 0 \\ c & 0 \end{bmatrix} \right\} \quad \text{and} \quad U + W = \left\{ \begin{bmatrix} a & b \\ c & 0 \end{bmatrix} \right\}, \quad U \cap W = \left\{ \begin{bmatrix} a & 0 \\ 0 & 0 \end{bmatrix} \right\}$$

That is, $U + W$ consists of those matrices whose lower right entry is 0, and $U \cap W$ consists of those matrices whose second row and second column are zero. Note that $\dim U = 2$, $\dim W = 2$, $\dim(U \cap W) = 1$. Also, $\dim(U + W) = 3$, which is expected from [Theorem 4.20](#). That is,

$$\dim(U + W) = \dim U + \dim W - \dim(U \cap W) = 2 + 2 - 1 = 3$$

Direct Sums

The vector space V is said to be the *direct sum* of its subspaces U and W , denoted by

$$V = U \oplus W$$

if every $v \in V$ can be written in one and only one way as $v = u + w$ where $u \in U$ and $w \in W$.

The following theorem (proved in [Problem 4.59](#)) characterizes such a decomposition.

THEOREM 4.21: The vector space V is the direct sum of its subspaces U and W if and only if:
(i) $V = U + W$, (ii) $U \cap W = \{0\}$.

EXAMPLE 4.15 Consider the vector space $V = \mathbf{R}^3$.

(a) Let U be the xy -plane and let W be the yz -plane; that is,

$$U = \{(a, b, 0) : a, b \in \mathbf{R}\} \quad \text{and} \quad W = \{(0, b, c) : b, c \in \mathbf{R}\}$$

Then $\mathbf{R}^3 = U + W$, because every vector in $\mathbf{R}^3$ is the sum of a vector in U and a vector in W . However, $\mathbf{R}^3$ is not the direct sum of U and W , because such sums are not unique. For example,

$$(3, 5, 7) = (3, 1, 0) + (0, 4, 7) \quad \text{and also} \quad (3, 5, 7) = (3, -4, 0) + (0, 9, 7)$$

(b) Let U be the xy -plane and let W be the z -axis; that is,

$$U = \{(a, b, 0) : a, b \in \mathbf{R}\} \quad \text{and} \quad W = \{(0, 0, c) : c \in \mathbf{R}\}$$

Now any vector $(a, b, c) \in \mathbf{R}^3$ can be written as the sum of a vector in U and a vector in W in one and only one way:

$$(a, b, c) = (a, b, 0) + (0, 0, c)$$

Accordingly, $\mathbf{R}^3$ is the direct sum of U and W ; that is,
 $\mathbf{R}^3 = U \oplus W$.

General Direct Sums

The notion of a direct sum is extended to more than one factor in the obvious way. That is, V is the *direct sum* of

subspaces $W_1, W_2, \dots, W_r$ written

$$V = W_1 \oplus W_2 \oplus \cdots \oplus W_r$$

if every vector $v \in V$ can be written in one and only one way as

$$v = w_1 + w_2 + \cdots + w_r$$

where $w_1 \in W_1, w_2 \in W_2, \dots, w_r \in W_r$.

The following theorems hold.

THEOREM 4.22: Suppose $V = W_1 \oplus W_2 \oplus \cdots \oplus W_r$. Also, for each k , suppose S_k is a linearly independent subset of W_k . Then

- (a) The union $S = \cup_k S_k$ is linearly independent in V .
- (b) If each S_k is a basis of W_k , then $\cup_k S_k$ is a basis of V .
- (c) $\dim V = \dim W_1 + \dim W_2 + \cdots + \dim W_r$.

THEOREM 4.23: Suppose $V = W_1 + W_2 + \cdots + W_r$ and $\dim V = \sum_k \dim W_k$. Then $V = W_1 \oplus W_2 \oplus \cdots \oplus W_r$.

4.11 Coordinates

Let V be an n -dimensional vector space over K with basis $S = \{u_1, u_2, \dots, u_n\}$. Then any vector $v \in V$ can be expressed uniquely as a linear combination of the basis vectors in S , say

$$v = a_1u_1 + a_2u_2 + \cdots + a_nu_n$$

These n scalars $a_1, a_2, \dots, a_n$ are called the *coordinates* of v relative to the basis S , and they form a vector $[a_1, a_2, \dots, a_n]$ in K^n called the *coordinate vector* of v relative to S . We denote this vector by $[v]_S$, or simply $[v]$, when S is understood. Thus,

$$[v]_S = [a_1, a_2, \dots, a_n]$$

For notational convenience, brackets [...], rather than parentheses (...), are used to denote the coordinate vector.

Remark: The above n scalars $a_1, a_2, \dots, a_n$ also form the *coordinate column vector* $[a_1, a_2, \dots, a_n]^T$ of v relative to S . The choice of the column vector rather than the row vector to represent v depends on the context in which it is used. The use of such column vectors will become clear later in [Chapter 6](#).

EXAMPLE 4.16 Consider the vector space $\mathbf{P}_2(t)$ of polynomials of degree ≤ 2 . The polynomials

$$p_1 = t + 1, \quad p_2 = t - 1, \quad p_3 = (t - 1)^2 = t^2 - 2t + 1$$

form a basis S of $\mathbf{P}_2(t)$. The coordinate vector $[v]$ of $v = 2t^2 - 5t + 9$ relative to S is obtained as follows.

Set $v = xp_1 + yp_2 + zp_3$ using unknown scalars x, y, z , and simplify:

$$\begin{aligned} 2t^2 - 5t + 9 &= x(t + 1) + y(t - 1) + z(t^2 - 2t + 1) \\ &= xt + x + yt - y + zt^2 - 2zt + z \\ &= zt^2 + (x + y - 2z)t + (x - y + z) \end{aligned}$$

Then set the coefficients of the same powers of t equal to each other to obtain the system

$$z = 2, \quad x + y - 2z = -5, \quad x - y + z = 9$$

The solution of the system is $x = 3$, $y = -4$, $z = 2$. Thus,

$$v = 3p_1 - 4p_2 + 2p_3, \text{ and hence, } [v] = [3, -4, 2]$$

EXAMPLE 4.17 Consider real space $\mathbf{R}^3$. The following vectors form a basis S of $\mathbf{R}^3$:

$$u_1 = (1, -1, 0), \quad u_2 = (1, 1, 0), \quad u_3 = (0, 1, 1)$$

The coordinates of $v = (5, 3, 4)$ relative to the basis S are obtained as follows.

Set $v = xv_1 + yv_2 + zv_3$; that is, set v as a linear combination of the basis vectors using unknown scalars x , y , z . This yields

$$\begin{bmatrix} 5 \\ 3 \\ 4 \end{bmatrix} = x \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} + y \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + z \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

The equivalent system of linear equations is as follows:

$$x + y = 5, \quad -x + y + z = 3, \quad z = 4$$

The solution of the system is $x = 3$, $y = 2$, $z = 4$. Thus,

$$v = 3u_1 + 2u_2 + 4u_3, \quad \text{and so} \quad [v]_s = [3, 2, 4]$$

Remark 1: There is a geometrical interpretation of the coordinates of a vector v relative to a basis S for the real space $\mathbf{R}^n$, which we illustrate using the basis S of $\mathbf{R}^3$ in [Example 4.17](#). First consider the space $\mathbf{R}^3$ with the usual x ,

y, z axes. Then the basis vectors determine a new coordinate system of $\mathbf{R}^3$, say with x', y', z' axes, as shown in Fig. 4-4. That is,

- (1) The x' -axis is in the direction of u_1 with unit length $\|u_1\|$.
- (2) The y' -axis is in the direction of u_2 with unit length $\|u_2\|$.
- (3) The z' -axis is in the direction of u_3 with unit length $\|u_3\|$.

Then each vector $v = (a, b, c)$ or, equivalently, the point $P(a, b, c)$ in $\mathbf{R}^3$ will have new coordinates with respect to the new x', y', z' axes. These new coordinates are precisely $[v]_S$, the coordinates of v with respect to the basis S . Thus, as shown in Example 4.17, the coordinates of the point $P(5, 3, 4)$ with the new axes form the vector $[3, 2, 4]$.

Remark 2: Consider the usual basis $E = \{e_1, e_2, \dots, e_n\}$ of K^n defined by

$$e_1 = (1, 0, 0, \dots, 0, 0), \quad e_2 = (0, 1, 0, \dots, 0, 0), \quad \dots, \quad e_n = (0, 0, 0, \dots, 0, 1)$$

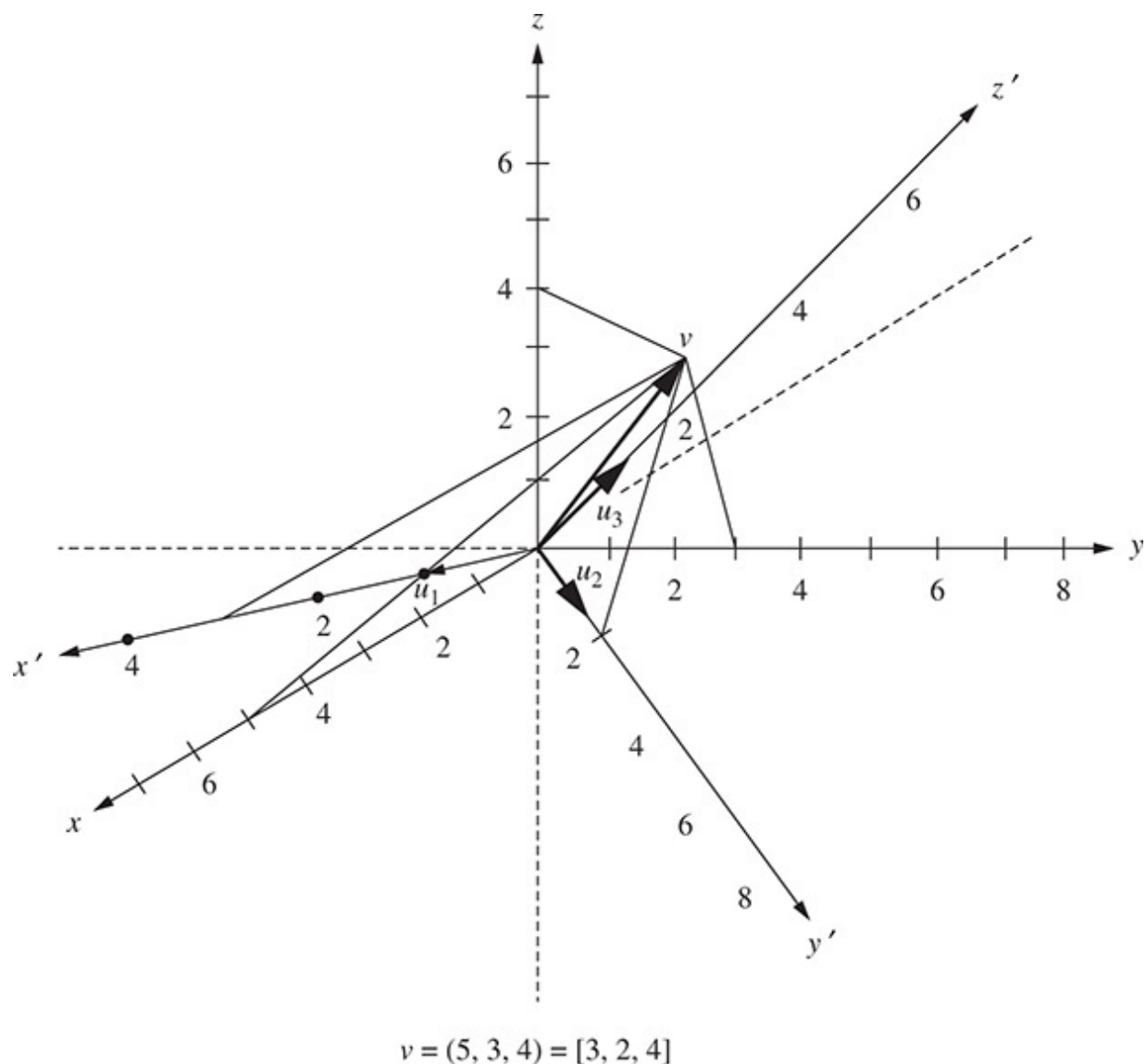


Figure 4-4

Let $v = (a_1, a_2, \dots, a_n)$ be any vector in K^n . Then one can easily show that

$$v = a_1 e_1 + a_2 e_2 + \dots + a_n e_n, \quad \text{and so} \quad [v]_E = [a_1, a_2, \dots, a_n]$$

That is, the coordinate vector $[v]_E$ of any vector v relative to the usual basis E of K^n is identical to the original vector v .

4.12 Isomorphism of V and K^n

Let V be a vector space of dimension n over K , and suppose $S = \{u_1, u_2, \dots, u_n\}$ is a basis of V . Then each vector $v \in V$ corresponds to a unique n -tuple $[v]_S$ in K^n . On the other hand, each n -tuple $[c_1, c_2, \dots, c_n]$ in K^n corresponds to a unique vector $c_1u_1 + c_2u_2 + \dots + c_nu_n$ in V . Thus, the basis S induces a one-to-one correspondence between V and K^n . Furthermore, suppose

$$v = a_1u_1 + a_2u_2 + \dots + a_nu_n \quad \text{and} \quad w = b_1u_1 + b_2u_2 + \dots + b_nu_n$$

Then

$$\begin{aligned} v + w &= (a_1 + b_1)u_1 + (a_2 + b_2)u_2 + \dots + (a_n + b_n)u_n \\ kv &= (ka_1)u_1 + (ka_2)u_2 + \dots + (ka_n)u_n \end{aligned}$$

where k is a scalar. Accordingly,

$$\begin{aligned} [v + w]_S &= [a_1 + b_1, \dots, a_n + b_n] = [a_1, \dots, a_n] + [b_1, \dots, b_n] = [v]_S + [w]_S \\ [kv]_S &= [ka_1, ka_2, \dots, ka_n] = k[a_1, a_2, \dots, a_n] = k[v]_S \end{aligned}$$

Thus, the above one-to-one correspondence between V and K^n preserves the vector space operations of vector addition and scalar multiplication. We then say that V and K^n are isomorphic, written

$$V \cong K^n$$

We state this result formally.

THEOREM 4.24: Let V be an n -dimensional vector space over a field K . Then V and K^n are isomorphic.

The next example gives a practical application of the above result.

EXAMPLE 4.18 Suppose we want to determine whether or not the following matrices in $V = \mathbf{M}_{2,3}$ are linearly dependent:

$$A = \begin{bmatrix} 1 & 2 & -3 \\ 4 & 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 3 & -4 \\ 6 & 5 & 4 \end{bmatrix}, \quad C = \begin{bmatrix} 3 & 8 & -11 \\ 16 & 10 & 9 \end{bmatrix}$$

The coordinate vectors of the matrices in the usual basis of $\mathbf{M}_{2,3}$ are as follows:

$$[A] = [1, 2, -3, 4, 0, 1], \quad [B] = [1, 3, -4, 6, 5, 4], \quad [C] = [3, 8, -11, 16, 10, 9]$$

Form the matrix M whose rows are the above coordinate vectors and reduce M to an echelon form:

$$M = \begin{bmatrix} 1 & 2 & -3 & 4 & 0 & 1 \\ 1 & 3 & -4 & 6 & 5 & 4 \\ 3 & 8 & -11 & 16 & 10 & 9 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & -3 & 4 & 0 & 1 \\ 0 & 1 & -1 & 2 & 5 & 3 \\ 0 & 2 & -2 & 4 & 10 & 6 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & -3 & 4 & 0 & 1 \\ 0 & 1 & -1 & 2 & 5 & 3 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Because the echelon matrix has only two nonzero rows, the coordinate vectors $[A]$, $[B]$, $[C]$ span a subspace of dimension 2 and so are linearly dependent. Accordingly, the original matrices A , B , C are linearly dependent.

4.13 Full Rank Factorization

A matrix B is said to have *full row rank* r if B has r rows that are linearly independent, and a matrix C is said to have *full column rank* r if C has r columns that are linearly independent.

DEFINITION 4.6: Let A be a $m \times n$ matrix of rank r . Then A is said to have the *full rank factorization*

$$A = BC$$

where B has full-column rank r and C has full-row rank r .

THEOREM 4.25: Every matrix A with rank $r > 0$ has a full rank factorization.

There are many full rank factorizations of a matrix A . Fig. 4-5 gives an algorithm to find one such factorization.

Algorithm 4-1: The input is a matrix A of rank $r > 0$. The output is a full rank factorization of A .

Step 1. Find the row canonical form M of A .

Step 2. Let B be the matrix whose columns are the columns of A corresponding to the columns of M with pivots.

Step 3. Let C be the matrix whose rows are the nonzero rows of M .

Then $A = BC$ is a full rank factorization of A .

Figure 4-5

EXAMPLE 4.19 Let $A = \begin{bmatrix} 1 & 1 & -1 & 2 \\ 2 & 2 & -1 & 3 \\ -1 & -1 & 2 & -3 \end{bmatrix}$ where $M = \begin{bmatrix} 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ is the row canonical form of A . We set

$$B = \begin{bmatrix} 1 & -1 \\ 2 & -1 \\ -1 & 2 \end{bmatrix} \quad \text{and} \quad C = \begin{bmatrix} 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & -1 \end{bmatrix}$$

Then $A = BC$ is a full rank factorization of A .

4.14 Generalized (Moore-Penrose) Inverse

Here we assume that the field of scalars is the complex field $\mathbf{C}$ where the matrix A^H is the conjugate transpose of a matrix A . [If A is a real matrix, then $A^H = A^T$.]

DEFINITION 4.7: Let A be an $m \times n$ matrix over $\mathbf{C}$. A matrix, denoted by A^+ , is called the pseudoinverse or Morre–Penrose inverse or MP-inverse of A if A^+ satisfies the following four equations:

$$\begin{aligned} [\text{MP1}] \quad AXA &= A, & [\text{MP3}] \quad (AX)^H &= AX, \\ [\text{MP2}] \quad XAX &= X, & [\text{MP4}] \quad (XA)^H &= XA, \end{aligned}$$

Clearly, A^+ is an $n \times m$ matrix. Also, $A^+ = A^{-1}$ if A is nonsingular.

LEMMA 4.26: A^+ is unique (when it exists).

Proof. Suppose X and Y satisfy the four MP equations. Then

$$AY = (AY)^H = (AXAY)^H = (AY)^H (AX)^H = AYAX = (AYA)X = AX$$

The first and fourth equations use **[MP3]**, and the second and last equations use **[MP1]**. Similarly, $YA = XA$ (which uses **[MP4]** and **[MP1]**). Then,

$$Y = YAY = (YA)Y = (XA)Y = X(AY) = X(AX) = X$$

where the first equation uses **[MP2]**.

LEMMA 4.27: A^+ exists for any matrix A .

[Fig. 4-6](#) gives an algorithm that finds an MP-inverse for any matrix A .

Algorithm 4-2: Input is an $m \times n$ matrix A over $\mathbf{C}$ of rank r . Output is A^+ .

Step 1. Interchange rows and columns of A so that $PAQ = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}$ where A_{11} is a nonsingular $r \times r$ block. [Here P and Q are the products of elementary matrices corresponding to the interchanges of the rows and columns.]

Step 2. Set $B = \begin{bmatrix} A_{11} \\ A_{21} \end{bmatrix}$ and $C = [I_r, A_{11}^{-1} A_{12}]$ where I_r is the $r \times r$ identity matrix.

Step 3. Set $A^+ = Q \left[C^H (CC^H)^{-1} (B^H B)^{-1} B^{11} \right] P$.

Figure 4-6

Combining the above two lemmas we obtain:

THEOREM 4.28: Every matrix A over $\mathbf{C}$ has a unique Moore–Penrose matrix A^+ .

There are special cases when A has full-row rank or full-column rank.

THEOREM 4.29: Let A be a matrix over $\mathbf{C}$.

- (a) If A has full column rank (columns are linearly independent), then $A^+ = (A^H A)^{-1} A^H$.
- (b) If A has full row rank (rows are linearly independent), then $A^+ = A^H (A A^H)^{-1}$.

THEOREM 4.30: Let A be a matrix over $\mathbf{C}$. Suppose $A = BC$ is a full rank factorization of A . Then

$$A^+ = C^+ B^+ = C^H (C C^H)^{-1} (B^H B)^{-1} B^H$$

Moreover, $AA^+ = BB^+$ and $A^+A = C^+C$.

EXAMPLE 4.20 Consider the full rank factorization $A = BC$ in Example D.1; that is,

$$A = \begin{bmatrix} 1 & 1 & -1 & 2 \\ 2 & 2 & -1 & 3 \\ -1 & -1 & 2 & -3 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & -1 \end{bmatrix} = BC$$

Then

$$(CC^H)^{-1} = \frac{1}{5} \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}, C(CC^H)^{-1} = \frac{1}{5} \begin{bmatrix} 2 & 1 \\ 1 & 3 \\ 1 & -2 \end{bmatrix}, (B^HB)^{-1} = \frac{1}{11} \begin{bmatrix} 6 & 5 \\ 5 & 6 \end{bmatrix}, B(B^HB)^{-1} = \frac{1}{11} \begin{bmatrix} 1 & 7 & 4 \\ -1 & 4 & 7 \end{bmatrix}$$

Accordingly, the following is the Moore–Penrose inverse of A :

$$A^+ = \frac{1}{55} \begin{bmatrix} 1 & 18 & 15 \\ 1 & 18 & 15 \\ -2 & 19 & 25 \\ 3 & -1 & -10 \end{bmatrix}$$

4.15 Least-Square Solution

Consider a system $AX = B$ of linear equations. A *least-square solution* of $AX = B$ is the vector of smallest Euclidean norm that minimizes $\|AX - B\|_2$. That vector is

$$X = A^+B$$

[In case A is invertible, so $A^+ = A^{-1}$, then $X = A^{-1}B$, which is the unique solution of the system.]

EXAMPLE 4.21 Consider the following system $AX = B$ of linear equations:

$$\begin{aligned} x + y - z + 2t &= 1 \\ 2x + 2y - z + 3t &= 3 \\ -x - y + 2z - 3t &= 2 \end{aligned}$$

Then, using [Example 4.20](#),

$$A = \begin{bmatrix} 1 & 1 & -1 & 2 \\ 2 & 2 & -1 & 3 \\ -1 & -1 & 2 & -3 \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix}, \quad A^+ = \frac{1}{55} \begin{bmatrix} 1 & 18 & 15 \\ 1 & 18 & 15 \\ -2 & 19 & 25 \\ 3 & -1 & -10 \end{bmatrix}$$

Accordingly,

$$X = A^+B = (1/55)[85, 85, 105, -20]^T = [17/11, 17/11, 21/11, -4/11]^T$$

is the vector of smallest Euclidean norm which minimizes $\|AX - B\|_2$.

SOLVED PROBLEMS

Vector Spaces, Linear Combinations

4.1. Suppose u and v belong to a vector space V . Simplify each of the following expressions:

(a) $E_1 = 3(2u - 4v) + 5u + 7v$,

(b) $E_2 = 3u + 6(3u - 5v) + 7u$,

(c) $E_3 = 2uv + 3(2u + 4v)$

(d) $E_4 = 5u - \frac{3}{v} + 5u$

Multiply out and collect terms:

(a) $E_1 = 6u - 12v + 5u + 7v = 11u - 5v$

(b) $E_2 = 3u + 18u + 30v + 7u = -8u + 30v$

(c) E_3 is not defined because the product uv of vectors is not defined.

(d) E_4 is not defined because division by a vector is not defined.

4.2. Prove [Theorem 4.1](#): Let V be a vector space over a field K .

(i) $k0 = 0$.

(ii) $0u = 0$.

(iii) If $ku = 0$, then $k = 0$ or $u = 0$.

(iv) $(-k)u = k(-u) = -ku$.

(i) By Axiom $[A_2]$ with $u = 0$, we have $0 + 0 = 0$. Hence, by Axiom $[M_1]$, we have

$$k0 = k(0 + 0) = k0 + k0$$

Adding $k0$ to both sides gives the desired result.

(ii) For scalars, $0 + 0 = 0$. Hence, by Axiom $[M_2]$, we have

$$0u = (0 + 0)u = 0u + 0u$$

Adding $-0u$ to both sides gives the desired result.

(iii) Suppose $ku = 0$ and $k \neq 0$. Then there exists a scalar k^{-1} such that $k^{-1}k = 1$. Thus,

$$u = 1u = (k^{-1}k)u = k^{-1}(ku) = k^{-1}0 = 0$$

(iv) Using $u + (-u) = 0$ and $k + (-k) = 0$ yields

$$0 = k0 = k[u + (-u)] = ku + k(-u) \quad \text{and} \quad 0 = 0u = [k + (-k)]u = ku + (-k)u$$

Adding $-ku$ to both sides of the first equation gives $-ku = k(-u)$, and adding $-ku$ to both sides of the

second equation gives $-ku = (-k)u$. Thus, $(-k)u = k(-u) = -ku$.

4.3. Show that (a) $k(u - v) = ku - kv$, (b) $u + u = 2u$.

(a) Using the definition of subtraction, that $u - v = u + (-v)$, and [Theorem 4.1\(iv\)](#), that $k(-v) = -kv$, we have

$$k(u - v) = k[u + (-v)] = ku + k(-v) = ku + (-kv) = ku - kv$$

(b) Using Axiom $[M_4]$ and then Axiom $[M_2]$, we have

$$u + u = 1u + 1u = (1 + 1)u = 2u$$

4.4. Express $v = (1, -2, 5)$ in $\mathbf{R}^3$ as a linear combination of the vectors

$$u_1 = (1, 1, 1), \quad u_2 = (1, 2, 3), \quad u_3 = (2, -1, 1)$$

We seek scalars x, y, z , as yet unknown, such that $v = xu_1 + yu_2 + zu_3$. Thus, we require

$$\begin{bmatrix} 1 \\ -2 \\ 5 \end{bmatrix} = x \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + y \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + z \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix} \quad \text{or} \quad \begin{array}{rcl} x + y + 2z & = & 1 \\ x + 2y - z & = & -2 \\ x + 3y + z & = & 5 \end{array}$$

(For notational convenience, we write the vectors in $\mathbf{R}^3$ as columns, because it is then easier to find the equivalent system of linear equations.) Reducing the system to echelon form yields the triangular system

$$x + y + 2z = 1, \quad y - 3z = -3, \quad 5z = 10$$

The system is consistent and has a solution. Solving by back-substitution yields the solution $x = -6, y = 3, z = 2$. Thus, $v = -6u_1 + 3u_2 + 2u_3$.

Alternatively, write down the augmented matrix M of the equivalent system of linear equations, where u_1, u_2, u_3 are the first three columns of M and v is the last column, and then reduce M to echelon form:

$$M = \begin{bmatrix} 1 & 1 & 2 & 1 \\ 1 & 2 & -1 & -2 \\ 1 & 3 & 1 & 5 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 2 & 1 \\ 0 & 1 & -3 & -3 \\ 0 & 2 & -1 & 4 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 2 & 1 \\ 0 & 1 & -3 & -3 \\ 0 & 0 & 5 & 10 \end{bmatrix}$$

The last matrix corresponds to a triangular system, which has a solution. Solving the triangular system by back-substitution yields the solution $x = -6, y = 3, z = 2$. Thus, $v = -6u_1 + 3u_2 + 2u_3$.

4.5. Express $v = (2, -5, 3)$ in $\mathbf{R}^3$ as a linear combination of the vectors

$$u_1 = (1, -3, 2), \quad u_2 = (2, -4, -1), \quad u_3 = (1, -5, 7)$$

We seek scalars x, y, z , as yet unknown, such that $v = xu_1 + yu_2 + zu_3$. Thus, we require

$$\begin{bmatrix} 2 \\ -5 \\ 3 \end{bmatrix} = x \begin{bmatrix} 1 \\ -3 \\ 2 \end{bmatrix} + y \begin{bmatrix} 2 \\ -4 \\ -1 \end{bmatrix} + z \begin{bmatrix} 1 \\ -5 \\ 7 \end{bmatrix} \quad \text{or} \quad \begin{array}{rcl} x + 2y + z & = & 2 \\ -3x - 4y - 5z & = & -5 \\ 2x - y + 7z & = & 3 \end{array}$$

Reducing the system to echelon form yields the system

$$x + 2y + z = 2, \quad 2y - 2z = 1, \quad 0 = 3$$

The system is inconsistent and so has no solution. Thus, v cannot be written as a linear combination of u_1, u_2, u_3 .

4.6. Express the polynomial $v = t^2 + 4t - 3$ in $\mathbf{P}(t)$ as a linear combination of the polynomials

$$p_1 = t^2 - 2t + 5, \quad p_2 = 2t^2 - 3t, \quad p_3 = t + 1$$

Set v as a linear combination of p_1, p_2, p_3 using unknowns x, y, z to obtain

$$t^2 + 4t - 3 = x(t^2 - 2t + 5) + y(2t^2 - 3t) + z(t + 1) \quad (*)$$

We can proceed in two ways.

Method 1. Expand the right side of (*) and express it in terms of powers of t as follows:

$$\begin{aligned} t^2 + 4t - 3 &= xt^2 - 2xt + 5x + 2yt^2 - 3yt + zt + z \\ &= (x + 2y)t^2 + (-2x - 3y + z)t + (5x + 3z) \end{aligned}$$

Set coefficients of the same powers of t equal to each other, and reduce the system to echelon form. This yields

$$\begin{array}{ccc} \begin{array}{l} x + 2y = 1 \\ -2x - 3y + z = 4 \\ 5x + 3z = -3 \end{array} & \text{or} & \begin{array}{l} x + 2y = 1 \\ y + z = 6 \\ -10y + 3z = -8 \end{array} & \text{or} & \begin{array}{l} x + 2y = 1 \\ y + z = 6 \\ 13z = 52 \end{array} \end{array}$$

The system is consistent and has a solution. Solving by back-substitution yields the solution $x = -3, y = 2, z = 4$. Thus, $v = -3p_1 + 2p_2 + 4p_3$.

Method 2. The equation (*) is an identity in t ; that is, the equation holds for any value of t . Thus, we can set t equal to any numbers to obtain equations in the unknowns.

(a) Set $t = 0$ in (*) to obtain the equation $-3 = 5x + z$.

(b) Set $t = 1$ in (*) to obtain the equation $2 = 4x - y + 2z$.

(c) Set $t = -1$ in (*) to obtain the equation $-6 = 8x + 5y$.

Solve the system of the three equations to again obtain the solution $x = -3$, $y = 2$, $z = 4$. Thus, $v = -3p_1 + 2p_2 + 4p_3$.

4.7. Express M as a linear combination of the matrices A , B , C , where

$$M = \begin{bmatrix} 4 & 7 \\ 7 & 9 \end{bmatrix}, \quad \text{and} \quad A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 1 \\ 4 & 5 \end{bmatrix}$$

Set M as a linear combination of A , B , C using unknown scalars x , y , z ; that is, set $M = xA + yB + zC$. This yields

$$\begin{bmatrix} 4 & 7 \\ 7 & 9 \end{bmatrix} = x \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} + y \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} + z \begin{bmatrix} 1 & 1 \\ 4 & 5 \end{bmatrix} = \begin{bmatrix} x+y+z & x+2y+z \\ x+3y+4z & x+4y+5z \end{bmatrix}$$

Form the equivalent system of equations by setting corresponding entries equal to each other:

$$x + y + z = 4, \quad x + 2y + z = 7, \quad x + 3y + 4z = 7, \quad x + 4y + 5z = 9$$

Reducing the system to echelon form yields

$$x + y + z = 4, \quad y = 3, \quad 3z = -3, \quad 4z = -4$$

The last equation drops out. Solving the system by back-substitution yields $z = -1$, $y = 3$, $x = 2$. Thus, $M = 2A + 3B - C$.

Subspaces

4.8. Prove [Theorem 4.2](#): W is a subspace of V if the following two conditions hold:

(a) $0 \in W$. (b) If $u, v \in W$, then $u + v, ku \in W$.

By (a), W is nonempty, and, by (b), the operations of vector addition and scalar multiplication are well defined for W . Axioms $[A_1]$, $[A_4]$, $[M_1]$, $[M_2]$, $[M_3]$, $[M_4]$ hold in W because the vectors in W belong to V . Thus, we need only show that $[A_2]$ and $[A_3]$ also hold in W . Now $[A_2]$ holds because the zero vector in V belongs to W by (a). Finally, if $v \in W$, then $(-1)v = -v \in W$, and $v + (-v) = 0$. Thus $[A_3]$ holds.

4.9. Let $V = \mathbf{R}^3$. Show that W is not a subspace of V , where

(a) $W = \{(a, b, c) : a \geq 0\}$, (b) $W = \{(a, b, c) : a^2 + b^2 + c^2 \geq 1\}$.

In each case, show that [Theorem 4.2](#) does not hold.

(a) W consists of those vectors whose first entry is nonnegative. Thus, $v = (1, 2, 3)$ belongs to W . Let $k = -3$. Then $kv = (-3, -6, -9)$ does not belong to W , because -3 is negative. Thus, W is not a subspace of V .

(b) W consists of vectors whose length does not exceed 1. Hence, $u = (1, 0, 0)$ and $v = (0, 1, 0)$ belong to W , but $u + v = (1, 1, 0)$ does not belong to W , because $1^2 + 1^2 + 0^2 = 2 > 1$. Thus, W is not a subspace of V .

4.10. Let $V = \mathbf{P}(t)$, the vector space of real polynomials. Determine whether or not W is a subspace of V , where

(a) W consists of all polynomials with integral coefficients.

(b) W consists of all polynomials with degree ≥ 6 and the zero polynomial.

(c) W consists of all polynomials with only even powers of t .

(a) No, because scalar multiples of polynomials in W do not always belong to W . For example,

$$f(t) = 3 + 6t + 7t^2 \in W \quad \text{but} \quad \frac{1}{2}f(t) = \frac{3}{2} + 3t + \frac{7}{2}t^2 \notin W$$

(b and c) Yes. In each case, W contains the zero polynomial, and sums and scalar multiples of polynomials in W belong to W .

4.11. Let V be the vector space of functions $f: \mathbf{R} \rightarrow \mathbf{R}$.

Show that W is a subspace of V , where

(a) $W = \{f(x) : f(1) = 0\}$, all functions whose value at 1 is 0.

(b) $W = \{f(x) : f(3) = f(1)\}$, all functions assigning the same value to 3 and 1.

(c) $W = \{f(t) : f(-x) = f(x)\}$, all *odd functions*.

Let $\hat{0}$ denote the zero function, so $\hat{0}(x) = 0$ for every value of x .

(a) $\hat{0} \in W$, because $\hat{0}(1) = 0$. Suppose $f, g \in W$. Then $f(1) = 0$ and $g(1) = 0$. Also, for scalars a and b , we have

$$(af + bg)(1) = af(1) + bg(1) = a0 + b0 = 0$$

Thus, $af + bg \in W$, and hence W is a subspace.

(b) $\hat{0} \in W$, because $\hat{0}(3) = 0 = \hat{0}(1)$. Suppose $f, g \in W$. Then $f(3) = f(1)$ and $g(3) = g(1)$. Thus, for any scalars a and b , we have

$$(af + bg)(3) = af(3) + bg(3) = af(1) + bg(1) = (af + bg)(1)$$

Thus, $af + bg \in W$, and hence W is a subspace.

(c) $\hat{0} \in W$, because $\hat{0}(-x) = 0 = -0 = -\hat{0}(x)$. Suppose $f, g \in W$. Then $f(-x) = -f(x)$ and $g(-x) = -g(x)$. Also, for scalars a and b ,

$$(af + bg)(-x) = af(-x) + bg(-x) = -af(x) - bg(x) = -(af + bg)(x)$$

Thus, $af + bg \in W$, and hence W is a subspace of V .

4.12. Prove [Theorem 4.3](#): The intersection of any number of subspaces of V is a subspace of V .

Let $\{W_i : i \in I\}$ be a collection of subspaces of V and let $W = \cap (W_i : i \in I)$. Because each W_i is a subspace of V , we have $0 \in W_i$ for every $i \in I$. Hence, $0 \in W$. Suppose $u, v \in W$. Then $u, v \in W_i$ for every $i \in I$. Because each W_i is a subspace, $au + bv \in W_i$ for every $i \in I$. Hence, $au + bv \in W$. Thus, W is a subspace of V .

Linear Spans

4.13. Show that the vectors $u_1 = (1, 1, 1)$, $u_2 = (1, 2, 3)$, $u_3 = (1, 5, 8)$ span $\mathbf{R}^3$.

We need to show that an arbitrary vector $v = (a, b, c)$ in $\mathbf{R}^3$ is a linear combination of u_1, u_2, u_3 . Set $v = xu_1 + yu_2 + zu_3$; that is, set

$$(a, b, c) = x(1, 1, 1) + y(1, 2, 3) + z(1, 5, 8) = (x + y + z, \quad x + 2y + 5z, \quad x + 3y + 8z)$$

Form the equivalent system and reduce it to echelon form:

$$\begin{array}{rcl}
 x + y + z = a & & x + y + z = a \\
 x + 2y + 5z = b & \text{or} & y + 4z = b - a \\
 x + 3y + 8z = c & & 2y + 7z = c - a
 \end{array}
 \quad \text{or} \quad
 \begin{array}{rcl}
 x + y + z = a & & x + y + z = a \\
 y + 4z = b - a & & y + 4z = b - a \\
 -z = c - 2b + a
 \end{array}$$

The above system is in echelon form and is consistent; in fact,

$$x = -a + 5b - 3c, \quad y = 3a - 7b + 4c, \quad z = a + 2b - c$$

is a solution. Thus, u_1, u_2, u_3 span $\mathbf{R}^3$.

4.14. Find conditions on a, b, c so that $v = (a, b, c)$ in $\mathbf{R}^3$ belongs to $W = \text{span}(u_1, u_2, u_3)$, where

$$u_1 = (1, 2, 0), \quad u_2 = (-1, 1, 2), \quad u_3 = (3, 0, -4)$$

Set v as a linear combination of u_1, u_2, u_3 using unknowns x, y, z ; that is, set $v = xu_1 + yu_2 + zu_3$. This yields

$$(a, b, c) = x(1, 2, 0) + y(-1, 1, 2) + z(3, 0, -4) = (x - y + 3z, \quad 2x + y, \quad 2y - 4z)$$

Form the equivalent system of linear equations and reduce it to echelon form:

$$\begin{array}{rcl}
 x - y + 3z = a & & x - y + 3z = a \\
 2x + y = b & \text{or} & 3y - 6z = b - 2a \\
 2y - 4z = c & & 2y - 4z = c
 \end{array}
 \quad \text{or} \quad
 \begin{array}{rcl}
 x - y + 3z = a & & x - y + 3z = a \\
 3y - 6z = b - 2a & & 3y - 6z = b - 2a \\
 0 = 4a - 2b + 3c
 \end{array}$$

The vector $v = (a, b, c)$ belongs to W if and only if the system is consistent, and it is consistent if and only if $4a - 2b + 3c = 0$. Note, in particular, that u_1, u_2, u_3 do not span the whole space $\mathbf{R}^3$.

4.15. Show that the vector space $V = \mathbf{P}(t)$ of real polynomials cannot be spanned by a finite number of polynomials.

Any finite set S of polynomials contains a polynomial of maximum degree, say m . Then the linear span $\text{span}(S)$ of S cannot contain a polynomial of degree greater than m . Thus, $\text{span}(S) \neq V$, for any finite set S .

4.16. Prove [Theorem 4.5](#): Let S be a subset of V . (i) Then $\text{span}(S)$ is a subspace of V containing S . (ii) If W is a subspace of V containing S , then $\text{span}(S) \subseteq W$.

(i) Suppose S is empty. By definition, $\text{span}(S) = \{0\}$. Hence $\text{span}(S) = \{0\}$ is a subspace of V and $S \subseteq \text{span}(S)$. Suppose S is not empty and $v \in S$. Then $v = 1v \in \text{span}(S)$; hence, $S \subseteq \text{span}(S)$. Also

$0 = 0v \in \text{span}(S)$. Now suppose $u, w \in \text{span}(S)$, say

$$u = a_1u_1 + \cdots + a_ru_r = \sum_i a_iu_i \quad \text{and} \quad w = b_1w_1 + \cdots + b_sw_s = \sum_j b_jw_j$$

where $u_i, w_j \in S$ and $a_i, b_j \in K$. Then

$$u + v = \sum_i a_iu_i + \sum_j b_jw_j \quad \text{and} \quad ku = k\left(\sum_i a_iu_i\right) = \sum_i ka_iu_i$$

belong to $\text{span}(S)$ because each is a linear combination of vectors in S . Thus, $\text{span}(S)$ is a subspace of V .

(ii) Suppose $u_1, u_2, \dots, u_r \in S$. Then all the u_i belong to W . Thus, all multiples $a_1u_1, a_2u_2, \dots, a_ru_r \in W$, and so the sum $a_1u_1 + a_2u_2 + \dots + a_ru_r \in W$. That is, W contains all linear combinations of elements in S , or, in other words, $\text{span}(S) \subseteq W$, as claimed.

Linear Dependence

4.17. Determine whether or not u and v are linearly dependent, where

(a) $u = (1, 2), v = (3, -5),$

(b) $u = (1, -3), v = (-2, 6),$

(c) $u = (1, 2, -3), v = (4, 5, -6)$

(d) $u = (2, 4, -8), v = (3, 6, -12)$

Two vectors u and v are linearly dependent if and only if one is a multiple of the other.

(a) No.

(b) Yes; for $v = -2u.$

(c) No.

(d) Yes, for $v = \frac{3}{2}u.$

4.18. Determine whether or not u and v are linearly dependent, where

(a) $u = 2t^2 + 4t - 3, v = 4t^2 + 8t - 6,$

(b) $u = 2t^2 - 3t + 4, v = 4t^2 - 3t + 2,$

(c) $u = \begin{bmatrix} 1 & 3 & -4 \\ 5 & 0 & -1 \end{bmatrix}, v = \begin{bmatrix} -4 & -12 & 16 \\ -20 & 0 & 4 \end{bmatrix},$

(d) $u = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \end{bmatrix}, v = \begin{bmatrix} 2 & 2 & 2 \\ 3 & 3 & 3 \end{bmatrix}$

Two vectors u and v are linearly dependent if and only if one is a multiple of the other.

(a) Yes; for $v = 2u.$

(b) No.

(c) Yes, for $v = -4u.$

(d) No.

4.19. Determine whether or not the vectors $u = (1, 1, 2)$, $v = (2, 3, 1)$, $w = (4, 5, 5)$ in $\mathbf{R}^3$ are linearly dependent.

Method 1. Set a linear combination of u , v , w equal to the zero vector using unknowns x , y , z to obtain the equivalent homogeneous system of linear equations and then reduce the system to echelon form. This yields

$$x \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + y \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix} + z \begin{bmatrix} 4 \\ 5 \\ 5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \text{or} \quad \begin{array}{l} x + 2y + 4z = 0 \\ x + 3y + 5z = 0 \\ 2x + y + 5z = 0 \end{array} \quad \text{or} \quad \begin{array}{l} x + 2y + 4z = 0 \\ y + z = 0 \end{array}$$

The echelon system has only two nonzero equations in three unknowns; hence, it has a free variable and a nonzero solution. Thus, u , v , w are linearly dependent.

Method 2. Form the matrix A whose columns are u , v , w and reduce to echelon form:

$$A = \begin{bmatrix} 1 & 2 & 4 \\ 1 & 3 & 5 \\ 2 & 1 & 5 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 4 \\ 0 & 1 & 1 \\ 0 & -3 & -3 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 4 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

The third column does not have a pivot; hence, the third vector w is a linear combination of the first two vectors u and v . Thus, the vectors are linearly dependent. (Observe that the matrix A is also the coefficient matrix in Method 1. In other words, this method is essentially the same as the first method.)

Method 3. Form the matrix B whose rows are u , v , w , and reduce to echelon form:

$$B = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 3 & 1 \\ 4 & 5 & 5 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & -3 \\ 0 & 1 & -3 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & -3 \\ 0 & 0 & 0 \end{bmatrix}$$

Because the echelon matrix has only two nonzero rows, the three vectors are linearly dependent. (The three given vectors span a space of dimension 2.)

4.20. Determine whether or not each of the following lists of vectors in $\mathbf{R}^3$ is linearly dependent:

(a) $u_1 = (1, 2, 5)$, $u_2 = (1, 3, 1)$, $u_3 = (2, 5, 7)$, $u_4 = (3, 1, 4)$,

(b) $u = (1, 2, 5)$, $v = (2, 5, 1)$, $w = (1, 5, 2)$,

(c) $u = (1, 2, 3)$, $v = (0, 0, 0)$, $w = (1, 5, 6)$.

(a) Yes, because any four vectors in $\mathbf{R}^3$ are linearly dependent.

(b) Use Method 2 above; that is, form the matrix A whose columns are the given vectors, and reduce the matrix to echelon form:

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 5 & 5 \\ 5 & 1 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 3 \\ 0 & -9 & -3 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 24 \end{bmatrix}$$

Every column has a pivot entry; hence, no vector is a linear combination of the previous vectors. Thus, the vectors are linearly independent.

(c) Because $0 = (0, 0, 0)$ is one of the vectors, the vectors are linearly dependent.

4.21. Show that the functions $f(t) = \sin t$, $g(t) = \cos t$, $h(t) = t$ from $\mathbf{R}$ into $\mathbf{R}$ are linearly independent.

Set a linear combination of the functions equal to the zero function $\mathbf{0}$ using unknown scalars x, y, z ; that is, set $xf + yg + zh = \mathbf{0}$. Then show $x = 0$, $y = 0$, $z = 0$. We emphasize that $xf + yg + zh = \mathbf{0}$ means that, for every value of t , we have $xf(t) + yg(t) + zh(t) = 0$.

Thus, in the equation $x \sin t + y \cos t + zt = 0$:

- | | | | | | |
|-------|-----------------|-----------|-----------------------------|----|--------------------|
| (i) | Set $t = 0$ | to obtain | $x(0) + y(1) + z(0) = 0$ | or | $y = 0.$ |
| (ii) | Set $t = \pi/2$ | to obtain | $x(1) + y(0) + z\pi/2 = 0$ | or | $x + \pi z/2 = 0.$ |
| (iii) | Set $t = \pi$ | to obtain | $x(0) + y(-1) + z(\pi) = 0$ | or | $-y + \pi z = 0.$ |

The three equations have only the zero solution; that is, $x = 0, y = 0, z = 0$. Thus, f, g, h are linearly independent.

4.22. Suppose the vectors u, v, w are linearly independent. Show that the vectors $u + v, u - v, u - 2v + w$ are also linearly independent.

Suppose $x(u + v) + y(u - v) + z(u - 2v + w) = 0$.
Then

$$xu + xv + yu - yv + zu - 2zv + zw = 0$$

or

$$(x + y + z)u + (x - y - 2z)v + zw = 0$$

Because u, v, w are linearly independent, the coefficients in the above equation are each 0; hence,

$$x + y + z = 0, \quad x - y - 2z = 0, \quad z = 0$$

The only solution to the above homogeneous system is $x = 0, y = 0, z = 0$. Thus, $u + v, u - v, u - 2v + w$ are linearly independent.

4.23. Show that the vectors $u = (1 + i, 2i)$ and $w = (1, 1 + i)$ in $\mathbf{C}^2$ are linearly dependent over the complex field $\mathbf{C}$ but linearly independent over the real field $\mathbf{R}$.

Recall that two vectors are linearly dependent (over a field K) if and only if one of them is a multiple of the other (by an element in K). Because

$$(1 + i)w = (1 + i)(1, 1 + i) = (1 + i, 2i) = u$$

u and w are linearly dependent over $\mathbf{C}$. On the other hand, u and w are linearly independent over $\mathbf{R}$, as no real multiple of w can equal u . Specifically, when k is real, the first component of $kw = (k, k + ki)$ must be real, and it can never equal the first component $1 + i$ of u , which is complex.

Basis and Dimension

4.24. Determine whether or not each of the following form a basis of $\mathbf{R}^3$:

- (a) $(1, 1, 1), (1, 0, 1)$;
- (b) $(1, 2, 3), (1, 3, 5), (1, 0, 1), (2, 3, 0)$;
- (c) $(1, 1, 1), (1, 2, 3), (2, -1, 1)$;
- (d) $(1, 1, 2), (1, 2, 5), (5, 3, 4)$.

(a and b) No, because a basis of $\mathbf{R}^3$ must contain exactly three elements because $\dim \mathbf{R}^3 = 3$.

(c) The three vectors form a basis if and only if they are linearly independent. Thus, form the matrix whose rows are the given vectors, and row reduce the matrix to echelon form:

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 2 & -1 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & -3 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 5 \end{bmatrix}$$

The echelon matrix has no zero rows; hence, the three vectors are linearly independent, and so they do form a basis of $\mathbf{R}^3$.

(d) Form the matrix whose rows are the given vectors, and row reduce the matrix to echelon form:

$$\begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 5 \\ 5 & 3 & 4 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 3 \\ 0 & -2 & -6 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix}$$

The echelon matrix has a zero row; hence, the three vectors are linearly dependent, and so they do not form a basis of $\mathbf{R}^3$.

- 4.25.** Determine whether $(1, 1, 1, 1)$, $(1, 2, 3, 2)$, $(2, 5, 6, 4)$, $(2, 6, 8, 5)$ form a basis of $\mathbf{R}^4$. If not, find the dimension of the subspace they span.

Form the matrix whose rows are the given vectors, and row reduce to echelon form:

$$B = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 2 \\ 2 & 5 & 6 & 4 \\ 2 & 6 & 8 & 5 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 1 \\ 0 & 3 & 4 & 2 \\ 0 & 4 & 6 & 3 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & -2 & -1 \\ 0 & 0 & -2 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

The echelon matrix has a zero row. Hence, the four vectors are linearly dependent and do not form a basis of $\mathbf{R}^4$. Because the echelon matrix has three nonzero rows, the four vectors span a subspace of dimension 3.

- 4.26.** Extend $\{u_1 = (1, 1, 1, 1), u_2 = (2, 2, 3, 4)\}$ to a basis of $\mathbf{R}^4$.

First form the matrix with rows u_1 and u_2 , and reduce to echelon form:

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 2 & 2 & 3 & 4 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 2 \end{bmatrix}$$

Then $w_1 = (1, 1, 1, 1)$ and $w_2 = (0, 0, 1, 2)$ span the same set of vectors as spanned by u_1 and u_2 . Let $u_3 = (0, 1, 0, 0)$ and $u_4 = (0, 0, 0, 1)$. Then w_1, u_3, w_2, u_4 form a matrix in echelon form. Thus, they are linearly

independent, and they form a basis of $\mathbf{R}^4$. Hence, u_1, u_2, u_3, u_4 also form a basis of $\mathbf{R}^4$.

4.27. Consider the complex field $\mathbf{C}$, which contains the real field $\mathbf{R}$, which contains the rational field $\mathbf{Q}$. (Thus, $\mathbf{C}$ is a vector space over $\mathbf{R}$, and $\mathbf{R}$ is a vector space over $\mathbf{Q}$.)

(a) Show that $\{1, i\}$ is a basis of $\mathbf{C}$ over $\mathbf{R}$; hence, $\mathbf{C}$ is a vector space of dimension 2 over $\mathbf{R}$.

(b) Show that $\mathbf{R}$ is a vector space of infinite dimension over $\mathbf{Q}$.

(a) For any $v \in \mathbf{C}$, we have $v = a + bi = a(1) + b(i)$, where $a, b \in \mathbf{R}$. Hence, $\{1, i\}$ spans $\mathbf{C}$ over $\mathbf{R}$. Furthermore, if $x(1) + y(i) = 0$ or $x + yi = 0$, where $x, y \in \mathbf{R}$, then $x = 0$ and $y = 0$. Hence, $\{1, i\}$ is linearly independent over $\mathbf{R}$. Thus, $\{1, i\}$ is a basis for $\mathbf{C}$ over $\mathbf{R}$.

(b) It can be shown that π is a transcendental number; that is, π is not a root of any polynomial over $\mathbf{Q}$. Thus, for any n , the $n + 1$ real numbers $1, \pi, \pi^2, \dots, \pi^n$ are linearly independent over $\mathbf{Q}$. $\mathbf{R}$ cannot be of dimension n over $\mathbf{Q}$. Accordingly, $\mathbf{R}$ is of infinite dimension over $\mathbf{Q}$.

4.28. Suppose $S = \{u_1, u_2, \dots, u_n\}$ is a subset of V . Show that the following Definitions A and B of a basis of V are equivalent:

(A) S is linearly independent and spans V .

(B) Every $v \in V$ is a unique linear combination of vectors in S .

Suppose (A) holds. Because S spans V , the vector v is a linear combination of the u_i , say

$$v = a_1u_1 + a_2u_2 + \cdots + a_nu_n \quad \text{and} \quad v = b_1u_1 + b_2u_2 + \cdots + b_nu_n$$

Subtracting, we get

$$0 = v - v = (a_1 - b_1)u_1 + (a_2 - b_2)u_2 + \cdots + (a_n - b_n)u_n$$

But the u_i are linearly independent. Hence, the coefficients in the above relation are each 0:

$$a_1 - b_1 = 0, \quad a_2 - b_2 = 0, \quad \dots, \quad a_n - b_n = 0$$

Therefore, $a_1 = b_1$, $a_2 = b_2$, ..., $a_n = b_n$. Hence, the representation of v as a linear combination of the u_i is unique. Thus, (A) implies (B).

Suppose (B) holds. Then S spans V . Suppose

$$0 = c_1u_1 + c_2u_2 + \cdots + c_nu_n$$

However, we do have

$$0 = 0u_1 + 0u_2 + \cdots + 0u_n$$

By hypothesis, the representation of 0 as a linear combination of the u_i is unique. Hence, each $c_i = 0$ and the u_i are linearly independent. Thus, (B) implies (A).

Dimension and Subspaces

4.29. Find a basis and dimension of the subspace W of $\mathbf{R}^3$ where

(a) $W = \{(a, b, c) : a + b + c = 0\},$

(b) $W = \{(a, b, c) : (a = b = c)\}$

(a) Note that $W \neq \mathbf{R}^3$, because, for example, $(1, 2, 3) \notin W$. Thus, $\dim W < 3$. Note that $u_1 = (1, 0, -1)$

and $u_2 = (0, 1, -1)$ are two independent vectors in W . Thus, $\dim W = 2$, and so u_1 and u_2 form a basis of W .

- (b) The vector $u = (1, 1, 1) \in W$. Any vector $w \in W$ has the form $w = (k, k, k)$. Hence, $w = ku$. Thus, u spans W and $\dim W = 1$.

4.30. Let W be the subspace of $\mathbf{R}^4$ spanned by the vectors

$$u_1 = (1, -2, 5, -3), \quad u_2 = (2, 3, 1, -4), \quad u_3 = (3, 8, -3, -5)$$

- (a) Find a basis and dimension of W .
 (b) Extend the basis of W to a basis of $\mathbf{R}^4$.
 (a) Apply Algorithm 4.1, the row space algorithm. Form the matrix whose rows are the given vectors, and reduce it to echelon form:

$$A = \begin{bmatrix} 1 & -2 & 5 & -3 \\ 2 & 3 & 1 & -4 \\ 3 & 8 & -3 & -5 \end{bmatrix} \sim \begin{bmatrix} 1 & -2 & 5 & -3 \\ 0 & 7 & -9 & 2 \\ 0 & 14 & -18 & 4 \end{bmatrix} \sim \begin{bmatrix} 1 & -2 & 5 & -3 \\ 0 & 7 & -9 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

The nonzero rows $(1, -2, 5, -3)$ and $(0, 7, -9, 2)$ of the echelon matrix form a basis of the row space of A and hence of W . Thus, in particular, $\dim W = 2$.

- (b) We seek four linearly independent vectors, which include the above two vectors. The four vectors $(1, -2, 5, -3)$, $(0, 7, -9, 2)$, $(0, 0, 1, 0)$, and $(0, 0, 0, 1)$ are linearly independent (because they form an echelon matrix), and so they form a basis of $\mathbf{R}^4$, which is an extension of the basis of W .

4.31. Let W be the subspace of $\mathbf{R}^5$ spanned by $u_1 = (1, 2, -1, 3, 4)$, $u_2 = (2, 4, -2, 6, 8)$, $u_3 = (1, 3, 2, 2, 6)$, $u_4 =$

$(1, 4, 5, 1, 8)$, $u_5 = (2, 7, 3, 3, 9)$. Find a subset of the vectors that form a basis of W .

Here we use Algorithm 4.2, the casting-out algorithm. Form the matrix M whose columns (not rows) are the given vectors, and reduce it to echelon form:

$$M = \begin{bmatrix} 1 & 2 & 1 & 1 & 2 \\ 2 & 4 & 3 & 4 & 7 \\ -1 & -2 & 2 & 5 & 3 \\ 3 & 6 & 2 & 1 & 3 \\ 4 & 8 & 6 & 8 & 9 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 1 & 1 & 2 \\ 0 & 0 & 1 & 2 & 3 \\ 0 & 0 & 3 & 6 & 5 \\ 0 & 0 & -1 & -2 & -3 \\ 0 & 0 & 2 & 4 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 1 & 1 & 2 \\ 0 & 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 & -4 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

The pivot positions are in columns C_1, C_3, C_5 . Hence, the corresponding vectors u_1, u_3, u_5 form a basis of W , and $\dim W = 3$.

4.32. Let V be the vector space of 2×2 matrices over K . Let W be the subspace of symmetric matrices. Show that $\dim W = 3$, by finding a basis of W .

Recall that a matrix $A = [a_{ij}]$ is symmetric if $A^T = A$, or, equivalently, each $a_{ij} = a_{ji}$. Thus, $A = \begin{bmatrix} a & b \\ b & d \end{bmatrix}$ denotes an arbitrary 2×2 symmetric matrix. Setting (i) $a = 1, b = 0, d = 0$; (ii) $a = 0, b = 1, d = 0$; (iii) $a = 0, b = 0, d = 1$, we obtain the respective matrices:

$$E_1 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad E_2 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad E_3 = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

We claim that $S = \{E_1, E_2, E_3\}$ is a basis of W ; that is, (a) S spans W and (b) S is linearly independent.

(a) The above matrix $A = \begin{bmatrix} a & b \\ b & d \end{bmatrix} = aE_1 + bE_2 + dE_3$. Thus, S spans W .

(b) Suppose $x\mathbf{E}_1 + y\mathbf{E}_2 + z\mathbf{E}_3 = \mathbf{0}$, where x, y, z are unknown scalars. That is, suppose

$$x \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + y \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} + z \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \quad \text{or} \quad \begin{bmatrix} x & y \\ y & z \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Setting corresponding entries equal to each other yields $x = 0, y = 0, z = 0$. Thus, S is linearly independent. Therefore, S is a basis of W , as claimed.

Theorems on Linear Dependence, Basis, and Dimension

4.33. Prove Lemma 4.10: Suppose two or more nonzero vectors $v_1, v_2, \dots, v_m$ are linearly dependent. Then one of them is a linear combination of the preceding vectors.

Because the v_i are linearly dependent, there exist scalars $a_1, \dots, a_m$, not all 0, such that $a_1 v_1 + \dots + a_m v_m = \mathbf{0}$. Let k be the largest integer such that $a_k \neq 0$. Then

$$a_1 v_1 + \dots + a_k v_k + 0v_{k+1} + \dots + 0v_m = \mathbf{0} \quad \text{or} \quad a_1 v_1 + \dots + a_k v_k = \mathbf{0}$$

Suppose $k = 1$; then $a_1 v_1 = \mathbf{0}$, $a_1 \neq 0$, and so $v_1 = \mathbf{0}$. But the v_i are nonzero vectors. Hence, $k > 1$ and

$$v_k = -a_k^{-1} a_1 v_1 - \dots - a_k^{-1} a_{k-1} v_{k-1}$$

That is, v_k is a linear combination of the preceding vectors.

4.34. Suppose $S = \{v_1, v_2, \dots, v_m\}$ spans a vector space V .

(a) If $w \in V$, then $\{w, v_1, \dots, v_m\}$ is linearly dependent and spans V .

(b) If v_i is a linear combination of $v_1, \dots, v_{i-1}$, then S without v_i spans V .

(a) The vector w is a linear combination of the v_i , because $\{v_i\}$ spans V . Accordingly, $\{w, v_1, \dots, v_m\}$ is linearly dependent. Clearly, w with the v_i span V , as the v_i by themselves span V , that is, $\{w, v_1, \dots, v_m\}$ spans V .

(b) Suppose $v_i = k_1 v_1 + \dots + k_{i-1} v_{i-1}$. Let $u \in V$. Because $\{v_i\}$ spans V , u is a linear combination of the v_j 's, say $u = a_1 v_1 + \dots + a_m v_m$. Substituting for v_i , we obtain

$$\begin{aligned} u &= a_1 v_1 + \dots + a_{i-1} v_{i-1} + a_i (k_1 v_1 + \dots + k_{i-1} v_{i-1}) + a_{i+1} v_{i+1} + \dots + a_m v_m \\ &= (a_1 + a_i k_1) v_1 + \dots + (a_{i-1} + a_i k_{i-1}) v_{i-1} + a_{i+1} v_{i+1} + \dots + a_m v_m \end{aligned}$$

Thus, $\{v_1, \dots, v_{i-1}, v_{i+1}, \dots, v_m\}$ spans V . In other words, we can delete v_i from the spanning set and still retain a spanning set.

4.35. Prove Lemma 4.13: Suppose $\{v_1, v_2, \dots, v_n\}$ spans V , and suppose $\{w_1, w_2, \dots, w_m\}$ is linearly independent. Then $m \leq n$, and V is spanned by a set of the form $\{w_1, w_2, \dots, w_m, v_{i_1}, v_{i_2}, \dots, v_{i_{n-m}}\}$

Thus, any $n + 1$ or more vectors in V are linearly dependent.

It suffices to prove the lemma in the case that the v_i are all not 0. (Prove!) Because $\{v_i\}$ spans V , we have by [Problem 4.34](#) that

$$\{w_1, v_1, \dots, v_n\} \tag{1}$$

is linearly dependent and also spans V . By Lemma 4.10, one of the vectors in (1) is a linear combination of the preceding vectors. This vector cannot be w_1 , so it must be one of the v 's, say v_j . Thus by [Problem 4.34](#), we can delete v_j from the spanning set (1) and obtain the spanning set

$$\{w_1, v_1, \dots, v_{j-1}, v_{j+1}, \dots, v_n\} \quad (2)$$

Now we repeat the argument with the vector w_2 . That is, because (2) spans V , the set

$$\{w_1, w_2, v_1, \dots, v_{j-1}, v_{j+1}, \dots, v_n\} \quad (3)$$

is linearly dependent and also spans V . Again by Lemma 4.10, one of the vectors in (3) is a linear combination of the preceding vectors. We emphasize that this vector cannot be w_1 or w_2 , because $\{w_1, \dots, w_m\}$ is independent; hence, it must be one of the v 's, say v_k . Thus, by [Problem 4.34](#), we can delete v_k from the spanning set (3) and obtain the spanning set

$$\{w_1, w_2, v_1, \dots, v_{j-1}, v_{j+1}, \dots, v_{k-1}, v_{k+1}, \dots, v_n\}$$

We repeat the argument with w_3 , and so forth. At each step, we are able to add one of the w 's and delete one of the v 's in the spanning set. If $m \geq n$, then we finally obtain a spanning set of the required form:

$$\{w_1, \dots, w_m, v_{i_1}, \dots, v_{i_{n-m}}\}$$

Finally, we show that $m > n$ is not possible. Otherwise, after n of the above steps, we obtain the spanning set $\{w_1, \dots, w_n\}$. This implies that w_{n+1} is a

linear combination of $w_1, \dots, w_n$, which contradicts the hypothesis that $\{w_i\}$ is linearly independent.

4.36. Prove [Theorem 4.12](#): Every basis of a vector space V has the same number of elements.

Suppose $\{u_1, u_2, \dots, u_n\}$ is a basis of V , and suppose $\{v_1, v_2, \dots\}$ is another basis of V . Because $\{u_i\}$ spans V , the basis $\{v_1, v_2, \dots\}$ must contain n or less vectors, or else it is linearly dependent by [Problem 4.35](#)—Lemma 4.13. On the other hand, if the basis $\{v_1, v_2, \dots\}$ contains less than n elements, then $\{u_1, u_2, \dots, u_n\}$ is linearly dependent by [Problem 4.35](#). Thus, the basis $\{v_1, v_2, \dots\}$ contains exactly n vectors, and so the theorem is true.

4.37. Prove [Theorem 4.14](#): Let V be a vector space of finite dimension n . Then

- (i) Any $n + 1$ or more vectors must be linearly dependent.
- (ii) Any linearly independent set $S = \{u_1, u_2, \dots, u_n\}$ with n elements is a basis of V .
- (iii) Any spanning set $T = \{v_1, v_2, \dots, v_n\}$ of V with n elements is a basis of V .

Suppose $B = \{w_1, w_2, \dots, w_n\}$ is a basis of V .

- (i) Because B spans V , any $n + 1$ or more vectors are linearly dependent by Lemma 4.13.
- (ii) By Lemma 4.13, elements from B can be adjoined to S to form a spanning set of V with n elements. Because S already has n elements, S itself is a spanning set of V . Thus, S is a basis of V .

(iii) Suppose T is linearly dependent. Then some v_i is a linear combination of the preceding vectors. By [Problem 4.34](#), V is spanned by the vectors in T without v_i and there are $n - 1$ of them. By Lemma 4.13, the independent set B cannot have more than $n - 1$ elements. This contradicts the fact that B has n elements. Thus, T is linearly independent, and hence T is a basis of V .

4.38. Prove [Theorem 4.15](#): Suppose S spans a vector space V . Then

- (i) Any maximum number of linearly independent vectors in S form a basis of V .
- (ii) Suppose one deletes from S every vector that is a linear combination of preceding vectors in S . Then the remaining vectors form a basis of V .
- (i) Suppose $\{v_1, \dots, v_m\}$ is a maximum linearly independent subset of S , and suppose $w \in S$. Accordingly, $\{v_1, \dots, v_m, w\}$ is linearly dependent. No v_k can be a linear combination of preceding vectors. Hence, w is a linear combination of the v_i . Thus, $w \in \text{span}(v_i)$, and hence $S \subseteq \text{span}(v_i)$. This leads to

$$V = \text{span}(S) \subseteq \text{span}(v_i) \subseteq V$$

Thus, $\{v_i\}$ spans V , and, as it is linearly independent, it is a basis of V .

- (ii) The remaining vectors form a maximum linearly independent subset of S ; hence, by (i), it is a basis of V .

4.39. Prove [Theorem 4.16](#): Let V be a vector space of finite dimension and let $S = \{u_1, u_2, \dots, u_r\}$ be a set of linearly independent vectors in V . Then S is part of a basis of V ; that is, S may be extended to a basis of V .

Suppose $B = \{w_1, w_2, \dots, w_n\}$ is a basis of V . Then B spans V , and hence V is spanned by

$$S \cup B = \{u_1, u_2, \dots, u_r, w_1, w_2, \dots, w_n\}$$

By [Theorem 4.15](#), we can delete from $S \cup B$ each vector that is a linear combination of preceding vectors to obtain a basis B' for V . Because S is linearly independent, no u_k is a linear combination of preceding vectors. Thus, B' contains every vector in S , and S is part of the basis B' for V .

4.40. Prove [Theorem 4.17](#): Let W be a subspace of an n -dimensional vector space V . Then $\dim W \leq n$. In particular, if $\dim W = n$, then $W = V$.

Because V is of dimension n , any $n + 1$ or more vectors are linearly dependent. Furthermore, because a basis of W consists of linearly independent vectors, it cannot contain more than n elements. Accordingly, $\dim W \leq n$.

In particular, if $\{w_1, \dots, w_n\}$ is a basis of W , then, because it is an independent set with n elements, it is also a basis of V . Thus, $W = V$ when $\dim W = n$.

Rank of a Matrix, Row and Column Spaces

4.41. Find the rank and basis of the row space of each of the following matrices:

(a)

$$A = \begin{bmatrix} 1 & 2 & 0 & -1 \\ 2 & 6 & -3 & -3 \\ 3 & 10 & -6 & -5 \end{bmatrix}, \quad (b) \quad B = \begin{bmatrix} 1 & 3 & 1 & -2 & -3 \\ 1 & 4 & 3 & -1 & -4 \\ 2 & 3 & -4 & -7 & -3 \\ 3 & 8 & 1 & -7 & -8 \end{bmatrix}.$$

(a) Row reduce A to echelon form:

$$A \sim \begin{bmatrix} 1 & 2 & 0 & -1 \\ 0 & 2 & -3 & -1 \\ 0 & 4 & -6 & -2 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 0 & -1 \\ 0 & 2 & -3 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

The two nonzero rows $(1, 2, 0, -1)$ and $(0, 2, -3, -1)$ of the echelon form of A form a basis for $\text{rowsp}(A)$. In particular, $\text{rank}(A) = 2$.

(b) Row reduce B to echelon form:

$$B \sim \begin{bmatrix} 1 & 3 & 1 & -2 & -3 \\ 0 & 1 & 2 & 1 & -1 \\ 0 & -3 & -6 & -3 & 3 \\ 0 & -1 & -2 & -1 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 3 & 1 & -2 & -3 \\ 0 & 1 & 2 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

The two nonzero rows $(1, 3, 1, -2, -3)$ and $(0, 1, 2, 1, -1)$ of the echelon form of B form a basis for $\text{rowsp}(B)$. In particular, $\text{rank}(B) = 2$.

4.42. Show that $U = W$, where U and W are the following subspaces of $\mathbf{R}^3$:

$$U = -\text{span}(u_1, u_2, u_3) = \text{span}(1, 1, -1), (2, 3, -1), (3, 1, -5)\}$$

$$W = \text{span}(w_1, w_2, w_3) = \text{span}(1, -1, -3), (3, -2, -8), (2, 1, -3)\}$$

Form the matrix A whose rows are the u_i , and row reduce A to row canonical form:

$$A = \begin{bmatrix} 1 & 1 & -1 \\ 2 & 3 & -1 \\ 3 & 1 & -5 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & -1 \\ 0 & 1 & 1 \\ 0 & -2 & -2 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

Next form the matrix B whose rows are the w_j , and row reduce B to row canonical form:

$$B = \begin{bmatrix} 1 & -1 & -3 \\ 3 & -2 & -8 \\ 2 & 1 & -3 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & -3 \\ 0 & 1 & 1 \\ 0 & 3 & 3 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

Because A and B have the same row canonical form, the row spaces of A and B are equal, and so $U = W$.

4.43. Let $A = \begin{bmatrix} 1 & 2 & 1 & 2 & 3 & 1 \\ 2 & 4 & 3 & 7 & 7 & 4 \\ 1 & 2 & 2 & 5 & 5 & 6 \\ 3 & 6 & 6 & 15 & 14 & 15 \end{bmatrix}$.

- Find $\text{rank}(M_k)$, for $k = 1, 2, \dots, 6$, where M_k is the submatrix of A consisting of the first k columns $C_1, C_2, \dots, C_k$ of A .
 - Which columns C_{k+1} are linear combinations of preceding columns $C_1, \dots, C_k$?
 - Find columns of A that form a basis for the column space of A .
 - Express column C_4 as a linear combination of the columns in part (c).
- (a) Row reduce A to echelon form:

$$A \sim \begin{bmatrix} 1 & 2 & 1 & 2 & 3 & 1 \\ 0 & 0 & 1 & 3 & 1 & 2 \\ 0 & 0 & 1 & 3 & 2 & 5 \\ 0 & 0 & 3 & 9 & 5 & 12 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 1 & 2 & 3 & 1 \\ 0 & 0 & 1 & 3 & 1 & 2 \\ 0 & 0 & 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Observe that this simultaneously reduces all the matrices M_k to echelon form; for example, the first four columns of the echelon form of A are an echelon form of M_4 . We know that $\text{rank}(M_k)$ is equal to the number of pivots or, equivalently, the number of nonzero rows in an echelon form of M_k . Thus,

$$\begin{aligned} \text{rank}(M_1) = \text{rank}(M_2) = 1, & \quad \text{rank}(M_3) = \text{rank}(M_4) = 2 \\ \text{rank}(M_5) = \text{rank}(M_6) = 3 \end{aligned}$$

- (b) The vector equation $x_1 C_1 + x_2 C_2 + \dots + x_k C_k = C_{k+1}$ yields the system with coefficient matrix M_k and augmented M_{k+1} . Thus, C_{k+1} is a linear combination of $C_1, \dots, C_k$ if and only if $\text{rank}(M_k) = \text{rank}(M_{k+1})$ or, equivalently, if C_{k+1} does not contain a pivot. Thus, each of C_2, C_4, C_6 is a linear combination of preceding columns.
- (c) In the echelon form of A , the pivots are in the first, third, and fifth columns. Thus, columns C_1, C_3, C_5 of A form a basis for the columns space of A . Alternatively, deleting columns C_2, C_4, C_6 from the spanning set of columns (they are linear combinations of other columns), we obtain, again, C_1, C_3, C_5 .
- (d) The echelon matrix tells us that C_4 is a linear combination of columns C_1 and C_3 . The augmented

matrix M of the vector equation $C_4 = xC_1 + yC_2$ consists of the columns C_1, C_3, C_4 of A which, when reduced to echelon form, yields the matrix (omitting zero rows)

$$\begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 3 \end{bmatrix} \quad \text{or} \quad \begin{array}{l} x + y = 2 \\ y = 3 \end{array} \quad \text{or} \quad \begin{array}{l} x = -1, \\ y = 3 \end{array}$$

Thus, $C_4 = -C_1 + 3C_3 = -C_1 + 3C_3 + 0C_5$.

4.44. Suppose $u = (a_1, a_2, \dots, a_n)$ is a linear combination of the rows $R_1, R_2, \dots, R_m$ of a matrix $B = [b_{ij}]$, say $u = k_1R_1 + k_2R_2 + \dots + k_mR_m$. Prove that

$$a_i = k_1b_{1i} + k_2b_{2i} + \dots + k_mb_{mi}, \quad i = 1, 2, \dots, n$$

where $b_{1i}, b_{2i}, \dots, b_{mi}$ are the entries in the i th column of B .

We are given that $u = k_1R_1 + k_2R_2 + \dots + k_mR_m$.

Hence,

$$\begin{aligned} (a_1, a_2, \dots, a_n) &= k_1(b_{11}, \dots, b_{1n}) + \dots + k_m(b_{m1}, \dots, b_{mn}) \\ &= (k_1b_{11} + \dots + k_mb_{m1}, \dots, k_1b_{1n} + \dots + k_mb_{mn}) \end{aligned}$$

Setting corresponding components equal to each other, we obtain the desired result.

4.45. Prove [Theorem 4.7](#): Suppose $A = [a_{ij}]$ and $B = [b_{ij}]$ are row equivalent echelon matrices with respective pivot entries

$$a_{1j_1}, a_{2j_2}, \dots, a_{rj_r} \quad \text{and} \quad b_{1k_1}, b_{2k_2}, \dots, b_{sk_s}$$

(pictured in [Fig. 4-7](#)). Then A and B have the same number of nonzero rows—that is, $r = s$ —and their pivot

entries are in the same positions; that is, $j_1 = k_1, j_2 = k_2, \dots, j_r = k_r$

$$A = \begin{bmatrix} a_{1j_1} & * & * & * & * & * & * \\ & a_{2j_2} & * & * & * & * & * \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ & & & a_{rj_r} & * & * & * \end{bmatrix}, \quad B = \begin{bmatrix} b_{1k_1} & * & * & * & * & * & * \\ & b_{2k_2} & * & * & * & * & * \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ & & & & b_{sk_s} & * & * \end{bmatrix}$$

Figure 4-7

Clearly $A = 0$ if and only if $B = 0$, and so we need only prove the theorem when $r \geq 1$ and $s \geq 1$. We first show that $j_1 = k_1$. Suppose $j_1 < k_1$. Then the j_1 th column of B is zero. Because the first row R^* of A is in the row space of B , we have $R^* = c_1 R_1 + c_2 R_2 + \dots + c_m R_m$, where the R_i are the rows of B . Because the j_1 th column of B is zero, we have

$$a_{1j_1} = c_1 0 + c_2 0 + \dots + c_m 0 = 0$$

But this contradicts the fact that the pivot entry $a_{1j_1} \neq 0$. Hence, $j_1 \geq k_1$ and, similarly, $k_1 \geq j_1$. Thus $j_1 = k_1$.

Now let A' be the submatrix of A obtained by deleting the first row of A , and let B' be the submatrix of B obtained by deleting the first row of B . We prove that A' and B' have the same row space. The theorem will then follow by induction, because A' and B' are also echelon matrices.

Let $R = (a_1, a_2, \dots, a_n)$ be any row of A' and let $R_1, \dots, R_m$ be the rows of B . Because R is in the row space of B , there exist scalars $d_1, \dots, d_m$ such that $R = d_1 R_1 + d_2 R_2 + \dots + d_m R_m$. Because A is in echelon form and R is not the first row of A , the j_1 th entry of R is zero: $a_{j_1} = 0$.

$= 0$ for $i = j_1 = k_1$. Furthermore, because B is in echelon form, all the entries in the k_1 th column of B are 0 except the first: $b_{1k_1} \neq 0$, but $b_{2k_1} = 0, \dots, b_{mk_1} = 0$. Thus,

$$0 = a_{k_1} = d_1 b_{1k_1} + d_2 0 + \dots + d_m 0 = d_1 b_{1k_1}$$

Now $b_{1k_1} \neq 0$ and so $d_1 = 0$. Thus, R is a linear combination of $R_2, \dots, R_m$ and so is in the row space of B' . Because R was any row of A' , the row space of A' is contained in the row space of B' . Similarly, the row space of B' is contained in the row space of A' . Thus, A' and B' have the same row space, and so the theorem is proved.

4.46. Prove [Theorem 4.8](#): Suppose A and B are row canonical matrices. Then A and B have the same row space if and only if they have the same nonzero rows.

Obviously, if A and B have the same nonzero rows, then they have the same row space. Thus we only have to prove the converse.

Suppose A and B have the same row space, and suppose $R \neq 0$ is the i th row of A . Then there exist scalars $c_1, \dots, c_s$ such that

$$R = c_1 R_1 + c_2 R_2 + \dots + c_s R_s \tag{1}$$

where the R_j are the nonzero rows of B . The theorem is proved if we show that $R = R_i$; that is, that $c_i = 1$ but $c_k = 0$ for $k \neq i$.

Let a_{ij} be the pivot entry in R —that is, the first nonzero entry of R . By (1) and [Problem 4.44](#),

$$a_{ij} = c_1 b_{1j} + c_2 b_{2j} + \dots + c_s b_{sj} \tag{2}$$

But, by [Problem 4.45](#), b_{iji} is a pivot entry of B , and, as B is row reduced, it is the only nonzero entry in the j th column of B . Thus, from (2), we obtain $a_{iji} = c_i b_{iji}$. However, $a_{iji} = 1$ and $b_{iji} = 1$, because A and B are row reduced; hence, $c_i = 1$.

Now suppose $k \neq i$, and b_{kjk} is the pivot entry in R_k . By (1) and [Problem 4.44](#),

$$a_{ijk} = c_1 b_{1jk} + c_2 b_{2jk} + \cdots + c_s b_{sjk} \quad (3)$$

Because B is row reduced, b_{kjk} is the only nonzero entry in the j th column of B . Hence, by (3), $a_{ijk} = c_k b_{kjk}$. Furthermore, by [Problem 4.45](#), a_{kjk} is a pivot entry of A , and because A is row reduced, $a_{ijk} = 0$. Thus, $c_k b_{kjk} = 0$, and as $b_{kjk} = 1$, $c_k = 0$. Accordingly $R = R_i$, and the theorem is proved.

4.47. Prove Corollary 4.9: Every matrix A is row equivalent to a unique matrix in row canonical form.

Suppose A is row equivalent to matrices A_1 and A_2 , where A_1 and A_2 are in row canonical form. Then $\text{rowsp}(A) = \text{rowsp}(A_1)$ and $\text{rowsp}(A) = \text{rowsp}(A_2)$. Hence, $\text{rowsp}(A_1) = \text{rowsp}(A_2)$. Because A_1 and A_2 are in row canonical form, $A_1 = A_2$ by [Theorem 4.8](#). Thus, the corollary is proved.

4.48. Suppose RB and AB are defined, where R is a row vector and A and B are matrices. Prove

- (a) RB is a linear combination of the rows of B .
- (b) The row space of AB is contained in the row space of B .

- (c) The column space of AB is contained in the column space of A .
- (d) If C is a column vector and AC is defined, then AC is a linear combination of the columns of A .
- (e) $\text{rank}(AB) \leq \text{rank}(B)$ and $\text{rank}(AB) \geq \text{rank}(A)$.
- (a) Suppose $R = (a_1, a_2, \dots, a_m)$ and $B = [b_{ij}]$. Let $B_1, \dots, B_m$ denote the rows of B and $B^1, \dots, B^n$ its columns. Then

$$\begin{aligned}
 RB &= (RB^1, RB^2, \dots, RB^n) \\
 &= (a_1b_{11} + a_2b_{21} + \dots + a_mb_{m1}, \dots, a_1b_{1n} + a_2b_{2n} + \dots + a_mb_{mn}) \\
 &= a_1(b_{11}, b_{12}, \dots, b_{1n}) + a_2(b_{21}, b_{22}, \dots, b_{2n}) + \dots + a_m(b_{m1}, b_{m2}, \dots, b_{mn}) \\
 &= a_1B_1 + a_2B_2 + \dots + a_mB_m
 \end{aligned}$$

Thus, RB is a linear combination of the rows of B , as claimed.

- (b) The rows of AB are R_iB , where R_i is the i th row of A . Thus, by part (a), each row of AB is in the row space of B . Thus, $\text{rowsp}(AB) \subseteq \text{rowsp}(B)$, as claimed.
- (c) Using part (b), we have $\text{colsp}(AB) = \text{rowsp}(AB)^T = \text{rowsp}(B^T A^T) \subseteq \text{rowsp}(A^T) = \text{colsp}(A)$.
- (d) Follows from (c) where C replaces B .
- (e) The row space of AB is contained in the row space of B ; hence, $\text{rank}(AB) \leq \text{rank}(B)$. Furthermore, the column space of AB is contained in the column space of A ; hence, $\text{rank}(AB) \leq \text{rank}(A)$.

4.49. Let A be an n -square matrix. Show that A is invertible if and only if $\text{rank}(A) = n$.

Note that the rows of the n -square identity matrix I_n are linearly independent, because I_n is in echelon form; hence, $\text{rank}(I_n) = n$. Now if A is invertible, then A is row equivalent to I_n ; hence, $\text{rank}(A) = n$. But if A is not invertible, then A is row equivalent to a matrix with a zero row; hence, $\text{rank}(A) < n$; that is, A is invertible if and only if $\text{rank}(A) = n$.

Applications to Linear Equations

4.50. Find the dimension and a basis of the solution space W of each homogeneous system:

$$x + 2y + 2z - s + 3t = 0$$

$$x + 2y + 3z + s + t = 0$$

$$3x + 6y + 8z + s + 5t = 0$$

(a)

$$x + 2y + z - 2t = 0$$

$$2x + 4y + 4z - 3t = 0$$

$$3x + 6y + 7z - 4t = 0$$

(b)

$$x + y + 2z = 0$$

$$2x + 3y + 3z = 0$$

$$x + 3y + 5z = 0$$

(c)

(a) Reduce the system to echelon form:

$$x + 2y + 2z - s + 3t = 0$$

$$z + 2s - 2t = 0$$

$$2z + 4s - 4t = 0$$

$$x + 2y + 2z - s + 3t = 0$$

$$z + 2s - 2t = 0$$

or

The system in echelon form has two (nonzero) equations in five unknowns. Hence, the system has $5 - 2 = 3$ free variables, which are y, s, t . Thus, $\dim W = 3$. We obtain a basis for W :

(1) Set $y = 1, s = 0, t = 0$ to obtain the solution $v_1 = (-2, 1, 0, 0, 0)$.

(2) Set $y = 0, s = 1, t = 0$ to obtain the solution $v_2 = (5, 0, -2, 1, 0)$.

(3) Set $y = 0, s = 0, t = 1$ to obtain the solution $v_3 = (-7, 0, 2, 0, 1)$.

The set $\{v_1, v_2, v_3\}$ is a basis of the solution space W .

(b) (Here we use the matrix format of our homogeneous system.) Reduce the coefficient matrix A to echelon form:

$$A = \begin{bmatrix} 1 & 2 & 1 & -2 \\ 2 & 4 & 4 & -3 \\ 3 & 6 & 7 & -4 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 1 & -2 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 4 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 1 & -2 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

This corresponds to the system

$$\begin{aligned} x + 2y + 2z - 2t &= 0 \\ 2z + t &= 0 \end{aligned}$$

The free variables are y and t , and $\dim W = 2$.

(i) Set $y = 1$, $z = 0$ to obtain the solution $u_1 = (-2, 1, 0, 0)$.

(ii) Set $y = 0$, $z = 2$ to obtain the solution $u_2 = (6, 0, -1, 2)$.

Then $\{u_1, u_2\}$ is a basis of W .

(c) Reduce the coefficient matrix A to echelon form:

$$A = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 3 & 3 \\ 1 & 3 & 5 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & -1 \\ 0 & 2 & 3 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & -1 \\ 0 & 0 & 5 \end{bmatrix}$$

This corresponds to a triangular system with no free variables. Thus, 0 is the only solution; that is, $W = \{0\}$. Hence, $\dim W = 0$.

4.51. Find a homogeneous system whose solution set W is spanned by

$$\{u_1, u_2, u_3\} = \{(1, -2, 0, 3), \quad (1, -1, -1, 4), \quad (1, 0, -2, 5)\}$$

Let $v = (x, y, z, t)$. Then $v \in W$ if and only if v is a linear combination of the vectors u_1, u_2, u_3 that span W . Thus, form the matrix M whose first columns are u_1, u_2, u_3 and whose last column is v , and then row reduce M to echelon form. This yields

$$M = \begin{bmatrix} 1 & 1 & 1 & x \\ -2 & -1 & 0 & y \\ 0 & -1 & -2 & z \\ 3 & 4 & 5 & t \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 & x \\ 0 & 1 & 2 & 2x+y \\ 0 & -1 & -2 & z \\ 0 & 1 & 2 & -3x+t \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 & x \\ 0 & 1 & 2 & 2x+y \\ 0 & 0 & 0 & 2x+y+z \\ 0 & 0 & 0 & -5x-y+t \end{bmatrix}$$

Then v is a linear combination of u_1, u_2, u_3 if $\text{rank}(M) = \text{rank}(A)$, where A is the submatrix without column v . Thus, set the last two entries in the fourth column on the right equal to zero to obtain the required homogeneous system:

$$\begin{aligned} 2x + y + z &= 0 \\ 5x + y &- t = 0 \end{aligned}$$

4.52. Let $x_{i_1}, x_{i_2}, \dots, x_{i_k}$ be the free variables of a homogeneous system of linear equations with n unknowns. Let v_j be the solution for which $x_{i_j} = 1$, and all other free variables equal 0. Show that the solutions $v_1, v_2, \dots, v_k$ are linearly independent.

Let A be the matrix whose rows are the v_i . We interchange column 1 and column i_1 , then column 2 and column $i_2, \dots$, then column k and column i_k , and we obtain the $k \times n$ matrix

$$B = [I, C] = \begin{bmatrix} 1 & 0 & 0 & \dots & 0 & 0 & c_{1,k+1} & \dots & c_{1n} \\ 0 & 1 & 0 & \dots & 0 & 0 & c_{2,k+1} & \dots & c_{2n} \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & \dots & 0 & 1 & c_{k,k+1} & \dots & c_{kn} \end{bmatrix}$$

The above matrix B is in echelon form, and so its rows are independent; hence, $\text{rank}(B) = k$. Because A and B are column equivalent, they have the same rank— $\text{rank}(A) = k$. But A has k rows; hence, these rows (i.e., the v_i) are linearly independent, as claimed.

Sums, Direct Sums, Intersections

4.53. Let U and W be subspaces of a vector space V . Show that

- (a) $U + W$ is a subspace of V .
 - (b) U and W are contained in $U + W$.
 - (c) $U + W$ is the smallest subspace containing U and W ; that is, $U + W = \text{span}(U, W)$.
 - (d) $W + W = W$.
- (a) Because U and W are subspaces, $0 \in U$ and $0 \in W$. Hence, $0 = 0 + 0$ belongs to $U + W$. Now suppose $u, u' \in U + W$. Then $u = u_1 + w_1$ and $u' = u_2 + w_2$, where $u_1, u_2 \in U$ and $w_1, w_2 \in W$. Then

$$au + bu' = (au_1 + bu_2) + (aw_1 + bw_2) \in U + W$$

Thus, $U + W$ is a subspace of V .

- (b) Let $u \in U$. Because W is a subspace, $0 \in W$. Hence, $u = u + 0$ belongs to $U + W$. Thus, $U \subseteq U + W$. Similarly, $W \subseteq U + W$.
- (c) Because $U + W$ is a subspace of V containing U and W , it must also contain the linear span of U and W . That is, $\text{span}(U, W) \subseteq U + W$.

On the other hand, if $v \in U + W$, then $v = u + w = 1u + 1w$, where $u \in U$ and $w \in W$. Thus, v is a

linear combination of elements in $U \cup W$, and so $v \in \text{span}(U, W)$. Hence, $U + W \subseteq \text{span}(U, W)$.

The two inclusion relations give the desired result.

- (d) Because W is a subspace of V , we have that W is closed under vector addition; hence, $W + W \subseteq W$. By part (a), $W \subseteq W + W$. Hence, $W + W = W$.

4.54. Consider the following subspaces of $\mathbf{R}^5$:

$$U = \text{span}(u_1, u_2, u_3) = \text{span}\{(1, 3, -2, 2, 3), (1, 4, -3, 4, 2), (2, 3, -1, -2, 9)\}$$

$$W = \text{span}(w_1, w_2, w_3) = \text{span}\{(1, 3, 0, 2, 1), (1, 5, -6, 6, 3), (2, 5, 3, 2, 1)\}$$

Find a basis and the dimension of (a) $U + W$, (b) $U \cap W$.

- (a) $U + W$ is the space spanned by all six vectors.

Hence, form the matrix whose rows are the given six vectors, and then row reduce to echelon form:

$$\begin{bmatrix} 1 & 3 & -2 & 2 & 3 \\ 1 & 4 & -3 & 4 & 2 \\ 2 & 3 & -1 & -2 & 9 \\ 1 & 3 & 0 & 2 & 1 \\ 1 & 5 & -6 & 6 & 3 \\ 2 & 5 & 3 & 2 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 3 & -2 & 2 & 3 \\ 0 & 1 & -1 & 2 & -1 \\ 0 & -3 & 3 & -6 & 3 \\ 0 & 0 & 2 & 0 & -2 \\ 0 & 2 & -4 & 4 & 0 \\ 0 & -1 & 7 & -2 & -5 \end{bmatrix} \sim \begin{bmatrix} 1 & 3 & -2 & 2 & 3 \\ 0 & 1 & -1 & 2 & -1 \\ 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

The following three nonzero rows of the echelon matrix form a basis of $U \cap W$:

$$(1, 3, -2, 2, 3), \quad (0, 1, -1, 2, -1), \quad (0, 0, 1, 0, -1)$$

Thus, $\dim(U + W) = 3$.

- (b) Let $v = (x, y, z, s, t)$ denote an arbitrary element in $\mathbf{R}^5$. First find, say as in [Problem 4.49](#), homogeneous systems whose solution sets are U and W , respectively.

Let M be the matrix whose columns are the u_i and v , and reduce M to echelon form:

$$M = \begin{bmatrix} 1 & 1 & 2 & x \\ 3 & 4 & 3 & y \\ -2 & -3 & -1 & z \\ 2 & 4 & -2 & s \\ 3 & 2 & 9 & t \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 2 & x \\ 0 & 1 & -3 & -3x + y \\ 0 & 0 & 0 & -x + y + z \\ 0 & 0 & 0 & 4x - 2y + s \\ 0 & 0 & 0 & -6x + y + t \end{bmatrix}$$

Set the last three entries in the last column equal to zero to obtain the following homogeneous system whose solution set is U :

$$-x + y + z = 0, \quad 4x - 2y + s = 0, \quad -6x + y + t = 0$$

Now let M' be the matrix whose columns are the w_i and v , and reduce M' to echelon form:

$$M' = \begin{bmatrix} 1 & 1 & 2 & x \\ 3 & 5 & 5 & y \\ 0 & -6 & 3 & z \\ 2 & 6 & 2 & s \\ 1 & 3 & 1 & t \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 2 & x \\ 0 & 2 & -1 & -3x + y \\ 0 & 0 & 0 & -9x + 3y + z \\ 0 & 0 & 0 & 4x - 2y + s \\ 0 & 0 & 0 & 2x - y + t \end{bmatrix}$$

Again set the last three entries in the last column equal to zero to obtain the following homogeneous system whose solution set is W :

$$-9x + 3y + z = 0, \quad 4x - 2y + s = 0, \quad 2x - y + t = 0$$

Combine both of the above systems to obtain a homogeneous system, whose solution space is $U \cap W$, and reduce the system to echelon form, yielding

$$\begin{aligned}
 -x + y + z &= 0 \\
 2y + 4z + s &= 0 \\
 8z + 5s + 2t &= 0 \\
 s - 2t &= 0
 \end{aligned}$$

There is one free variable, which is t ; hence, $\dim(U \cap W) = 1$. Setting $t = 2$, we obtain the solution $u = (1, 4, -3, 4, 2)$, which forms our required basis of $U \cap W$.

4.55. Suppose U and W are distinct four-dimensional subspaces of a vector space V , where $\dim V = 6$. Find the possible dimensions of $U \cap W$.

Because U and W are distinct, $U + W$ properly contains U and W ; consequently, $\dim(U + W) > 4$. But $\dim(U + W)$ cannot be greater than 6, as $\dim V = 6$. Hence, we have two possibilities: (a) $\dim(U + W) = 5$ or (b) $\dim(U + W) = 6$. By [Theorem 4.20](#),

$$\dim(U \cap W) = \dim U + \dim W - \dim(U + W) = 8 - \dim(U + W)$$

Thus (a) $\dim(U \cap W) = 3$ or (b) $\dim(U \cap W) = 2$.

4.56. Let U and W be the following subspaces of $\mathbf{R}^3$:

$$U = \{(a, b, c) : a = b = c\} \quad \text{and} \quad W = \{(0, b, c)\}$$

(Note that W is the yz -plane.) Show that $\mathbf{R}^3 = U \oplus W$.

First we show that $U \cap W = \{0\}$. Suppose $v = (a, b, c) \in U \cap W$. Then $a = b = c$ and $a = 0$. Hence, $a = 0$, $b = 0$, $c = 0$. Thus, $v = 0 = (0, 0, 0)$.

Next we show that $\mathbf{R}^3 = U + W$. For, if $v = (a, b, c) \in \mathbf{R}^3$, then

$$v = (a, a, a) + (0, b - a, c - a) \quad \text{where} \quad (a, a, a) \in U \quad \text{and} \quad (0, b - a, c - a) \in W$$

Both conditions $U \cap W = \{0\}$ and $U + W = \mathbf{R}^3$ imply that $\mathbf{R}^3 = U \oplus W$.

4.57. Suppose that U and W are subspaces of a vector space V and that $S = \{u_i\}$ spans U and $S' = \{w_j\}$ spans W . Show that $S \cup S'$ spans $U + W$. (Accordingly, by induction, if S_i spans W_i , for $i = 1, 2, \dots, n$, then $S_1 \cup \dots \cup S_n$ spans $W_1 + \dots + W_n$.)

Let $v \in U + W$. Then $v = u + w$, where $u \in U$ and $w \in W$. Because S spans U , u is a linear combination of u_i , and as S' spans W , w is a linear combination of w_j ; say

$$u = a_1 u_{i_1} + a_2 u_{i_2} + \dots + a_r u_{i_r} \quad \text{and} \quad v = b_1 w_{j_1} + b_2 w_{j_2} + \dots + b_s w_{j_s}$$

where $a_i, b_j \in K$. Then

$$v = u + w = a_1 u_{i_1} + a_2 u_{i_2} + \dots + a_r u_{i_r} + b_1 w_{j_1} + b_2 w_{j_2} + \dots + b_s w_{j_s}$$

Accordingly, $S \cup S' = \{u_i, w_j\}$ spans $U + W$.

4.58. Prove [Theorem 4.20](#): Suppose U and V are finite-dimensional subspaces of a vector space V . Then $U + W$ has finite dimension and

$$\dim(U + W) = \dim U + \dim W - \dim(U \cap W)$$

Observe that $U \cap W$ is a subspace of both U and W . Suppose $\dim U = m$, $\dim W = n$, $\dim(U \cap W) = r$. Suppose $\{v_1, \dots, v_r\}$ is a basis of $U \cap W$. By [Theorem 4.16](#), we can extend $\{v_i\}$ to a basis of U and to a basis of W ; say

$$\{v_1, \dots, v_r, u_1, \dots, u_{m-r}\} \quad \text{and} \quad \{v_1, \dots, v_r, w_1, \dots, w_{n-r}\}$$

are bases of U and W , respectively. Let

$$B = \{v_1, \dots, v_r, u_1, \dots, u_{m-r}, w_1, \dots, w_{n-r}\}$$

Note that B has exactly $m + n - r$ elements. Thus, the theorem is proved if we can show that B is a basis of $U + W$. Because $\{v_i, u_j\}$ spans U and $\{v_i, w_k\}$ spans W , the union $B = \{v_i, u_j, w_k\}$ spans $U + W$. Thus, it suffices to show that B is independent.

Suppose

$$a_1 v_1 + \cdots + a_r v_r + b_1 u_1 + \cdots + b_{m-r} u_{m-r} + c_1 w_1 + \cdots + c_{n-r} w_{n-r} = 0 \quad (1)$$

where a_i, b_j, c_k are scalars. Let

$$v = a_1 v_1 + \cdots + a_r v_r + b_1 u_1 + \cdots + b_{m-r} u_{m-r} \quad (2)$$

By (1), we also have

$$v = -c_1 w_1 - \cdots - c_{n-r} w_{n-r} \quad (3)$$

Because $\{v_i, u_j\} \subseteq U$, $v \in U$ by (2); and as $\{w_k\} \subseteq W$, $v \in W$ by (3). Accordingly, $v \in U \cap W$. Now $\{v_i\}$ is a basis of $U \cap W$, and so there exist scalars $d_1, \dots, d_r$ for which $v = d_1 v_1 + \dots + d_r v_r$. Thus, by (3), we have

$$d_1 v_1 + \cdots + d_r v_r + c_1 w_1 + \cdots + c_{n-r} w_{n-r} = 0$$

But $\{v_i, w_k\}$ is a basis of W , and so is independent. Hence, the above equation forces $c_1 = 0, \dots, c_{n-r} = 0$. Substituting this into (1), we obtain

$$a_1 v_1 + \cdots + a_r v_r + b_1 u_1 + \cdots + b_{m-r} u_{m-r} = 0$$

But $\{v_i, u_j\}$ is a basis of U , and so is independent. Hence, the above equation forces $a_1 = 0, \dots, a_r = 0, b_1 = 0, \dots, b_{m-r} = 0$.

Because (1) implies that the a_i, b_j, c_k are all 0, $B = \{v_i, u_j, w_k\}$ is independent, and the theorem is proved.

4.59. Prove [Theorem 4.21](#): $V = U \oplus W$ if and only if (i) $V = U + W$, (ii) $U \cap W = \{0\}$.

Suppose $V = U \oplus W$. Then any $v \in V$ can be uniquely written in the form $v = u + w$, where $u \in U$ and $w \in W$. Thus, in particular, $V = U + W$. Now suppose $v \in U \cap W$. Then

$$(1) \quad v = v + 0, \text{ where } v \in U, 0 \in W, \quad (2) \quad v = 0 + v, \text{ where } 0 \in U, v \in W.$$

Thus, $v = 0 + 0 = 0$ and $U \cap W = \{0\}$.

On the other hand, suppose $V = U + W$ and $U \cap W = \{0\}$. Let $v \in V$. Because $V = U + W$, there exist $u \in U$ and $w \in W$ such that $v = u + w$. We need to show that such a sum is unique. Suppose also that $v = u' + w'$, where $u' \in U$ and $w' \in W$. Then

$$u + w = u' + w', \quad \text{and so} \quad u - u' = w' - w$$

But $u - u' \in U$ and $w' - w \in W$; hence, by $U \cap W = \{0\}$,

$$u - u' = 0, \quad w' - w = 0, \quad \text{and so} \quad u = u', \quad w = w'$$

Thus, such a sum for $v \in V$ is unique, and $V = U \oplus W$.

4.60. Prove [Theorem 4.22](#) (for two factors): Suppose $V = U \oplus W$. Also, suppose $S = \{u_1, \dots, u_m\}$ and $S' = \{w_1, \dots, w_n\}$ are linearly independent subsets of U and W , respectively. Then

(a) The union $S \cup S'$ is linearly independent in V .

(b) If S and S' are bases of U and W , respectively, then $S \cup S'$ is a basis of V .

(c) $\dim V = \dim U + \dim W$.

(a) Suppose $a_1u_1 + \dots + a_mu_m + b_1w_1 + \dots + b_nw_n = 0$, where a_i, b_j are scalars. Then

$$(a_1u_1 + \dots + a_mu_m) + (b_1w_1 + \dots + b_nw_n) = 0 = 0 + 0$$

where $0, a_1u_1 + \dots + a_mu_m \in U$ and $0, b_1w_1 + \dots + b_nw_n \in W$. Because such a sum for 0 is unique, this leads to

$$a_1u_1 + \dots + a_mu_m = 0 \quad \text{and} \quad b_1w_1 + \dots + b_nw_n = 0$$

Because S_1 is linearly independent, each $a_i = 0$, and because S_2 is linearly independent, each $b_j = 0$. Thus, $S = S_1 \cup S_2$ is linearly independent.

(b) By part (a), $S = S_1 \cup S_2$ is linearly independent, and, by [Problem 4.55](#), $S = S_1 \cup S_2$ spans $V = U + W$. Thus, $S = S_1 \cup S_2$ is a basis of V .

(c) This follows directly from part (b).

Coordinates

4.61. Relative to the basis $S = \{u_1, u_2\} = \{(1, 1), (2, 3)\}$ of $\mathbf{R}^2$, find the coordinate vector of v , where

(a) $v = (4, -3)$, (b) $v = (a, b)$.

In each case, set

$$v = xu_1 + yu_2 = x(1, 1) + y(2, 3) = (x + 2y, x + 3y)$$

and then solve for x and y .

(a) We have

$$(4, -3) = (x + 2y, x + 3y) \quad \text{or} \quad \begin{cases} x + 2y = 4 \\ x + 3y = -3 \end{cases}$$

The solution is $x = 18$, $y = -7$. Hence, $[v] = [18, -7]$.

(b) We have

$$(a, b) = (x + 2y, x + 3y) \quad \text{or} \quad \begin{cases} x + 2y = a \\ x + 3y = b \end{cases}$$

The solution is $x = 3a - 2b$, $y = -a + b$. Hence, $[v] = [3a - 2b, a + b]$.

4.62. Find the coordinate vector of $v = (a, b, c)$ in $\mathbf{R}^3$ relative to

(a) the usual basis $E = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$,

(b) the basis $S = \{u_1, u_2, u_3\} = \{(1, 1, 1), (1, 1, 0), (1, 0, 0)\}$.

(a) Relative to the usual basis E , the coordinates of $[v]_E$ are the same as v . That is, $[v]_E = [a, b, c]$.

(b) Set v as a linear combination of u_1, u_2, u_3 using unknown scalars x, y, z . This yields

$$\begin{bmatrix} a \\ b \\ c \end{bmatrix} = x \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + y \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + z \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad \text{or} \quad \begin{cases} x + y + z = a \\ x + y = b \\ x = c \end{cases}$$

Solving the system yields $x = c$, $y = b - c$, $z = a - b$. Thus, $[v]_S = [c, b - c, a - b]$.

4.63. Consider the vector space $\mathbf{P}_3(t)$ of polynomials of degree ≤ 3 .

(a) Show that $S = \{(t - 1)^3, (t - 1)^2, t - 1, 1\}$ is a basis of $\mathbf{P}_3(t)$.

(b) Find the coordinate vector $[v]$ of $v = 3t^3 - 4t^2 + 2t - 5$ relative to S .

- (a) The degree of $(t - 1)^k$ is k ; writing the polynomials of S in reverse order, we see that no polynomial is a linear combination of preceding polynomials. Thus, the polynomials are linearly independent, and, because $\dim \mathbf{P}_3(t) = 4$, they form a basis of $\mathbf{P}_3(t)$.
- (b) Set v as a linear combination of the basis vectors using unknown scalars x, y, z, s . We have

$$\begin{aligned} v &= 3t^3 + 4t^2 + 2t - 5 = x(t - 1)^3 + y(t - 1)^2 + z(t - 1) + s(1) \\ &= x(t^3 - 3t^2 + 3t - 1) + y(t^2 - 2t + 1) + z(t - 1) + s(1) \\ &= xt^3 - 3xt^2 + 3xt - x + yt^2 - 2yt + y + zt - z + s \\ &= xt^3 + (-3x + y)t^2 + (3x - 2y + z)t + (-x + y - z + s) \end{aligned}$$

Then set coefficients of the same powers of t equal to each other to obtain

$$x = 3, \quad -3x + y = 4, \quad 3x - 2y + z = 2, \quad -x + y - z + s = -5$$

Solving the system yields $x = 3, y = 13, z = 19, s = 4$. Thus, $[v] = [3, 13, 19, 4]$.

4.64. Find the coordinate vector of $A = \begin{bmatrix} 2 & 3 \\ 4 & -7 \end{bmatrix}$ in the real vector space $\mathbf{M} = \mathbf{M}_{2,2}$ relative to

- (a) the basis

$$S = \left\{ \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \right\},$$

- (b) the usual basis

$$E = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$$

- (a) Set A as a linear combination of the basis vectors using unknown scalars x, y, z, t as follows:

$$A = \begin{bmatrix} 2 & 3 \\ 4 & -7 \end{bmatrix} = x \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} + y \begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix} + z \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix} + t \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} x+y+z+t & x-y-z \\ x+y & x \end{bmatrix}$$

Set corresponding entries equal to each other to obtain the system

$$x + y + z + t = 2, \quad x - y - z = 3, \quad x + y = 4, \quad x = -7$$

Solving the system yields $x = -7$, $y = 11$, $z = -21$, $t = 19$. Thus, $[A]_S = [-7, 11, -21, 19]$. (Note that the coordinate vector of A is a vector in $\mathbf{R}^4$, because $\dim \mathbf{M} = 4$.)

(b) Expressing A as a linear combination of the basis matrices yields

$$\begin{bmatrix} 2 & 3 \\ 4 & -7 \end{bmatrix} = x \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + y \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + z \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + t \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} x & y \\ z & t \end{bmatrix}$$

Thus, $x = 2$, $y = 3$, $z = 4$, $t = -7$. Hence, $[A] = [2, 3, 4, -7]$, whose components are the elements of A written row by row.

Remark: This result is true in general; that is, if A is any $m \times n$ matrix in $\mathbf{M} = \mathbf{M}_{m,n}$, then the coordinates of A relative to the usual basis of $\mathbf{M}$ are the elements of A written row by row.

4.65. In the space $\mathbf{M} = \mathbf{M}_{2,3}$, determine whether or not the following matrices are linearly dependent:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 0 & 5 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 4 & 7 \\ 10 & 1 & 13 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 2 & 5 \\ 8 & 2 & 11 \end{bmatrix}$$

If the matrices are linearly dependent, find the dimension and a basis of the subspace W of $\mathbf{M}$ spanned by the matrices.

The coordinate vectors of the above matrices relative to the usual basis of $\mathbf{M}$ are as follows:

$$[A] = [1, 2, 3, 4, 0, 5], \quad [B] = [2, 4, 7, 10, 1, 13], \quad [C] = [1, 2, 5, 8, 2, 11]$$

Form the matrix M whose rows are the above coordinate vectors, and reduce M to echelon form:

$$M = \begin{bmatrix} 1 & 2 & 3 & 4 & 0 & 5 \\ 2 & 4 & 7 & 10 & 1 & 13 \\ 1 & 2 & 5 & 8 & 2 & 11 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 3 & 4 & 0 & 5 \\ 0 & 0 & 1 & 2 & 1 & 3 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Because the echelon matrix has only two nonzero rows, the coordinate vectors $[A]$, $[B]$, $[C]$ span a space of dimension two, and so they are linearly dependent. Thus, A , B , C are linearly dependent. Furthermore, $\dim W = 2$, and the matrices

$$w_1 = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 0 & 5 \end{bmatrix} \quad \text{and} \quad w_2 = \begin{bmatrix} 0 & 0 & 1 \\ 2 & 1 & 3 \end{bmatrix}$$

corresponding to the nonzero rows of the echelon matrix form a basis of W .

Miscellaneous Problems

4.66. Consider a finite sequence of vectors $S = \{\nu_1, \nu_2, \dots, \nu_n\}$. Let T be the sequence of vectors obtained from S by one of the following “elementary operations”: (i) interchange two vectors, (ii) multiply a vector by a nonzero scalar, (iii) add a multiple of one vector to another. Show that S and T span the same space W . Also show that T is independent if and only if S is independent.

Observe that, for each operation, the vectors in T are linear combinations of vectors in S . On the other

hand, each operation has an inverse of the same type (Prove!); hence, the vectors in S are linear combinations of vectors in T . Thus S and T span the same space W . Also, T is independent if and only if $\dim W = n$, and this is true if and only if S is also independent.

4.67. Let $A = [a_{ij}]$ and $B = [b_{ij}]$ be row equivalent $m \times n$ matrices over a field K , and let $v_1, \dots, v_n$ be any vectors in a vector space V over K . Let

$$\begin{array}{ll} u_1 = a_{11}v_1 + a_{12}v_2 + \cdots + a_{1n}v_n & w_1 = b_{11}v_1 + b_{12}v_2 + \cdots + b_{1n}v_n \\ u_2 = a_{21}v_1 + a_{22}v_2 + \cdots + a_{2n}v_n & w_2 = b_{21}v_1 + b_{22}v_2 + \cdots + b_{2n}v_n \\ \cdots & \cdots \\ u_m = a_{m1}v_1 + a_{m2}v_2 + \cdots + a_{mn}v_n & w_m = b_{m1}v_1 + b_{m2}v_2 + \cdots + b_{mn}v_n \end{array}$$

Show that $\{u_i\}$ and $\{w_i\}$ span the same space.

Applying an “elementary operation” of [Problem 4.66](#) to $\{u_i\}$ is equivalent to applying an elementary row operation to the matrix A . Because A and B are row equivalent, B can be obtained from A by a sequence of elementary row operations; hence, $\{w_i\}$ can be obtained from $\{u_i\}$ by the corresponding sequence of operations. Accordingly, $\{u_i\}$ and $\{w_i\}$ span the same space.

4.68. Let $v_1, \dots, v_n$ belong to a vector space V over K , and let $P = [a_{ij}]$ be an n -square matrix over K . Let

$$w_1 = a_{11}v_1 + a_{12}v_2 + \cdots + a_{1n}v_n, \quad \dots, \quad w_n = a_{n1}v_1 + a_{n2}v_2 + \cdots + a_{nn}v_n$$

(a) Suppose P is invertible. Show that $\{w_i\}$ and $\{v_i\}$ span the same space; hence, $\{w_i\}$ is independent if and only if $\{v_i\}$ is independent.

- (b) Suppose P is not invertible. Show that $\{w_i\}$ is dependent.
- (c) Suppose $\{w_i\}$ is independent. Show that P is invertible.
- (a) Because P is invertible, it is row equivalent to the identity matrix I . Hence, by [Problem 4.67](#), $\{w_i\}$ and $\{v_i\}$ span the same space. Thus, one is independent if and only if the other is.
- (b) Because P is not invertible, it is row equivalent to a matrix with a zero row. This means that $\{w_i\}$ spans a space that has a spanning set of less than n elements. Thus, $\{w_i\}$ is dependent.
- (c) This is the contrapositive of the statement of (b), and so it follows from (b).

4.69. Find a homogeneous system whose solution space is spanned by

$$v_1 = (1, 2, 1, 2, 1), \quad v_2 = (1, 3, 2, 5, 3), \quad v_3 = (1, 3, 3, 6, 7)$$

First we seek the orthogonal complement to the v 's, that is, the set of vectors $w = (a, b, c, d, e)$ orthogonal to v_1, v_2, v_3 . Accordingly, we seek the solution to the system

$$\begin{array}{rcl} a + 2b + c + 2d + e = 0 & & a + 2b + c + 2d + e = 0 \\ a + 3b + 2c + 5d + 3e = 0 & \text{or} & b + c + 3d + 2e = 0 \\ a + 3b + 3c + 6d + 7e = 0 & & c + d + 4e = 0 \end{array}$$

Here d and e are free variables. Setting (d, e) equal to $(1, 0)$ and then $(0, 1)$ yields the following two solutions of the system:

$$w_1 = (3, -2, -1, 1, 0) \quad \text{and} \quad w_2 = (5, 2, -4, 0, 1)$$

Thus the homogeneous system follows:

$$3x - 2y - z + s = 0 \quad \text{and} \quad 5x + 2y - 4z + t = 0$$

(Clearly the solution is not unique.)

4.70. Let K be a subfield of a field L , and let L be a subfield of a field E . (Thus, $K \subseteq L \subseteq E$, and K is a subfield of E .) Suppose E is of dimension n over L , and L is of dimension m over K . Show that E is of dimension mn over K .

Suppose $\{\nu_1, \dots, \nu_n\}$ is a basis of E over L and $\{a_1, \dots, a_m\}$ is a basis of L over K . We claim that $\{a_i\nu_j : i = 1, \dots, m, j = 1, \dots, n\}$ is a basis of E over K . Note that $\{a_i\nu_j\}$ contains mn elements.

Let w be any arbitrary element in E . Because $\{\nu_1, \dots, \nu_n\}$ spans E over L , w is a linear combination of the ν_i with coefficients in L :

$$w = b_1 v_1 + b_2 v_2 + \cdots + b_n v_n, \quad b_i \in L \quad (1)$$

Because $\{a_1, \dots, a_m\}$ spans L over K , each $b_i \in L$ is a linear combination of the a_j with coefficients in K :

$$\begin{aligned} b_1 &= k_{11}a_1 + k_{12}a_2 + \cdots + k_{1m}a_m \\ b_2 &= k_{21}a_1 + k_{22}a_2 + \cdots + k_{2m}a_m \\ &\dots\dots\dots \\ b_n &= k_{n1}a_1 + k_{n2}a_2 + \cdots + k_{nm}a_m \end{aligned}$$

where $k_{ji} \in K$. Substituting in (1), we obtain

$$\begin{aligned}
w &= (k_{11}a_1 + \cdots + k_{1m}a_m)v_1 + (k_{21}a_1 + \cdots + k_{2m}a_m)v_2 + \cdots + (k_{n1}a_1 + \cdots + k_{nm}a_m)v_n \\
&= k_{11}a_1v_1 + \cdots + k_{1m}a_mv_1 + k_{21}a_1v_2 + \cdots + k_{2m}a_mv_2 + \cdots + k_{n1}a_1v_n + \cdots + k_{nm}a_mv_n \\
&= \sum_{i,j} k_{ji}(a_iv_j)
\end{aligned}$$

where $k_{ji} \in K$. Thus, w is a linear combination of the a_iv_j with coefficients in K ; hence, $\{a_iv_j\}$ spans E over K .

The proof is complete if we show that $\{a_iv_j\}$ is linearly independent over K . Suppose, for scalars $x_{ji} \in K$, we have $\sum_{i,j} x_{ji}(a_iv_j) = 0$; that is,

$$(x_{11}a_1v_1 + x_{12}a_2v_1 + \cdots + x_{1m}a_mv_1) + \cdots + (x_{n1}a_1v_n + x_{n2}a_2v_n + \cdots + x_{nm}a_mv_n) = 0$$

or

$$(x_{11}a_1 + x_{12}a_2 + \cdots + x_{1m}a_m)v_1 + \cdots + (x_{n1}a_1 + x_{n2}a_2 + \cdots + x_{nm}a_m)v_n = 0$$

Because $\{v_1, \dots, v_n\}$ is linearly independent over L and the above coefficients of the v_i belong to L , each coefficient must be 0:

$$x_{11}a_1 + x_{12}a_2 + \cdots + x_{1m}a_m = 0, \quad \dots, \quad x_{n1}a_1 + x_{n2}a_2 + \cdots + x_{nm}a_m = 0$$

But $\{a_1, \dots, a_m\}$ is linearly independent over K ; hence, because the $x_{ji} \in K$,

$$x_{11} = 0, \ x_{12} = 0, \ \dots, \ x_{1m} = 0, \ \dots, \ x_{n1} = 0, \ x_{n2} = 0, \ \dots, \ x_{nm} = 0$$

Accordingly, $\{a_iv_j\}$ is linearly independent over K , and the theorem is proved.

SUPPLEMENTARY PROBLEMS

Vector Spaces

4.71. Suppose u and v belong to a vector space V . Simplify each of the following expressions:

(a) $E_1 = 4(5u - 6v) + 2(3u + v),$

(b) $E_2 = 5(2u - 3v) + 4(7v + 8),$

(c) $E_3 = 6(3u + 2v) + 5u - 7v,$

(d) $E_4 = 3(5u + 2v).$

4.72. Let V be the set of ordered pairs (a, b) of real numbers with addition in V and scalar multiplication on V defined by

$$(a, b) + (c, d) = (a + c, b + d) \quad \text{and} \quad k(a, b) = (ka, 0)$$

Show that V satisfies all the axioms of a vector space except $[M_4]$ —that is, except $1u = u$. Hence, $[M_4]$ is not a consequence of the other axioms.

4.73. Show that Axiom $[A_4]$ of a vector space V (that $u + v = v + u$) can be derived from the other axioms for V .

4.74. Let V be the set of ordered pairs (a, b) of real numbers. Show that V is not a vector space over $\mathbf{R}$ with addition and scalar multiplication defined by

(i) $(a, b) + (c, d) = (a + d, b + c)$ and $k(a, b) = (ka, kb),$

(ii) $(a, b) + (c, d) = (a + c, b + d)$ and $k(a, b) = (a, b),$

(iii) $(a, b) + (c, d) = (0, 0)$ and $k(a, b) = (ka, kb),$

(iv) $(a, b) + (c, d) = (ac, bd)$ and $k(a, b) = (ka, kb).$

4.75. Let V be the set of infinite sequences $(a_1, a_2, \dots)$ in a field K . Show that V is a vector space over K with addition and scalar multiplication defined by

$$(a_1, a_2, \dots) + (b_1, b_2, \dots) = (a_1 + b_1, a_2 + b_2, \dots) \quad \text{and} \quad k(a_1, a_2, \dots) = (ka_1, ka_2, \dots)$$

4.76. Let U and W be vector spaces over a field K . Let V be the set of ordered pairs (u, w) where $u \in U$ and $w \in W$. Show that V is a vector space over K with addition in V and scalar multiplication on V defined by

$$(u, w) + (u', w') = (u + u', w + w') \quad \text{and} \quad k(u, w) = (ku, kw)$$

(This space V is called the *external direct product* of U and W .)

Subspaces

4.77. Determine whether or not W is a subspace of $\mathbf{R}^3$ where W consists of all vectors (a, b, c) in $\mathbf{R}^3$ such that

- (a) $a = 3b$,
- (b) $a \leq b \leq c$,
- (c) $ab = 0$,
- (d) $a + b + c = 0$,
- (e) $b = a^2$,
- (f) $a = 2b = 3c$.

4.78. Let V be the vector space of n -square matrices over a field K . Show that W is a subspace of V if W consists of all matrices $A = [a_{ij}]$ that are

- (a) symmetric ($A^T = A$ or $a_{ij} = a_{ji}$),
- (b) (upper) triangular,
- (c) diagonal,
- (d) scalar.

4.79. Let $AX = B$ be a nonhomogeneous system of linear equations in n unknowns; that is, $B \neq 0$. Show that the solution set is not a subspace of K^n .

- 4.80.** Suppose U and W are subspaces of V for which $U \cup W$ is a subspace. Show that $U \subseteq W$ or $W \subseteq U$.
- 4.81.** Let V be the vector space of all functions from the real field $\mathbf{R}$ into $\mathbf{R}$. Show that W is a subspace of V where W consists of all: (a) bounded functions, (b) even functions. [Recall that $f: \mathbf{R} \rightarrow \mathbf{R}$ is *bounded* if $\exists M \in \mathbf{R}$ such that $\forall x \in \mathbf{R}$, we have $|f(x)| \leq M$; and $f(x)$ is *even* if $f(-x) = f(x)$, $\forall x \in \mathbf{R}$.]
- 4.82.** Let V be the vector space (Problem 4.75) of infinite sequences $(a_1, a_2, \dots)$ in a field K . Show that W is a subspace of V if W consists of all sequences with (a) 0 as the first element, (b) only a finite number of nonzero elements.

Linear Combinations, Linear Spans

- 4.83.** Consider the vectors $u = (1, 2, 3)$ and $v = (2, 3, 1)$ in $\mathbf{R}^3$.
- (a) Write $w = (1, 3, 8)$ as a linear combination of u and v .
 - (b) Write $w = (2, 4, 5)$ as a linear combination of u and v .
 - (c) Find k so that $w = (1, k, 4)$ is a linear combination of u and v .
 - (d) Find conditions on a, b, c so that $w = (a, b, c)$ is a linear combination of u and v .
- 4.84.** Write the polynomial $f(t) = at^2 + bt + c$ as a linear combination of the polynomials $p_1 = (t - 1)^2$, $p_2 = t - 1$, $p_3 = 1$. [Thus, p_1, p_2, p_3 span the space $\mathbf{P}_2(t)$ of polynomials of degree ≤ 2 .]

- 4.85.** Find one vector in $\mathbf{R}^3$ that spans the intersection of U and W where U is the xy -plane—that is, $U = \{(a, b, 0)\}$ —and W is the space spanned by the vectors $(1, 1, 1)$ and $(1, 2, 3)$.
- 4.86.** Prove that $\text{span}(S)$ is the intersection of all subspaces of V containing S .
- 4.87.** Show that $\text{span}(S) = \text{span}(S \cup \{0\})$. That is, by joining or deleting the zero vector from a set, we do not change the space spanned by the set.
- 4.88.** Show that (a) If $S \subseteq T$, then $\text{span}(S) \subseteq \text{span}(T)$. (b) $\text{span}[\text{span}(S)] = \text{span}(S)$.

Linear Dependence and Linear Independence

- 4.89.** Determine whether the following vectors in $\mathbf{R}^4$ are linearly dependent or independent:
- (a) $(1, 2, -3, 1), (3, 7, 1, -2), (1, 3, 7, -4)$;
- (b) $(1, 3, 1, -2), (2, 5, -1, 3), (1, 3, 7, -2)$.
- 4.90.** Determine whether the following polynomials u, v, w in $\mathbf{P}(t)$ are linearly dependent or independent:
- (a) $u = t^3 - 4t^2 + 3t + 3, v = t^3 + 2t^2 + 4t - 1, w = 2t^3 - t^2 - 3t + 5$;
- (b) $u = t^3 - 5t^2 - 2t + 3, v = t^3 - 4t^2 - 3t + 4, w = 2t^3 - 17t^2 - 7t + 9$.
- 4.91.** Show that the following functions f, g, h are linearly independent:
- (a) $f(t) = e^t, g(t) = \sin t, h(t) = t^2$;
- (b) $f(t) = e^t, g(t) = e^{2t}, h(t) = t$.

4.92. Show that $u = (a, b)$ and $v = (c, d)$ in K^2 are linearly dependent if and only if $ad - bc = 0$.

4.93. Suppose u, v, w are linearly independent vectors. Prove that S is linearly independent where

(a) $S = \{u + v, u + w, v + w\};$

(b) $S = \{u + v - 3w, u + 3v - w, v + w\}.$

4.94. Suppose $\{u_1, \dots, u_r, w_1, \dots, w_s\}$ is a linearly independent subset of V . Show that

$$\text{span}(u_i) \cap \text{span}(w_j) = \{0\}$$

4.95. Suppose $v_1, v_2, \dots, v_n$ are linearly independent. Prove that S is linearly independent where

(a) $S = \{a_1 v_1, a_2 v_2, \dots, a_n v_n\}$ and each $a_i \neq 0$.

(b) $S = \{v_1, \dots, v_{k-1}, w, v_{k+1}, \dots, v_n\}$ and $w = \sum_i b_i v_i$ and $b_k \neq 0$.

4.96. Suppose $(a_{11}, \dots, a_{1n}), (a_{21}, \dots, a_{2n}), \dots, (a_{m1}, \dots, a_{mn})$ are linearly independent vectors in K^n , and suppose $v_1, v_2, \dots, v_n$ are linearly independent vectors in a vector space V over K . Show that the following vectors are also linearly independent:

$$w_1 = a_{11}v_1 + \dots + a_{1n}v_n, \quad w_2 = a_{21}v_1 + \dots + a_{2n}v_n, \quad \dots, \quad w_m = a_{m1}v_1 + \dots + a_{mn}v_n$$

Basis and Dimension

4.97. Find a subset of u_1, u_2, u_3, u_4 that gives a basis for $W = \text{span}(u_i)$ of $\mathbf{R}^5$, where

- (a) $u_1 = (1, 1, 1, 2, 3)$, $u_2 = (1, 2, -1, -2, 1)$, $u_3 = (3, 5, -1, -2, 5)$, $u_4 = (1, 2, 1, -1, 4)$
 (b) $u_1 = (1, -2, 1, 3, -1)$, $u_2 = (-2, 4, -2, -6, 2)$, $u_3 = (1, -3, 1, 2, 1)$, $u_4 = (3, -7, 3, 8, -1)$
 (c) $u_1 = (1, 0, 1, 0, 1)$, $u_2 = (1, 1, 2, 1, 0)$, $u_3 = (2, 1, 3, 1, 1)$, $u_4 = (1, 2, 1, 1, 1)$
 (d) $u_1 = (1, 0, 1, 1, 1)$, $u_2 = (2, 1, 2, 0, 1)$, $u_3 = (1, 1, 2, 3, 4)$, $u_4 = (4, 2, 5, 4, 6)$

4.98. Consider the subspaces $U = \{(a, b, c, d) : b - 2c + d = 0\}$ and $W = \{(a, b, c, d) : a = d, b = 2c\}$ of $\mathbf{R}^4$. Find a basis and the dimension of (a) U , (b) W , (c) $U \cap W$.

4.99. Find a basis and the dimension of the solution space W of each of the following homogeneous systems:

$$\begin{array}{l} x + 2y - 2z + 2s - t = 0 \\ \text{(a) } x + 2y - z + 3s - 2t = 0 \\ \quad 2x + 4y - 7z + s + t = 0 \\ \quad x + 2y - z + 3s - 4t = 0 \\ \text{(b) } 2x + 4y - 2z - s + 5t = 0 \\ \quad 2x + 4y - 2z + 4s - 2t = 0 \end{array}$$

4.100. Find a homogeneous system whose solution space is spanned by the following sets of three vectors:

- (a) $(1, -2, 0, 3, -1)$, $(2, -3, 2, 5, -3)$, $(1, -2, 1, 2, -2)$;
 (b) $(1, 1, 2, 1, 1)$, $(1, 2, 1, 4, 3)$, $(3, 5, 4, 9, 7)$.

4.101. Determine whether each of the following is a basis of the vector space $\mathbf{P}_n(t)$:

- (a) $\{1, 1+t, 1+t+t^2, 1+t+t^2+t^3, \dots, 1+t+t^2+\dots+t^{n-1}+t^n\}$;
 (b) $\{1+t, t+t^2, t^2+t^3, \dots, t^{n-2}+t^{n-1}, t^{n-1}+t^n\}$.

4.102. Find a basis and the dimension of the subspace W of $\mathbf{P}(t)$ spanned by

- (a) $u = t^3 + 2t^2 - 2t + 1$, $v = t^3 + 3t^2 - 3t + 4$, $w = 2t^3 + 7t^2 - 7t + 11$,

(b)

$$u = t^3 + t^2 - 3t + 2, \quad v = 2t^3 + t^2 + t - 4, \quad w = 4t^3 + 3t^2 - 5t + 2.$$

4.103. Find a basis and the dimension of the subspace W of $V = \mathbf{M}_{2,2}$ spanned by

$$A = \begin{bmatrix} 1 & -5 \\ -4 & 2 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 1 \\ -1 & 5 \end{bmatrix}, \quad C = \begin{bmatrix} 2 & -4 \\ -5 & 7 \end{bmatrix}, \quad D = \begin{bmatrix} 1 & -7 \\ -5 & 1 \end{bmatrix}$$

Rank of a Matrix, Row and Column Spaces

4.104. Find the rank of each of the following matrices:

$$(a) \begin{bmatrix} 1 & 3 & -2 & 5 & 4 \\ 1 & 4 & 1 & 3 & 5 \\ 1 & 4 & 2 & 4 & 3 \\ 2 & 7 & -3 & 6 & 13 \end{bmatrix}, \quad (b) \begin{bmatrix} 1 & 2 & -3 & -2 \\ 1 & 3 & -2 & 0 \\ 3 & 8 & -7 & -2 \\ 2 & 1 & -9 & -10 \end{bmatrix}, \quad (c) \begin{bmatrix} 1 & 1 & 2 \\ 4 & 5 & 5 \\ 5 & 8 & 1 \\ -1 & -2 & 2 \end{bmatrix}$$

4.105. For $k = 1, 2, \dots, 5$, find the number n_k of linearly independent subsets consisting of k columns for each of the following matrices:

$$(a) A = \begin{bmatrix} 1 & 1 & 0 & 2 & 3 \\ 1 & 2 & 0 & 2 & 5 \\ 1 & 3 & 0 & 2 & 7 \end{bmatrix}, \quad (b) B = \begin{bmatrix} 1 & 2 & 1 & 0 & 2 \\ 1 & 2 & 3 & 0 & 4 \\ 1 & 1 & 5 & 0 & 6 \end{bmatrix}$$

4.106. Let

$$(a) A = \begin{bmatrix} 1 & 2 & 1 & 3 & 1 & 6 \\ 2 & 4 & 3 & 8 & 3 & 15 \\ 1 & 2 & 2 & 5 & 3 & 11 \\ 4 & 8 & 6 & 16 & 7 & 32 \end{bmatrix}, \quad (b) B = \begin{bmatrix} 1 & 2 & 2 & 1 & 2 & 1 \\ 2 & 4 & 5 & 4 & 5 & 5 \\ 1 & 2 & 3 & 4 & 4 & 6 \\ 3 & 6 & 7 & 7 & 9 & 10 \end{bmatrix}$$

,

For each matrix (where $C_1, \dots, C_6$ denote its columns):

- (i) Find its row canonical form M .
- (ii) Find the columns that are linear combinations of preceding columns.

(iii) Find columns (excluding C_6) that form a basis for the column space.

(iv) Express C_6 as a linear combination of the basis vectors obtained in (iii).

4.107. Determine which of the following matrices have the same row space:

$$A = \begin{bmatrix} 1 & -2 & -1 \\ 3 & -4 & 5 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & -1 & 2 \\ 2 & 3 & -1 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & -1 & 3 \\ 2 & -1 & 10 \\ 3 & -5 & 1 \end{bmatrix}$$

4.108. Determine which of the following subspaces of $\mathbf{R}^3$ are identical:

$$U_1 = \text{span}[(1, 1, -1), (2, 3, -1), (3, 1, -5)], \quad U_2 = \text{span}[(1, -1, -3), (3, -2, -8), (2, 1, -3)] \\ U_3 = \text{span}[(1, 1, 1), (1, -1, 3), (3, -1, 7)]$$

4.109. Determine which of the following subspaces of $\mathbf{R}^4$ are identical:

$$U_1 = \text{span}[(1, 2, 1, 4), (2, 4, 1, 5), (3, 6, 2, 9)], \quad U_2 = \text{span}[(1, 2, 1, 2), (2, 4, 1, 3)], \\ U_3 = \text{span}[(1, 2, 3, 10), (2, 4, 3, 11)]$$

4.110. Find a basis for (i) the row space and (ii) the column space of each matrix M :

$$(a) \quad M = \begin{bmatrix} 0 & 0 & 3 & 1 & 4 \\ 1 & 3 & 1 & 2 & 1 \\ 3 & 9 & 4 & 5 & 2 \\ 4 & 12 & 8 & 8 & 7 \end{bmatrix}, \quad (b) \quad M = \begin{bmatrix} 1 & 2 & 1 & 0 & 1 \\ 1 & 2 & 2 & 1 & 3 \\ 3 & 6 & 5 & 2 & 7 \\ 2 & 4 & 1 & -1 & 0 \end{bmatrix}.$$

4.111. Show that if any row is deleted from a matrix in echelon (respectively, row canonical) form, then the resulting matrix is still in echelon (respectively, row canonical) form.

4.112. Let A and B be arbitrary $m \times n$ matrices. Show that $\text{rank}(A + B) \leq \text{rank}(A) + \text{rank}(B)$.

4.113. Let $r = \text{rank}(A + B)$. Find 2×2 matrices A and B such that

(a) $r < \text{rank}(A), \text{rank}(B)$;

(b) $r = \text{rank}(A) = \text{rank}(B)$;

(c) $r > \text{rank}(A), \text{rank}(B)$.

Sums, Direct Sums, Intersections

4.114. Suppose U and W are two-dimensional subspaces of K^3 . Show that $U \cap W \neq \{0\}$.

4.115. Suppose U and W are subspaces of V such that $\dim U = 4$, $\dim W = 5$, and $\dim V = 7$. Find the possible dimensions of $U \cap W$.

4.116. Let U and W be subspaces of $\mathbf{R}^3$ for which $\dim U = 1$, $\dim W = 2$, and $U \subseteq W$. Show that $\mathbf{R}^3 = U \oplus W$.

4.117. Consider the following subspaces of $\mathbf{R}^5$:

$$U = \text{span}[(1, -1, -1, -2, 0), (1, -2, -2, 0, -3), (1, -1, -2, -2, 1)]$$

$$W = \text{span}[(1, -2, -3, 0, -2), (1, -1, -3, 2, -4), (1, -1, -2, 2, -5)]$$

(a) Find two homogeneous systems whose solution spaces are U and W , respectively.

(b) Find a basis and the dimension of $U \cap W$.

4.118. Let U_1, U_2, U_3 be the following subspaces of $\mathbf{R}^3$:

$$U_1 = \{(a, b, c) : a = c\}, \quad U_2 = \{(a, b, c) : a + b + c = 0\}, \quad U_3 = \{(0, 0, c)\}$$

Show that (a) $\mathbf{R}^3 = U_1 + U_2$, (b) $\mathbf{R}^3 = U_2 + U_3$, (c) $\mathbf{R}^3 = U_1 + U_3$. When is the sum direct?

4.119. Suppose U, W_1, W_2 are subspaces of a vector space V . Show that

$$(U \cap W_1) + (U \cap W_2) \subseteq U \cap (W_1 + W_2)$$

Find subspaces of $\mathbf{R}^2$ for which equality does not hold.

4.120. Suppose $W_1, W_2, \dots, W_r$ are subspaces of a vector space V . Show that

$$(a) \text{span}(W_1, W_2, \dots, W_r) = W_1 + W_2 + \dots + W_r$$

(b) If S_i spans W_i for $i = 1, \dots, r$, then $S_1 \cup S_2 \cup \dots \cup S_r$ spans $W_1 + W_2 + \dots + W_r$.

4.121. Suppose $V = U \oplus W$. Show that $\dim V = \dim U + \dim W$.

4.122. Let S and T be arbitrary nonempty subsets (not necessarily subspaces) of a vector space V and let k be a scalar. The sum $S + T$ and the scalar product kS are defined by

$$S + T = \{u + v : u \in S, v \in T\}, \quad kS = \{ku : u \in S\}$$

[We also write $w + S$ for $\{w\} + S$.] Let

$$S = \{(1, 2), (2, 3)\}, \quad T = \{(1, 4), (1, 5), (2, 5)\}, \quad w = (1, 1), \quad k = 3$$

Find: (a) $S + T$, (b) $w + S$, (c) kS , (d) kT , (e) $kS + kT$, (f) $k(S + T)$.

4.123. Show that the above operations of $S + T$ and kS satisfy

- (a) Commutative law: $S + T = T + S$.
- (b) Associative law: $(S_1 + S_2) + S_3 = S_1 + (S_2 + S_3)$.
- (c) Distributive law: $k(S + T) = kS + kT$.
- (d) $S + \{0\} = \{0\} + S = S$ and $S + V = V + S = V$.

4.124. Let V be the vector space of n -square matrices. Let U be the subspace of upper triangular matrices, and let W be the subspace of lower triangular matrices. Find (a) $U \cap W$, (b) $U + W$.

4.125. Let V be the external direct sum of vector spaces U and W over a field K . (See [Problem 4.76](#).) Let

$$\hat{U} = \{(u, 0) : u \in U\} \quad \text{and} \quad \hat{W} = \{(0, w) : w \in W\}$$

Show that (a) $\hat{U}$ and $\hat{W}$ are subspaces of V , (b) $V = \hat{U} \oplus \hat{W}$.

4.126. Suppose $V = U + W$. Let $\hat{V}$ be the external direct sum of U and W . Show that V is isomorphic to $\hat{V}$ under the correspondence $v = u + w \leftrightarrow (u, w)$.

4.127. Use induction to prove (a) [Theorem 4.22](#), (b) [Theorem 4.23](#).

Coordinates

4.128. The vectors $u_1 = (1, -2)$ and $u_2 = (4, -7)$ form a basis S of $\mathbf{R}^2$. Find the coordinate vector $[\nu]$ of ν relative to S where (a) $\nu = (5, 3)$, (b) $\nu = (a, b)$.

4.129. The vectors $u_1 = (1, 2, 0)$, $u_2 = (1, 3, 2)$, $u_3 = (0, 1, 3)$ form a basis S of $\mathbf{R}^3$. Find the coordinate vector $[\nu]$ of ν relative to S where (a) $\nu = (2, 7, -4)$, (b) $\nu = (a, b, c)$.

4.130. $S = \{t^3 + t^2, t^2 + t, t + 1, 1\}$ is a basis of $\mathbf{P}_3(t)$. Find the coordinate vector $[v]$ of v relative to S where (a) $v = 2t^3 + t^2 - 4t + 2$, (b) $v = at^3 + bt^2 + ct + d$.

4.131. Let $V = \mathbf{M}_{2,2}$. Find the coordinate vector $[A]$ of A relative to S where

$$S = \left\{ \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \right\} \quad \text{and} \quad \text{(a)} \quad A = \begin{bmatrix} 3 & -5 \\ 6 & 7 \end{bmatrix}, \quad \text{(b)} \quad A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

4.132. Find the dimension and a basis of the subspace W of $\mathbf{P}_3(t)$ spanned by

$$u = t^3 + 2t^2 - 3t + 4, \quad v = 2t^3 + 5t^2 - 4t + 7, \quad w = t^3 + 4t^2 + t + 2$$

4.133. Find the dimension and a basis of the subspace W of $\mathbf{M} = \mathbf{M}_{2,3}$ spanned by

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 3 & 1 & 2 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 4 & 3 \\ 7 & 5 & 6 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 2 & 3 \\ 5 & 7 & 6 \end{bmatrix}$$

Miscellaneous Problems

4.134. Answer true or false. If false, prove it with a counterexample.

(a) If u_1, u_2, u_3 span V , then $\dim V = 3$.

(b) If A is a 4×8 matrix, then any six columns are linearly dependent.

(c) If u_1, u_2, u_3 are linearly independent, then u_1, u_2, u_3, w are linearly dependent.

(d) If u_1, u_2, u_3, u_4 are linearly independent, then $\dim V \geq 4$.

(e) If u_1, u_2, u_3 span V , then w, u_1, u_2, u_3 span V .

- (f) If u_1, u_2, u_3, u_4 are linearly independent, then u_1, u_2, u_3 are linearly independent.

4.135. Answer true or false. If false, prove it with a counterexample.

- (a) If any column is deleted from a matrix in echelon form, then the resulting matrix is still in echelon form.
- (b) If any column is deleted from a matrix in row canonical form, then the resulting matrix is still in row canonical form.
- (c) If any column without a pivot is deleted from a matrix in row canonical form, then the resulting matrix is in row canonical form.

4.136. Determine the dimension of the vector space W of the following n -square matrices:

- (a) symmetric matrices,
- (b) antisymmetric matrices,
- (c) scalar matrices,
- (d) diagonal matrices.

4.137. Let $t_1, t_2, \dots, t_n$ be symbols, and let K be any field. Let V be the following set of expressions where $a_i \in K$:

$$a_1 t_1 + a_2 t_2 + \cdots + a_n t_n$$

Define addition in V and scalar multiplication on V by

$$(a_1 t_1 + \cdots + a_n t_n) + (b_1 t_1 + \cdots + b_n t_n) = (a_1 + b_1) t_1 + \cdots + (a_n + b_n) t_n$$

$$k(a_1 t_1 + a_2 t_2 + \cdots + a_n t_n) = k a_1 t_1 + k a_2 t_2 + \cdots + k a_n t_n$$

Show that V is a vector space over K with the above operations. Also, show that $\{t_1, \dots, t_n\}$ is a basis of V , where

$$t_j = 0t_1 + \cdots + 0t_{j-1} + 1t_j + 0t_{j+1} + \cdots + 0t_n$$

4.138. Suppose that $A_1, A_2, \dots$ are linearly independent sets of vectors and that $A_1 \subseteq A_2 \subseteq \dots$

Show that the union $A = A_1 \cup A_2 \cup \dots$ is also linearly independent.

ANSWERS TO SUPPLEMENTARY PROBLEMS

[Some answers, such as bases, need not be unique.]

4.71. (a) $E_1 = 26u + 22v$;

(b) The sum $7v + 8$ is not defined, so E_2 is not defined;

(c) $E_3 = 23u + 5v$;

(d) Division by v is not defined, so E_4 is not defined.

4.77. (a) Yes;

(b) No; e.g., $(1, 2, 3) \in W$ but $-2(1, 2, 3) \notin W$;

(c) No; e.g., $(1, 0, 0), (0, 1, 0) \in W$, but not their sum;

(d) Yes;

(e) No; e.g., $(1, 1, 1) \in W$, but $2(1, 1, 1) \notin W$;

(f) Yes

4.79. The zero vector 0 is not a solution.

4.83. (a) $w = 3u_1 + u_2$,

(b) Impossible,

(c) $k = \frac{11}{5}$,

(d) $7a - 5b + c = 0$

4.84. Using $f = xp_1 + yp_2 + zp_3$, we get $x = a$, $y = 2a + b$,
 $z = a + b + c$

4.85. $v = (2, 1, 0)$

4.89. (a) Dependent,

(b) Independent

4.90. (a) Independent,

(b) Dependent

4.97. (a) u_1, u_2, u_4 ;

(b) u_1, u_2, u_3 ;

(c) u_1, u_2, u_4 ;

(d) u_1, u_2, u_3

4.98. (a) $\dim U = 3$,

(b) $\dim W = 2$,

(c) $\dim(U \cap W) = 1$

4.99. (a) Basis: $\{(2, -1, 0, 0, 0), (4, 0, 1, -1, 0), (3, 0, 1, 0, 1)\}$; $\dim W = 3$;

(b) Basis: $\{(2, -1, 0, 0, 0), (1, 0, 1, 0, 0)\}$; $\dim W = 2$

4.100. (a) $5x + y - z - s = 0$, $x + y - z - t = 0$;

(b) $3x - y - z = 0$, $2x - 3y + s = 0$, $x - 2y + t = 0$

4.101. (a) Yes,

(b) No, because $\dim \mathbf{P}_n(t) = n + 1$, but the set contains only n elements.

4.102. (a) $\dim W = 2$,

(b) $\dim W = 3$

4.103. $\dim W = 2$

4.104. (a) 3,

(b) 2,

(c) 3

4.105. (a) $n_1 = 4, n_2 = 5, n_3 = n_4 = n_5 = 0$;

(b) $n_1 = 4, n_2 = 6, n_3 = 3, n_4 = n_5 = 0$

4.106. (a) (i) $M = [1, 2, 0, 1, 0, 3; 0, 0, 1, 2, 0, 1; 0, 0, 0, 0, 1, 2; 0]$;

(ii) C_2, C_4, C_6 ;

(iii) C_1, C_3, C_5 ;

(iv) $C_6 = 3C_1 + C_3 + 2C_5$.

(b) (i) $M = [1, 2, 0, 0, 3, 1; 0, 0, 1, 0, -1, -1; 0, 0, 0, 1, 1, 2; 0]$;

(ii) C_2, C_5, C_6 ;

(iii) C_1, C_3, C_4 ;

(iv) $C_6 = C_1 - C_3 + 2C_4$

4.107. A and C are row equivalent to $\begin{bmatrix} 1 & 0 & 7 \\ 0 & 1 & 4 \end{bmatrix}$, but not B

4.108. U_1 and U_2 are row equivalent to $\begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 1 \end{bmatrix}$, but not U_3

4.109. U_1 and U_3 are row equivalent to $\begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 3 \end{bmatrix}$ but not U_2

4.110. (a) (i) $(1, 3, 1, 2, 1), (0, 0, 1, -1, -1), (0, 0, 0, 4, 7)$;
(ii) C_1, C_3, C_4 ;
(b) (i) $(1, 2, 1, 0, 1), (0, 0, 1, 1, 2)$;
(ii) C_1, C_3

4.113. (a) $A = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}, B = \begin{bmatrix} -1 & -1 \\ 0 & 0 \end{bmatrix}$;

(b) $A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, B = \begin{bmatrix} 0 & 2 \\ 0 & 0 \end{bmatrix}$;

(c) $A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, B = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$

4.115. $\dim(U \cap W) = 2, 3, \text{ or } 4$

4.117. (a) (i) $\begin{matrix} 3x + 4y - z - t = 0 \\ 4x + 2y + s = 0 \end{matrix}$ (ii) $\begin{matrix} 4x + 2y - s = 0 \\ 9x + 2y + z + t = 0 \end{matrix}$;

(b) Basis: $\{(1, -2, -2, -5, 0, 0, 0), (0, 0, 1, 0, -1)\}$;
 $\dim(U \cap W) = 2$

4.118. The sum is direct in (b) and (c).

4.119. In $\mathbf{R}^2$, let U, V, W be, respectively, the line $y = x$, the x -axis, the y -axis.

4.122.

(a) $\{(2, 6), (2, 7), (3, 7), (3, 8), (4, 8)\}$; (b) $\{(2, 3), (3, 4)\}$;
(c) $\{(3, 6), (6, 9)\}$; (d) $\{(3, 12), (3, 15), (6, 15)\}$;
(e and f) $\{(6, 18), (6, 21), (9, 21), (9, 24), (12, 24)\}$

4.124. (a) Diagonal matrices,

(b) V

- 4.128.** (a) $[-47, 13]$,
 (b) $[-7a - 4b, 2a + b]$
- 4.129.** (a) $[-11, 13, -10]$,
 (b) $[c - 3b + 7a, -c + 3b - 6a, c - 2b + 4a]$
- 4.130.** (a) $[2, -1, -2, 2]$,
 (b) $[a, b - c, c - b + a, d - c + b - a]$
- 4.131.** (a) $[7, -1, -13, 10]$,
 (b) $[d, c - d, b + c - 2d, a - b - 2c + 2d]$
- 4.132.** $\dim W = 2$; basis: $\{t^3 + 2t^2 - 3t + 4, t^2 + 2t - 1\}$
- 4.133.** $\dim W = 2$; basis: $\{[1, 2, 1, 3, 1, 2], [0, 0, 1, 1, 3, 2]\}$
- 4.134.** (a) False; $(1, 1), (1, 2), (2, 1)$ span $\mathbf{R}^2$;
 (b) True;
 (c) False; $(1, 0, 0, 0), (0, 1, 0, 0), (0, 0, 1, 0), w = (0, 0, 0, 1)$;
 (d) True;
 (e) True;
 (f) True
- 4.135.** (a) True;
 (b) False; e.g. delete C_2 from $\begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 2 \end{bmatrix}$;
 (c) True
- 4.136.** (a) $\frac{1}{2}n(n+1)$, (b) $\frac{1}{2}n(n-1)$, (c) n , (d) 1